Quasipolynomial Cryptanalysis of the McEliece Cryptosystem (or: PIR Meets McEliece)*

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Abstract

The McEliece code-based cryptosystem, utilizing binary Goppa codes, is the earliest public-key encryption scheme that is still considered post-quantum secure. We present a simple, classical quasipolynomial-time distinguisher for Goppa–McEliece in the asymptotic "Classic McEliece" regime: for code length n, extension degree $m = \Theta(\log n)$, Goppa degree $t = \Theta(n/\log n)$, and public-code dimension $k = \Theta(n)$, the algorithm runs in time $n^{\mathcal{O}(\log n)}$ and distinguishes the McEliece public key from the uniform distribution over $\mathbb{F}_2^{k\times n}$ with advantage $1 - o(1)$. The distinguisher is not merely asymptotic: it applies to all Classic McEliece parameter sets considered in the NIST process and yields improved (though not yet practical) concrete attack estimates.

Our distinguishing attack originated from a failed attempt to construct doubly efficient private information retrieval (PIR) protocols from algebraic locally decodable codes, and can be intuitively explained from the PIR perspective. We extend this provable algorithm to a heuristic $n^{\mathcal{O}(\log n)}$-time ciphertext-decryption attack that recovers the message from a noisy codeword and a heuristic $n^{\mathcal{O}(\log n)}$-time key-recovery attack that outputs an equivalent decryption key. While the decryption attack is not concretely efficient, the key-recovery attack is much closer to the distinguisher and may be relevant to NIST security levels.

*This is an updated version of [GIJS26], which includes an extension of the distinguisher to a key-recovery algorithm. See Section 1.1 for an account of the changes and concurrent key-recovery algorithms.

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1 Introduction

In 1978, McEliece proposed an elegant method to construct public-key encryption from error-correcting codes [McE78]. The idea is to disguise an efficiently decodable code C by letting the public key include a random generator matrix of a secretly permuted version C' of C. If the public key is indistinguishable from a random matrix, then a noisy random codeword from C' is hard to decode given only the public key (under the currently standard "Learning Parity with Noise" assumption), but can be easily decoded using the secret decoder.

More concretely, the construction proceeds as follows. Let $C \subseteq \mathbb{F}_q^n$ be a K-dimensional linear code with generator matrix $\boldsymbol{G} \in \mathbb{F}_q^{K \times n}$ that determines an efficient algorithm for correcting every error vector $\boldsymbol{e}$ with support size $|\mathrm{supp}(\boldsymbol{e})| \leqslant t$. To hide the structure of C, the key-generation algorithm samples a full-rank matrix $\boldsymbol{S} \in \mathbb{F}_q^{k \times K}$ and a permutation matrix $\boldsymbol{P} \in \mathbb{F}_q^{n \times n}$. The public key is the disguised generator matrix

$$\boldsymbol{Y} = \boldsymbol{SGP}.$$

In the original construction, we have $k = K$ and the matrix $\boldsymbol{S}$ is invertible, but in this work we consider the more general version with a rectangular $\boldsymbol{S}$ to allow for random subcodes of the permuted error-correcting code. A message $\boldsymbol{u} \in \mathbb{F}_q^k$ is encrypted as

$$\boldsymbol{c} = \boldsymbol{uY} + \boldsymbol{e},$$

where $\boldsymbol{e} \in \mathbb{F}_q^n$ is sampled from a specified distribution supported on vectors satisfying $|\mathrm{supp}(\boldsymbol{e})| \leqslant t$. The holder of the secret key $(\boldsymbol{S}, \boldsymbol{G}, \boldsymbol{P})$ applies $\boldsymbol{P}^{-1}$ to $\boldsymbol{c}$, and then decodes the resulting $\boldsymbol{c}' = \boldsymbol{cP}^{-1}$ using $\boldsymbol{G}$ to recover $\boldsymbol{uS}$, and finally recovers $\boldsymbol{u}$ using $\boldsymbol{S}$; an adversary, by contrast, is faced with decoding a noisy codeword in an apparently unstructured public code.

The popular Goppa–McEliece instantiation, originally proposed by McEliece, uses binary irreducible Goppa codes defined as follows. Let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$ be a support of distinct elements of $\mathbb{F}_{2^m}$, let $\Gamma \in \mathbb{F}_{2^m}[Z]$ be an irreducible polynomial of degree t satisfying $\Gamma(\alpha_i) \neq 0$ for every $i \in [n]$, and set $\boldsymbol{\mu} := \big(\Gamma(\alpha_1)^{-1}, \ldots, \Gamma(\alpha_n)^{-1}\big)$. The corresponding binary Goppa code is

$$\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma) := \mathsf{GRS}_t(\boldsymbol{\alpha}, \boldsymbol{\mu})^{\perp} \cap \mathbb{F}_2^n,$$

where the dual is taken over $\mathbb{F}_{2^m}$. (By $\mathsf{GRS}_t(\boldsymbol{\alpha}, \boldsymbol{\mu})$ we denote a generalized Reed–Solomon code consisting of evaluations of polynomials of degree $< t$ over $\mathbb{F}_{2^m}$ on the support $\boldsymbol{\alpha}$, where the evaluation at α_i is multiplied by the scalar μ_i; see Definition 2.1.) The code $\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma)$ defined above has dimension at least $n - mt$ and supports efficient correction of up to t errors. In the asymptotic Classic McEliece regime considered in this work, $n \leqslant 2^m$, $m = \Theta(\log n)$, and $t = \Theta(n/\log n)$, with parameters chosen so that the designed dimension is $K := n - mt = \Theta(n)$; the actual Goppa-code dimension is at least K. The usual construction publishes $k = K$ rows, whereas we also consider public k-dimensional subcodes with $k \leqslant K$.

Historically, McEliece was the third concrete public-key encryption proposal in the open literature, after RSA and Diffie–Hellman, and is, in retrospect, the earliest one still regarded as a serious post-quantum candidate [McE78, Cla22]. It has since received nearly five decades of cryptanalysis, ranging from generic information-set decoding [Pra62, LB88, Ste89, CC98, CS98, BLP11, BJMM12, MO15] to structural attacks on algebraic-code instantiations and compact-key variants [SS92, FOPT10, Wie10, FGO$^+$13, CGG$^+$14, CMP17, COT17, CMT23, Ran25, HW25, Hem26].

Its modern KEM incarnation, Classic McEliece,[1] was selected as a finalist in the third round of NIST's post-quantum cryptography process and advanced to the fourth round; although NIST ultimately selected HQC as a code-based encryption scheme, Classic McEliece was incorporated into ISO/IEC 18033-2 in 2026 [ACD+22, ABC+25, ISO26]. Despite its large public keys, its smallest parameter set has a ciphertext of only 96 bytes, which is the smallest among the NIST PQC candidates. The construction is also deployed in practice, including as one component of the hybrid post-quantum key establishment used by Mullvad's quantum-resistant WireGuard tunnels and by Rosenpass [Mul25, Ros26].

Due to its foundational role in code-based cryptography and its deployment in real-world systems, understanding the security of Goppa–McEliece is of both theoretical and practical importance. We therefore ask:

How secure is the McEliece cryptosystem when instantiated with binary Goppa codes?

Until recently, the best attacks applicable to the Classic McEliece regime were generic decoding attacks running in time $2^{\Theta(t)} = 2^{\Theta(n/\log n)}$. The breakthrough *syzygy distinguisher* of Randriambololona [Ran25] showed, under a natural heuristic, how to distinguish a binary Goppa public key from a random linear code in time $2^{O\left(n\frac{(\log\log n)^3}{(\log n)^2}\right)}$, thereby breaking the exponential-in-t barrier. Although distinguishing does not by itself compromise security, it reveals non-random algebraic structure and is therefore considered a serious warning sign. More recently, Hemmert and Wiemers [HW25] gave a higher-order-vanishing distinguisher with a better concrete cost, and Hemmert [Hem26] extended this approach to key recovery, conjecturally with roughly the same complexity as the distinguisher. While these attacks reveal unexpected structure in the public keys and lead to meaningful improvements, the best previous provable running time was near-exponential. Our work fundamentally changes this asymptotic landscape. Our first result is a provable quasipolynomial distinguishing attack against Classic McEliece.[2]

Theorem 1.1 (Distinguishing attack, Informal)**.** *In the asymptotic Classic McEliece regime:*

$$m = \Theta(\log n), \qquad t = \Theta\left(\frac{n}{\log n}\right), \qquad k = \Theta(n),$$

there is an $n^{O(\log n)}$-time (classical) algorithm that distinguishes the Goppa–McEliece public-key distribution from the uniform distribution over $\mathbb{F}_2^{k\times n}$ with $1 - o(1)$ advantage.

Our distinguisher is remarkably simple to describe, and proceeds as follows. Choose a parameter $s = \Theta(\log n)$ and let $d = s + 1$. Write the input matrix as $\boldsymbol{Y} = [\boldsymbol{y}_1|\cdots|\boldsymbol{y}_n] \in \mathbb{F}_2^{k\times n}$, and hold out one column, say $\boldsymbol{y}_n$. Using linear algebra, compute the space of multilinear homogeneous degree-d polynomials $h \in \mathbb{F}_2[X_1, \ldots, X_k]$ that vanish on the union of the Hamming balls $B_{<s}(\boldsymbol{y}_j) := \{\boldsymbol{x} \in$

[1] Classic McEliece uses Niederreiter's dual formulation: its underlying ciphertext is a syndrome of a fixed-weight error vector, rather than a noisy codeword as in the original McEliece scheme. These are equivalent representations of the same decoding problem, and one can efficiently convert one public key to the other by computing the dual.

[2] More broadly, for n, m, t as in Theorem 1.1 and $k \geqslant n^{\Omega(1)}$, our technique implies a quasipolynomial-time distinguisher for any k-dimensional linear code $C \subseteq \mathbb{F}_2^n$ whose natural embedding into $\mathbb{F}_{2^m}^n$ is contained in an ambient GRS code $C \subseteq \mathsf{GRS}_{n-t}(\boldsymbol{\alpha}, \boldsymbol{\lambda}) \subseteq \mathbb{F}_{2^m}^n$. Here the entries of $\boldsymbol{\alpha}$ only need to be distinct and $\boldsymbol{\lambda} \in (\mathbb{F}_{2^m}^\times)^n$ is arbitrary. For a larger extension degree m, the exponent of the time complexity scales linearly with m; see Section 1.3.

parameter set	m	n	t	k	Dim	$\log_2 T_{\text{new}}$	$\log_2 T_{\text{ISD}}$	$\log_2 T_{\text{HW}}$
`mceliece348864/f`	12	3488	64	2720	$\binom{217}{8} \approx 2^{46.60}$	**114**	151	298
`mceliece460896/f`	13	4608	96	3360	$\binom{274}{8} \approx 2^{49.34}$	**120**	191	595
`mceliece6688128/f`	13	6688	128	5024	$\binom{315}{8} \approx 2^{50.97}$	**124**	257	691
`mceliece6960119/f`	13	6960	119	5413	$\binom{304}{8} \approx 2^{50.55}$	**123**	257	565
`mceliece8192128/f`	13	8192	128	6528	$\binom{315}{8} \approx 2^{50.97}$	**124**	287	551

Table 1: Estimated binary-operation costs of structural distinguishing attacks on the Classic McEliece parameter sets (see Section 5 for cost model). Here Dim is the dimension of the linear space used by our distinguisher, and T_{new} is the estimated cost of a single invocation; see Table 2 for more details and Remark 5.1 for further optimizations that are not reflected in the table and reduce the $\log_2 T_{\text{new}}$ estimates by roughly 20 bits. The quantity T_{ISD} is the estimated cost of the best generic decryption attack, based on information-set decoding [Cla22, BC24] (provided as a reference), while T_{HW} is the concrete leading-term cost of the higher-order-vanishing distinguisher of [HW25, Hem26].

$\mathbb{F}_2^k : d_{\text{H}}(\boldsymbol{x}, \boldsymbol{y}_j) < s\}$ around every non-held-out column $\boldsymbol{y}_j$, and sample a uniformly random nonzero polynomial h from this space.

Evaluate h at the held-out column $\boldsymbol{y}_n$. If $h(\boldsymbol{y}_n) \neq 0$, output random; otherwise, output Goppa. For a Goppa–McEliece public key, the vanishing constraints force $h(\boldsymbol{y}_n) = 0$, whereas for a uniformly random matrix, $h(\boldsymbol{y}_n) \neq 0$ with constant probability. For an appropriate choice of s, this gives a constant-advantage distinguisher, which can be amplified by holding out multiple columns simultaneously.

Concrete efficiency. The above distinguishing attack is not confined to the asymptotic setting. By applying a standard "shortening" optimization [Leo88, Ste89], which restricts the public code to codewords that vanish on a chosen set of coordinates, together with a "systematic-form" optimization (see Sections 5.1 and 5.2), the asymptotic distinguisher becomes relevant to the concrete Classic McEliece parameter sets considered in the NIST process. The resulting attack parameters and concrete cost estimates (using the binary-operations metric) are summarized in Table 1. For comparison, the previous best structural public-key distinguisher was estimated to require at least 2^{298} bit operations, with substantially higher costs for the larger Classic McEliece parameter sets [HW25, Hem26]. Our estimates range from 2^{114} to 2^{124} bit operations and from 2^{66} to 2^{72} bits of storage, depending on the parameter set; see Table 1 and Table 2. Further potential optimizations are discussed in Remark 5.1.

Our attack solves an $\mathbb{F}_2$-linear system whose polynomial-space dimension ranges from 2^{47} to 2^{51}. The sparsity of this system enables a relatively fast solution using the reliable small-field block Lanczos algorithm of [Ebe10]. The reported time and storage estimates were *not* fully optimized and can potentially be further improved by generating the matrix on demand or exploiting more structure from the linear systems that need to be solved. See Section 5 for more details on the cost model and the concrete efficiency analysis.

Beyond distinguishing attacks. While distinguishing the public key from random only suggests a structural weakness, our framework also yields plausible quasipolynomial-time attacks that directly apply to the semantic security of Classic McEliece. We present two such attacks.

First, we present a heuristic $n^{O(\log n)}$-time *decryption algorithm* that, given a public key $\boldsymbol{Y}$ and a noisy codeword $\boldsymbol{c} = \boldsymbol{u}\boldsymbol{Y} + \boldsymbol{e}$, recovers the encrypted message $\boldsymbol{u}$; see Section 6. This algorithm is conceptually simple and naturally builds on our distinguisher. The concrete cost of the algorithm is currently far beyond practical reach, and therefore it should only be viewed as an asymptotic proof of concept. However, this heuristic may already be enough to separate Classic McEliece from all other standard public-key cryptosystems based on number theory, lattices, or error-correcting codes, for which no (provable or heuristic) quasipolynomial-time decryption algorithms are known.

Then, we present a heuristic $n^{O(\log n)}$-time *key-recovery algorithm*, which simultaneously strengthens (single-ciphertext) decryption to output an equivalent decryption key and improves concrete efficiency. Note that unlike the information-set decoding attacks from the T_{ISD} column of Table 1, a key-recovery attack can be applied once per public key and used to decrypt any number of future ciphertexts. The key-recovery algorithm extends our distinguisher by following the high-level blueprint of the GRS–McEliece key-recovery technique of Sidelnikov and Shestakov [SS92]. Additional ideas are needed to accommodate multiplicities and the more challenging parameter regime. We describe this algorithm in Section 7 along with a technical overview. The concrete cost estimates for this algorithm are higher but much more closer to our distinguisher than our decryption. In particular, under the binary-operations metric, they remain close to NIST's baseline security target and fall strictly below the corresponding targets for the highest-security parameter sets. There seems to be room for further improvement (see Remark 7.3), which may also be relevant to other cost metrics. Our preliminary concrete estimates for this key-recovery attack can be found in Section 7.4 (Table 3).

1.1 Updates from Initial Version and Independent Key-Recovery Algorithms

The initial version of this work [GIJS26] included a provable quasipolynomial-time algorithm for distinguishing a Classic McEliece public key from random, as well as a heuristic quasipolynomial-time decryption algorithm. These algorithms and their analysis remain essentially unchanged in this version, except for adding Remark 5.1 on improving the distinguisher's concrete efficiency based on suggestions by Weis [Wei26b]. The initial version did not include a key-recovery algorithm. However, it identified the approach from [SS92] as a natural avenue for extending the distinguisher to a key-recovery algorithm with good concrete efficiency, and left this direction for future work. Most of the new material in the current version (Section 7, Appendix C) consists of the description and analysis of such an algorithm.

Building on the distinguisher from [GIJS26], independent key-recovery algorithms were reported in personal communication by Weis [Wei26b] (with the help of Anthropic's Claude) and by Vedenev [Ved26], with extensions, refinements and concrete analysis by Saarinen [Saa26]. These heuristic algorithms analyze tangent spaces, in the spirit of the recent technique of Lemoine [Lem25], whereas our algorithm extends the classical Sidelnikov–Shestakov technique [SS92]. By combining our key-recovery approach with the approach suggested in [Wei26b], Weis reported a successful key recovery of the TII-253 McEliece challenge [Wei26a]. A description of Weis's algorithm is planned for a forthcoming report.

1.2 Techniques

Our distinguisher originated from a failed attempt to construct doubly efficient private information retrieval (DEPIR) protocols by randomly permuting queries to an algebraic locally decodable code (LDC). We observe that the public keys of different McEliece-style cryptosystems correspond to query distributions of natural PIR protocol candidates. The linearity of the LDC gives rise to a simple span-based attack against the security of the PIR protocol. We then modify the span-based attack into an explicit distinguisher between the PIR query matrix and a random matrix. This distinguisher uses the specific algebraic structure of the underlying LDC, and serves as a basis for our Goppa–McEliece distinguisher.

The PIR viewpoint extends beyond Goppa–McEliece: it applies to public matrices $\boldsymbol{Y} = \boldsymbol{SGP}$ even when $\boldsymbol{S}$ is rectangular and only a proper subcode is revealed, and systematically captures subcodes of RS, GRS, alternant, and related algebraic-code families in regimes for which, to the best of our knowledge, no efficient attacks were previously known. The simplicity of the framework lets us prove the distinguishing guarantees unconditionally using linear algebra and interpolation, rather than relying on heuristics (such as those underlying recent subexponential distinguishing attacks). Similar ideas provide a natural route toward a decryption attack. While our attacks could be presented directly, we organize the technical overview around the PIR connection, aiming to make it as intuitive as possible. For a direct and self-contained exposition of the distinguisher and its analysis, we refer the reader to Section 3 and Section 4.

PIR and McEliece. A private information retrieval (PIR) protocol [CGKS98, KO97] allows a client to retrieve an entry $\mathsf{DB}[i]$ from a server holding a database DB, without revealing the requested index i. Moreover, the total communication between the server and the client should be sublinear in the database size N. In *doubly efficient PIR* (DEPIR) [BIM00, BIPW17, CHR17], the server may preprocess the database once, after which the online *computation* is required to be sublinear (and ideally polylogarithmic) in N. The feasibility of single-server DEPIR was a longstanding open problem, recently resolved under the Ring-LWE assumption [LMW23]. While this settles the theoretical question, even the best current solutions are still far from practical [OPPW24].

A natural route toward more efficient DEPIR, as well as standard single-server PIR protocols with attractive features, is to work directly with algebraic LDCs and try to make their local decoders hide the requested index. Concretely, one may start with a D-*smooth* LDC [KT00],[3] such as a suitable Reed–Muller code, and send to the server a secret random permutation of the query sequence in the hope that this achieves the required hiding property (setting $D = \omega(1)$ is helpful for eliminating simple local attacks). Different variations of this blueprint were successfully applied in alternative settings for DEPIR [IKOS06, BIPW17, CHR17, IKLM24, CIJ+26].

However, whenever the LDC is *linear* and *public*, each query specifies a publicly known linear functional of the database. The server can therefore test, for each candidate database entry, whether its coordinate functional lies in the linear span of the query functionals. If it does not, that entry cannot be the requested one. This span-based test is straightforward when taking the linear functional point of view, but can be much less obvious when just considering the PIR query

[3] In a D-smooth, n-query LDC, the i-th message entry is decoded by querying a D-wise independent sequence of coordinates $(i_1, \ldots, i_n)$ in the encoding. A polynomial-time encodable D-smooth LDC directly yields an n-*server* DEPIR with security against D colluding servers [BIM00, WY07, GLM+25].

distribution. Indeed, some previous proposals for code-based PIR were susceptible to such an attack [AIS26]. This is the starting point of our cryptanalysis: by tuning the underlying algebraic code and its parameters, such failed PIR query distributions lead to attacks on a broad family of McEliece-style assumptions, as we explain next.

RS–McEliece via Reed-Muller decoding. To see the connection with McEliece-style assumptions, consider the above LDC-based construction when the database is encoded by a Reed–Muller (RM) code, the canonical LDC example. Here the database is represented by a polynomial $F \in \mathbb{F}_q[X_1, \ldots, X_k]$ of total degree at most d, where each of the N database entries corresponds to the evaluation of F on some publicly known point. Let $\boldsymbol{u} \in \mathbb{F}_q^k$ be the target point corresponding to the desired database entry. An RM code supports the following simple curve-based local decoding algorithm: the client samples a random low-degree curve

$$\boldsymbol{f}(Z) = \big(f_1(Z), \ldots, f_k(Z)\big),$$

where each f_i is a univariate polynomial of degree $\leqslant D$, subject to $\boldsymbol{f}(0) = \boldsymbol{u}$. The client queries the points $\boldsymbol{f}(\alpha_1), \ldots, \boldsymbol{f}(\alpha_n)$ for distinct nonzero scalars $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$, in random order. The key observation is that restricting F to this curve gives the univariate polynomial $\phi(Z) := F\big(\boldsymbol{f}(Z)\big)$, which has degree at most dD. Consequently, whenever $n > dD$, the answers $\phi(\alpha_1), \ldots, \phi(\alpha_n)$ uniquely determine ϕ via standard Lagrange interpolation, allowing the client to recover $\phi(0) = F(\boldsymbol{u})$ from the answers given the permuted sequence of scalars α_i. The query matrix whose columns are the permuted points $\boldsymbol{f}(\alpha_1), \ldots, \boldsymbol{f}(\alpha_n)$ generates a subcode of a secretly permuted Reed–Solomon code. In particular, the unpermuted queries are $\boldsymbol{Q}_0 = (\boldsymbol{f}(\alpha_1), \ldots, \boldsymbol{f}(\alpha_n)) = \boldsymbol{S}\boldsymbol{G}_{\mathrm{RS}}$, where $\boldsymbol{S}$ contains the coefficients of the polynomials $\{f_j\}_{j\in[k]}$, and the server receives $\boldsymbol{Q} = \boldsymbol{S}\boldsymbol{G}_{\mathrm{RS}}\boldsymbol{P}$ for a random permutation matrix $\boldsymbol{P}$. Thus, up to the affine constraint $\boldsymbol{f}(0) = \boldsymbol{u}$, the PIR query is precisely a permuted low-dimensional subcode of a Reed–Solomon (RS) code. This is the same algebraic form as an RS-based McEliece public key.

From span attacks to hold-out distinguishers. To apply the span attack discussed above to the RM-based PIR, observe that the dimension of the space of polynomials F is $N = \binom{k+d}{d}$. This is also the number of field elements in a database that can be encoded by the scheme, while the server only sends n field elements. Restating the same attack differently, given the query points $\boldsymbol{p}_1 = \boldsymbol{f}(\alpha_1), \ldots, \boldsymbol{p}_n = \boldsymbol{f}(\alpha_n)$ and a target point $\boldsymbol{u}$, the attack checks whether the evaluation functional $F \mapsto F(\boldsymbol{u})$ lies in the span of the evaluation functionals $F \mapsto F(\boldsymbol{p}_i)$. This can be done via simple linear algebra in time $\mathrm{poly}(N)$ by checking whether there exist coefficients $c_1, \ldots, c_n$ such that $F(\boldsymbol{u}) = \sum_{i=1}^{n} c_i F(\boldsymbol{p}_i)$ simultaneously for every polynomial F of degree $\leqslant d$, or equivalently for every monomial of degree $\leqslant d$. Correctness of the PIR scheme guarantees such a relation for honestly generated queries, whereas independently random points are unlikely to satisfy it when the polynomial space has larger dimension than the number of query points, namely when $N \gg n$.

There are two discrepancies between this PIR attack and the McEliece setting. First, in a McEliece instance there is no database size or even a database; one is given only the parameters D, k, n, q and a matrix $\boldsymbol{Y}$. This causes no problem because the attacker can choose any d satisfying the PIR constraints. Second, the attacker does not know the target point $\boldsymbol{u}$. While the insecurity of the PIR implies that for most PIR indices, the resulting query distribution is *not* pseudorandom,

RM-based hold-out distinguisher for RS–McEliece.

Input: A matrix $\boldsymbol{Y} = [\boldsymbol{y}_1|\cdots|\boldsymbol{y}_n] \in \mathbb{F}_q^{k\times n}$. The matrix is sampled so that it is either a random k-dimensional subcode of an RS code of degree D or a truly random matrix.

1. Choose a degree parameter d such that $N_d \geqslant n > dD+1$, where $N_d := \binom{k+d}{d}$. Let $\boldsymbol{\nu}_d(\boldsymbol{x}) \in \mathbb{F}_q^{N_d}$ denote the vector of all monomials of total degree $\leqslant d$, evaluated at $\boldsymbol{x}$.
2. Choose a hold-out coordinate $\tau \in [n]$; by symmetry, one may fix $\tau = n$.
3. Compute $W_\tau := \mathrm{span}_{\mathbb{F}_q}\{\boldsymbol{\nu}_d(\boldsymbol{y}_j) : j \in [n] \setminus \{\tau\}\}$.
4. If $\boldsymbol{\nu}_d(\boldsymbol{y}_\tau) \in W_\tau$, output RS; otherwise output random.

Figure 1: The PIR-inspired hold-out distinguisher for RS–McEliece.

the distinguisher requires the unknown index as advice. It also does not apply to the exact RS–McEliece public key since the PIR query distribution $\boldsymbol{Q}$ is conditioned on the constraint $\boldsymbol{f}(0) = \boldsymbol{u}$.

To address this mismatch, we introduce a natural *hold-out distinguisher*. We consider the PIR query distribution $\boldsymbol{Q}$ for a *random* target point $\boldsymbol{u}$, as in the RS–McEliece public-key distribution $\boldsymbol{Y}$. To distinguish $\boldsymbol{Y}$ from random, we hold out one column of $\boldsymbol{Y}$ (say the n-th column corresponding to the unknown scalar α_n) and view it as the target point $\boldsymbol{u}$. We view the remaining columns as the PIR query points and apply the same span test to determine whether the value of F at the held-out column is determined by its values at all other columns. To make this attack work, we also slightly change the parameters by assuming $n > dD+1$ (instead of $n > dD$), which enables the client to interpolate $\phi(Z)$ from any $n-1$ of the n answers $\{\phi(\alpha_j)\}_{j\in[n]}$. From an interpolation perspective, there is nothing special about $\boldsymbol{f}(0) = \boldsymbol{u}$; one could set $\boldsymbol{u}$ as any point $\boldsymbol{f}(\alpha_j)$ on the curve (so long as all scalars $\alpha_1, \ldots, \alpha_n$ are distinct).

Put differently, the PIR-based hold-out distinguisher asks the following. Suppose we have an unknown database for which the answers to all of the non-held-out queries are 0. Does this imply that the evaluation at the held-out query must also be 0? By linear duality, this is an equivalent formulation of the span-based criterion.

The above hold-out distinguisher, applied to the RM-based PIR, is explicitly described in Fig. 1. By PIR correctness, when $\boldsymbol{Y}$ is taken from the RS–McEliece experiment, the algorithm always outputs RS. On the other hand, if the PIR has sublinear communication, then the $n-1$ answers reveal little information about F. This suggests that the answers typically do not determine the value of F at a random and independent held-out point, implying that on a random $\boldsymbol{Y}$ the algorithm will output random with high probability.

Relation with standard Schur-product attacks. While the above RM-based attack takes a new PIR-inspired perspective, it does not yield any new attacks on (subcode) RS–McEliece assumptions. A reader familiar with McEliece cryptanalysis would realize that the above attack is closely related to the basic Schur-product-based distinguisher [Wie10, FGO+13, CGG+14].

Recall that, for vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{F}_q^n$, their Schur product is the coordinatewise product $\boldsymbol{a} \star \boldsymbol{b} :=$

$(a_1b_1, \ldots, a_nb_n)$. For linear codes $\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathbb{F}_q^n$, define

$$\mathcal{C}_1 \star \mathcal{C}_2 := \mathrm{span}\{\boldsymbol{a} \star \boldsymbol{b} : \boldsymbol{a} \in \mathcal{C}_1,\, \boldsymbol{b} \in \mathcal{C}_2\},$$

and let $\mathcal{C}^{\star d}$ denote the d-th Schur power.

Let $\boldsymbol{M}_d = [\boldsymbol{\nu}_d(\boldsymbol{y}_1) \mid \cdots \mid \boldsymbol{\nu}_d(\boldsymbol{y}_n)]$ where $\boldsymbol{\nu}_d(\boldsymbol{x}) \in \mathbb{F}_q^{N_d}$ denotes the vector of all monomials of total degree $\leqslant d$, evaluated at $\boldsymbol{x}$. Its row space is the d-th Schur power of the public code, after adjoining the all-ones vector (which allows for monomials of degree smaller than d to be used). The usual Schur-power distinguisher tests whether $\mathsf{rank}(\boldsymbol{M}_d) = n$, which will typically hold in the random case but will not hold in the structured case when $n > dD + 1$. Indeed, in the structured case we get the d-th Schur power of a RS_{D+1} subcode, which is a RS_{dD+1} subcode. The hold-out test asks whether a selected column of $\boldsymbol{M}_d$ lies in the span of the others. Thus a successful hold-out test witnesses the same Schur-power rank defect. Conversely, a rank defect only guarantees that *some* column is redundant, but for an RS subcode this implies that every column is redundant. A similar Schur-product-based attack also applies in the more general case of GRS codes (see Definition 2.1).

Limitation of Schur-product attacks. There is a basic reason why Schur-product attacks, and equivalently the RM-based attack above, do not reach the parameter regimes relevant to Classic Goppa–McEliece or to high-rate GRS subcodes. Let the hidden ambient code be a length-n GRS code of dimension K. Its Schur square has dimension at most $\min\{n, 2K-1\}$, so the usual square-code distinguisher reveals a dimension defect only when $2K - 1 < n$, essentially when the ambient rate is below $1/2$. If the full GRS code is public and $K > n/2$, one can often bypass this barrier by applying the attack to the dual: the dual of a GRS code is again GRS, now of dimension $n - K < n/2$. This workaround breaks down if we are only given a public subcode $C \subsetneq \mathsf{GRS}_K$. Although $\mathsf{GRS}_K^{\perp} \subseteq C^{\perp}$, the public dual $C^{\perp}$ is only a supercode of the low-rate GRS dual and need not itself have a small Schur square. Thus, the use of the dual does not get around the rate-$1/2$ barrier in the case of subcodes.

Crossing the Schur barrier via multiplicities. To overcome this barrier, we switch back to the PIR perspective but replace the RM encoding by a different locally decodable code, namely an order-s multiplicity code [KSY14]; see Definition 2.10. Whereas an RM code stores only the value $F(\boldsymbol{x})$ at each point $\boldsymbol{x} \in \mathbb{F}_q^k$, an order-$s$ multiplicity code stores the entire *Hasse jet* of order $< s$,

$$J_{\boldsymbol{x}}^{<s}(F) := \big((\partial^{[\boldsymbol{a}]}F)(\boldsymbol{x})\big)_{|\boldsymbol{a}|<s},$$

consisting of the value of F together with all of its Hasse derivatives[4] of order smaller than s. Here $\boldsymbol{a} \in \mathbb{Z}_{\geqslant 0}^k$ specifies the multiplicity of each variable, where $|\boldsymbol{a}|$ denotes the ℓ_1-norm. If $J_{\boldsymbol{x}}^{<s}(F) = \boldsymbol{0}$, we say that F vanishes *with multiplicity* (at least) s at $\boldsymbol{x}$. For multilinear F over $\mathbb{F}_2$, this is equivalent to F vanishing on the Hamming ball $B_{<s}(\boldsymbol{x})$.

As in the RM-based scheme, the client chooses a random curve $\boldsymbol{f}(Z) = (f_1(Z), \ldots, f_k(Z))$ of degree at most D passing through the target point and queries $\boldsymbol{f}(\alpha_1), \ldots, \boldsymbol{f}(\alpha_n)$. Let $\phi(Z) := F(\boldsymbol{f}(Z))$. Since $\deg(F) \leqslant d$ and $\deg(f_j) \leqslant D$, we have $\deg(\phi) \leqslant dD$. By the Hasse-derivative chain rule, the order-$< s$ multivariate Hasse derivatives of F at $\boldsymbol{f}(\alpha_i)$, together with the curve known

[4] Hasse derivatives are the finite-field analogue of ordinary partial derivatives, characterized by the formal expansion $F(\boldsymbol{X} + \boldsymbol{Z}) = \sum_{\boldsymbol{a}} (\partial^{[\boldsymbol{a}]}F)(\boldsymbol{X})\boldsymbol{Z}^{\boldsymbol{a}}$. See Section 2.4 for more details, including a simpler definition over $\mathbb{F}_2$.

Multiplicity-based hold-out distinguisher for RS–McEliece.

Input: A matrix $\boldsymbol{Y} = [\boldsymbol{y}_1|\cdots|\boldsymbol{y}_n] \in \mathbb{F}_q^{k\times n}$. The matrix is sampled so that it is either a random k-dimensional subcode of an RS code of degree D or a truly random matrix.

1. Choose a degree $d < q$ and a multiplicity s such that

$$s(n-1) > dD \qquad \text{and} \qquad (n-1)\binom{k+s-1}{s-1} < N_d, \qquad N_d := \binom{k+d}{d}.$$

   For $\boldsymbol{x} \in \mathbb{F}_q^k$ and $\boldsymbol{a} \in \mathbb{Z}_{\geqslant 0}^k$, let $\boldsymbol{\nu}_{d,\boldsymbol{a}}(\boldsymbol{x}) \in \mathbb{F}_q^{N_d}$ denote the vector of the $\boldsymbol{a}$-th Hasse derivatives of all monomials of total degree at most d, evaluated at $\boldsymbol{x}$:

$$\boldsymbol{\nu}_{d,\boldsymbol{a}}(\boldsymbol{x}) := \left(\left(\partial^{[\boldsymbol{a}]}\boldsymbol{X}^{\boldsymbol{b}}\right)(\boldsymbol{x})\right)_{|\boldsymbol{b}|\leqslant d} \in \mathbb{F}_q^{N_d},$$

   where $\partial^{[\boldsymbol{a}]}$ denotes the corresponding Hasse derivative.

2. Choose a hold-out coordinate $\tau \in [n]$; by symmetry, one may fix $\tau = n$.

3. Compute

$$W_\tau := \operatorname{span}_{\mathbb{F}_q}\left\{\boldsymbol{\nu}_{d,\boldsymbol{a}}(\boldsymbol{y}_j) : j \in [n]\setminus\{\tau\},\ |\boldsymbol{a}| < s\right\}.$$

4. If $\boldsymbol{\nu}_{d,\boldsymbol{0}}(\boldsymbol{y}_\tau) \in W_\tau$, output RS; otherwise output random.

Figure 2: The multiplicity-based hold-out distinguisher for RS–McEliece.

to the client, determine the first s univariate Hasse derivatives of ϕ at α_i. Thus, each queried point supplies s linear constraints on ϕ (known as Hermite-interpolation constraints) rather than the single evaluation constraint supplied by an ordinary Reed–Muller encoding. Consequently, in the hold-out formulation, the Hasse jet information from the $n-1$ non-held-out columns determines $\phi(\alpha_\tau) = F(\boldsymbol{f}(\alpha_\tau))$ at the held-out column whenever $s(n-1) > dD$. In other words, multiplicities reduce the required number of queried points from roughly dD to roughly dD/s. Taking d and s to be comparable therefore permits $n \approx D$, which enables the resulting distinguisher to apply in regimes where the usual Schur-product dimension defect is not visible. This multiplicity-interpolation principle was used to get around similar barriers in the context of list-decoding algorithms for RS codes [GS99] and, more closely to the current context, in derivative-based PIR [WY07, GLM+25, HR26].

We illustrate the corresponding span-test distinguisher in Fig. 2. As in Fig. 1, here too the hidden curve and its parametrization are unknown to the distinguisher. The distinguisher only needs to test linear dependence among publicly computable derivative-evaluation functionals. The technical section will use a stronger variant of this test that achieves a bigger distinguishing advantage.

Analyzing the multiplicity-based distinguisher. For the distinguisher to work, we must show that the held-out evaluation functional is spanned in the structured case, namely for an RS subcode, but not in the unstructured case, namely for a random matrix. In the structured case, the span condition is implied as before by PIR correctness. Concretely, the Hasse jets at the $n-1$

non-held-out columns determine $\phi(\alpha_\tau)$ whenever $s(n-1) > dD$. It follows that the evaluation functional at the held-out column lies in the span of the order-$<$ s Hasse jet functionals at the remaining columns, and the distinguisher always outputs RS on the structured distribution.

For a random matrix, we use the corresponding dimension bound. Let $\mathcal{P}_{\leqslant d}$ denote the space of k-variate polynomials of total degree at most d. The polynomial space has dimension $\dim \mathcal{P}_{\leqslant d} = \binom{k+d}{d}$, whereas the $n-1$ non-held-out columns contribute at most $(n-1)\binom{k+s-1}{s-1}$ Hasse jet functionals. Thus, if $(n-1)\binom{k+s-1}{s-1} < \binom{k+d}{d}$, there exists a nonzero annihilating polynomial $H \in \mathcal{P}_{\leqslant d}$ whose Hasse jets of order-$<$ s vanish at all non-held-out columns. Conditioned on these columns, the held-out column remains uniform and independent, so the Schwartz–Zippel lemma gives $\mathbf{Pr}[H(\boldsymbol{y}_\tau) = 0] \leqslant d/q$. Note that whenever $H(\boldsymbol{y}_\tau) \neq 0$, the held-out evaluation vector cannot lie in W_τ, and hence the distinguisher must output random. Consequently, the distinguisher accepts a random matrix with probability at most d/q.

To demonstrate the effectiveness of this attack where Schur-based attacks fail, we consider the following running example.

Example 1.2 (Running example)**.** *Consider the parameter regime $k = \lfloor D^{0.9} \rfloor$ and $n = \lfloor 1.1D \rfloor$. A random k-dimensional code has generic Schur-square dimension $\min\left\{n, \binom{k+1}{2}\right\} = n$. On the other hand, the square of the published RS subcode is contained in RS_{2D+1}, which is already all of $\mathbb{F}_q^n$ because $2D+1 > n$. Thus, both the structured code and a random code may have full Schur square, and the usual dimension test does not distinguish them.*

Note that the Schur-based attack does not apply here because $D/n > 1/2$. Moreover, we release only a low-dimensional subcode, so neither dualization nor shortening restores a Schur-square defect in this regime. For the running Example 1.2, where $k = \lfloor D^{0.9} \rfloor$ and $n = \lfloor 1.1D \rfloor$, we may take $s = 11$ and $d = 12$. For all sufficiently large D, the interpolation condition holds because $11(n-1) > 12D$. Moreover, $(n-1)\binom{k+10}{10} = O(Dk^{10}) = O(D^{10})$, whereas $\binom{k+12}{12} = \Omega(k^{12}) = \Omega(D^{10.8})$, so the dimension condition also holds. The attack runs in time $\mathrm{poly}\left(\binom{k+12}{12}, n, \log q\right) = D^{O(1)}$, yielding a polynomial-time distinguisher in a regime beyond the usual Schur-product attack. This completes our overview of the core techniques. We next explain how these ideas extend beyond RS codes, in particular to GRS and alternant/Goppa codes, and then discuss candidate decryption and key-recovery algorithms.

Extending to GRS codes. A GRS subcode differs from an RS subcode by an unknown nonzero column multiplier vector $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n)$. After absorbing the hidden permutation into the indexing, its public columns have the form

$$\boldsymbol{y}_i = \lambda_i \boldsymbol{p}_i, \qquad \boldsymbol{p}_i = \big(f_1(\alpha_i), \ldots, f_k(\alpha_i)\big), \qquad \lambda_i \in \mathbb{F}_q^\times.$$

We remove these multipliers by restricting the test space to homogeneous polynomials. Indeed, if H is homogeneous of degree d, then for every multi-index $\boldsymbol{a}$ with $|\boldsymbol{a}| \leqslant d$,

$$\big(\partial^{[\boldsymbol{a}]} H\big)(\lambda_i \boldsymbol{p}_i) = \lambda_i^{d-|\boldsymbol{a}|} \big(\partial^{[\boldsymbol{a}]} H\big)(\boldsymbol{p}_i).$$

Since $\lambda_i \neq 0$, vanishing with multiplicity s at $\boldsymbol{y}_i$ is equivalent to vanishing with multiplicity s at $\boldsymbol{p}_i$. The same hold-out and Hermite-interpolation argument therefore applies to arbitrary published subcodes of GRS codes. The homogeneous polynomial space has dimension $\binom{k+d-1}{d}$, so the same argument for soundness applies as long as $(n-1)\binom{k+s-1}{s-1} < \binom{k+d-1}{d}$.

1.2.1 Distinguishing Other Families: Alternant and Goppa Codes

The above argument applies to McEliece instances generated from subcodes of random GRS codes. This does not yet cover binary Goppa codes as a special case. Indeed, when viewed as a subset of a GRS code over $\mathbb{F}_{2^m}$, a binary Goppa code is not closed under scalar multiplication in $\mathbb{F}_{2^m}$.

Towards capturing the Goppa case, as well as other classes of codes, we extend our distinguisher to an arbitrary public linear code $C \subseteq \mathbb{F}_2^n$ whose natural embedding into $\mathbb{F}_{2^m}^n$ is contained in an ambient GRS code $C \subseteq \mathsf{GRS}_K(\boldsymbol{\alpha}, \boldsymbol{\lambda}) \subseteq \mathbb{F}_{2^m}^n$. This includes both binary Goppa codes and alternant codes as special cases.[5]

For concreteness, consider the special case of binary Goppa codes:

$$\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma) = \mathsf{GRS}_t(\boldsymbol{\alpha}, \boldsymbol{\mu})^{\perp} \cap \mathbb{F}_2^n, \qquad \mu_i = \Gamma(\alpha_i)^{-1},$$

where Γ is a polynomial of degree t over $\mathbb{F}_{2^m}$. If Γ is square-free (and in particular if it is irreducible as in Classic McEliece), then there is an ambient GRS code of dimension $n - 2t$. The resulting public matrix $\boldsymbol{Y} \in \mathbb{F}_2^{k \times n}$ is over the binary base field.

Since the public code has an ambient GRS code over $\mathbb{F}_{2^m}$, it is tempting to apply our multiplicity-based distinguisher for GRS codes directly. There is, however, a subtlety: the random matrix is sampled from $\mathbb{F}_2^{k \times n}$, rather than uniformly from $\mathbb{F}_{2^m}^{k \times n}$. Thus, although the structured-side interpolation argument takes place over the extension field, the random-side analysis must account for the fact that the columns lie in $\mathbb{F}_2^k$. We therefore modify the previous strategy as follows:

- To obtain the desired soundness guarantee, we restrict the test space to *multilinear* homogeneous polynomials over $\mathbb{F}_2$. We present the resulting test in its dual vanishing-polynomial formulation, in which soundness follows from the minimum distance of binary RM codes. Namely, we sample a nonzero homogeneous multilinear polynomial $F \in \mathbb{F}_2[X_1, \ldots, X_k]$ of degree d such that F and all of its order-$< s$ Hasse jets vanish at the non-held-out columns.
- We then test whether $F(\boldsymbol{y}_\tau) = 0$. If yes, we output Goppa; otherwise, we output random.

The two required parameter conditions are $s(n-1) > d(n-2t-1)$ and $\binom{k}{d} > (n-1)\binom{k+s-1}{s-1}$.

The completeness analysis is essentially the same as before. For a structured public key, each column has the form

$$\boldsymbol{y}_i = \lambda_i \boldsymbol{p}_i, \qquad \boldsymbol{p}_i = \big(f_1(\alpha_i), \ldots, f_k(\alpha_i)\big).$$

Let $\widehat{F}$ denote the same polynomial as F, now viewed as a polynomial over $\mathbb{F}_{2^m}$. Since F is homogeneous of degree d, $F(\boldsymbol{y}_i) = \lambda_i^d \widehat{F}(\boldsymbol{p}_i)$. The analogous scaling relation holds for the Hasse derivatives of F. Because $\lambda_i \neq 0$, vanishing with multiplicity (at least) s at $\boldsymbol{y}_i$ therefore implies vanishing with the same multiplicity at $\boldsymbol{p}_i$.

Multiplicity-based interpolation, together with $s(n-1) > d(n-2t-1)$, then implies that $\widehat{F}(\boldsymbol{p}_\tau) = 0$, because this value is a linear combination of derivatives of order less than s at the points $\boldsymbol{p}_i$ for $i \neq \tau$. Consequently, $F(\boldsymbol{y}_\tau) = \lambda_\tau^d \widehat{F}(\boldsymbol{p}_\tau) = 0$. Thus, the distinguisher always accepts a structured public key.

[5] Alternant codes generalize Goppa codes by allowing an arbitrary ambient GRS code, without restricting $\boldsymbol{\alpha}$ and $\boldsymbol{\lambda}$; see Appendix B for definitions of these variants; here we focus on the binary case for simplicity, but our approach extends to linear codes $C \subseteq \mathbb{F}_q^n$ contained in an ambient $\mathsf{GRS}_K(\boldsymbol{\alpha}, \boldsymbol{\lambda}) \subseteq \mathbb{F}_{q^m}^n$.

For soundness, condition on the non-held-out columns and on a random nonzero polynomial F satisfying their vanishing constraints. If $\boldsymbol{Y}$ is uniformly random over $\mathbb{F}_2^{k\times n}$, then $\boldsymbol{y}_\tau$ remains uniform and independent in $\mathbb{F}_2^k$. The minimum-distance bound for binary RM codes gives $\mathbf{Pr}\left[F(\boldsymbol{y}_\tau)=0\right] \leqslant 1-2^{-d}$.

The space of degree-d homogeneous polynomials supported on multilinear (square-free) monomials has dimension $\binom{k}{d}$. Therefore, the second condition $\binom{k}{d} > (n-1)\binom{k+s-1}{s-1}$ ensures that such a nonzero F exists.

When $m=\Theta(\log n), k=\Theta(n), t=\Theta(n/\log n)$, both conditions can be satisfied with $s=\Theta(\log n)$ and $d=s+1$. This yields a quasipolynomial-time distinguishing algorithm with inverse-polynomial advantage. In Section 4, we show how to amplify this to distinguishing advantage $1-o(1)$. We give the full GRS and Goppa analyses in Sections 3 and 4, respectively.

1.2.2 Heuristic Decryption and Key-Recovery Attacks

The above attack yields a provable quasipolynomial-time distinguisher for the public key, but does not directly threaten the semantic security of the encryption scheme. To address this, we propose heuristic quasipolynomial-time decryption and key-recovery algorithms that build on our distinguisher.

The decryption algorithm, which extends the distinguisher in a conceptually simple way, decodes a random LPN sample generated with respect to the public key. This corresponds to decrypting an encryption of a random message.

We now describe the main ideas behind the decryption algorithm and the supporting evidence for its correctness. In the GRS case, the public key $\boldsymbol{Y}$ and ciphertext $\boldsymbol{c}=\boldsymbol{u}\boldsymbol{Y}+\boldsymbol{e}$ are defined over $\mathbb{F}_q$. In the binary Goppa case, they are defined over $\mathbb{F}_2$ and then viewed over the extension field $\mathbb{F}_q=\mathbb{F}_{2^m}$. In either case, we append the ciphertext to the public matrix and form

$$\boldsymbol{A} := \begin{pmatrix} \boldsymbol{Y} \\ \boldsymbol{c} \end{pmatrix}.$$

For every coordinate τ, we apply the homogeneous hold-out test over the corresponding ambient field $\mathbb{F}_q$: we compute the space of homogeneous degree-d polynomials that vanish to multiplicity s at every column $\boldsymbol{A}_j$ with $j\neq\tau$. We mark τ as clean when every polynomial in the hold-out kernel also vanishes at $\boldsymbol{A}_\tau$. Once the candidate clean set $\widehat{\mathcal{I}}_{\mathrm{cl}}$ has been identified, we recover the message by solving

$$\boldsymbol{u}\boldsymbol{Y}_{\widehat{\mathcal{I}}_{\mathrm{cl}}} = \boldsymbol{c}_{\widehat{\mathcal{I}}_{\mathrm{cl}}}.$$

The clean-coordinate guarantee is unconditional for both code families. Indeed, let $w := |\mathrm{supp}(\boldsymbol{e})|$, and let D be the degree bound of the ambient GRS representation, namely $D=K-1$ in the GRS case and $D=n-2t-1$ in the binary Goppa case. If $s(n-w-1) > dD$, then univariate Hermite interpolation forces every polynomial in the hold-out kernel to vanish at every clean held-out column. Thus, the attack never rejects a clean coordinate under this condition in either case.

For a dirty coordinate $e_\tau \neq 0$, we do not currently have a proof that the hold-out test rejects τ for either code family. We provide a heuristic analysis and supporting experiments in Appendix A, which also guides the choice of d and s. When $m=\Theta(\log n), k=\Theta(n), t=\Theta(n/\log n)$ for binary Goppa codes, we still choose $s,d=\Theta(\log n)$, which leads to quasipolynomial running time.

In addition to the above decryption algorithm, we propose a heuristic, quasipolynomial-time key-recovery algorithm that also significantly improves over the concrete cost of the decryption algorithm. The technical overview of this algorithm is deferred to Section 7.

1.3 Open Problems

Immediate open problems are to further improve the asymptotic and concrete complexity of our attacks, and obtain provable variants of the decryption and key-recovery algorithms. Below we discuss some aspects of these broad problems in more detail.

Can the dependence on the extension degree be improved? In more general Goppa parameter regimes, where the extension degree m may be larger while $mt = O(n)$, our current distinguishing attack runs in time $2^{\widetilde{O}(n/t)}$. In particular, when $mt = \Theta(n)$, the exponent scales essentially with the extension degree m. Can this be improved? Unless the dependence on m is improved, increasing m in the Classic McEliece parameterization could be a practical mitigation against the current attack, retaining some of the concrete advantage in ciphertext size.

How broad is the framework? Our present analysis applies to subcodes of RS, GRS, and alternant codes, including binary Goppa codes, and appears likely to extend to other algebraic evaluation codes, such as suitable algebraic-geometry codes [CMP17]. A particularly interesting test case is Wieschebrink's random-column proposal [Wie06], in which random columns are inserted into a generator matrix of a hidden GRS code. Although the resulting public matrix is not itself a GRS subcode, we expect our attack to extend by treating the inserted columns as outliers: the interpolation argument should survive as long as sufficiently many genuine GRS columns remain, while hold-out or shortening techniques may be used to identify and remove the random columns. Making this extension rigorous is an interesting open problem. Characterizing precisely which algebraic structures are captured by the PIR-inspired framework, and identifying natural code-based designs that genuinely evade it, remain open.

Can code-based PKE beat the quadratic ciphertext barrier? Alekhnovich's LPN-based public-key encryption [Ale03] provides a useful benchmark: it achieves ciphertext length $\widetilde{O}(\kappa^2)$ for κ-bit security under a standard conjecture on the security of LPN. In fact, a quadratic barrier is inherent to any Alekhnovich-style scheme that uses approximately-dual noise distributions [BBD+16, BLR14, GGMT25]. Before our work, Classic McEliece appeared to offer a qualitatively better tradeoff. In the usual regime

$$m = \Theta(\log n), \qquad t = \Theta\left(\frac{n}{\log n}\right),$$

the ciphertext length is $\Theta(mt) = \Theta(n)$, while generic information-set decoding takes time $2^{\widetilde{\Theta}(n)}$. Thus, κ-bit security appeared compatible with ciphertexts of length $\widetilde{O}(\kappa)$.

Our attacks change this picture. Generic decoding runs in time $2^{\widetilde{O}(t)}$, whereas our public-key distinguisher runs in time $2^{\widetilde{O}(n/t)}$. Under our decryption hypothesis, the same asymptotic bound applies to decryption. Consequently, under our conjecture, resistance to both types of attack is governed by

$$\min\left\{t, \frac{n}{t}\right\} \leqslant \sqrt{n}.$$

Requiring both quantities to be $\widetilde{\Omega}(\kappa)$ forces $n = \widetilde{\Omega}(\kappa^2)$. In the constant-rate Goppa regime, where $mt = \Theta(n)$, this corresponds to $\widetilde{\Omega}(\kappa^2)$-bit ciphertexts. Thus, at least for public-key pseudorandomness (and conditionally for decryption) Goppa–McEliece falls under the same qualitative quadratic tradeoff as Alekhnovich's cryptosystem. Whether any code-based encryption scheme based on noisy linear algebra over $\mathbb{F}_2$ can beat this apparent quadratic barrier is an interesting open question.

Can the decryption and key-recovery heuristics be proved or refuted? While our quasipolynomial distinguishing algorithm is unconditional, the decryption and the key-recovery algorithms are heuristic. Further validation of the underlying conjectures, or ideally an unconditional proof of correctness, are left for future work. For the decryption attack, we could not run an end-to-end test on binary Goppa codes because the resulting matrices were too large, but in Appendix A.4 we discuss evidence from small instances of GRS, alternant-code, "wild Goppa," and q-ary Goppa families over a small extension degree.

1.4 Related Work

In this section we survey prior related work, focusing mainly on *structural* cryptanalysis of McEliece-type cryptosystems. Generic message-recovery attacks, most notably information-set decoding (ISD) and its refinements, do not attempt to recognize or recover the hidden algebraic code; see [Pra62, LB88, Ste89, CC98, CS98, BLP11, BJMM12, MO15] and references therein.

Structural attacks on Reed–Solomon and related codes. The classical starting point for structural cryptanalysis is the Sidelnikov–Shestakov attack [SS92], which recovers the hidden support and multipliers of a full generalized Reed–Solomon code in polynomial time. This attack explains why GRS codes cannot be used directly in McEliece, and motivated the use of subcodes, subfield subcodes, and other transformations intended to conceal the ambient evaluation-code structure. A long line of work subsequently exploited additional algebraic structure in compact-key variants, GRS subcodes, algebraic-geometry codes, and special binary Goppa code families [FOPT10, Wie10, CGG$^+$14, CMP17, COT17]. These attacks are highly effective against their respective structured families, but do not directly recover the structure of the binary Goppa codes in the Classic McEliece parameter regime.

Schur-product and square-code distinguishers. Schur products were first used by Wieschebrink [Wie10] in his attack on GRS subcodes. Faugère, Gauthier-Umaña, Otmani, Perret, and Tillich [FGO$^+$13] gave the first polynomial-time distinguisher for alternant and Goppa codes in the very-high-rate regime by linearizing an algebraic key-recovery system and detecting a rank defect. Márquez-Corbella and Pellikaan [MP12] later reinterpreted in terms of the small dimension of the Schur square of the dual code; Mora and Tillich [MT23] subsequently proved rigorous upper bounds on this dimension by analyzing the structured side of the distinguisher.

Schur products became a standard tool for distinguishing and structurally attacking GRS-based cryptosystems, often in combination with shortening [CGG$^+$14]. Related techniques have yielded polynomial-time structural attacks on different variants of Goppa, alternant, and algebraic-geometry codes [COT17, CMP17, BMT24].

Recall from Section 1.2 that these methods are strongest in the very-high-rate regime or very-low-rate regime, where a Schur power of the public code or its dual exhibits a dimension defect. Our

hold-out test with multiplicity order $s = 1$ captures the same Schur-power rank defect. Our use of $s > 1$ multiplicities supplies s univariate Hermite-interpolation constraints per public coordinate. This allows the hold-out test to remain informative after the value-only Schur-power test has saturated; in particular, Corollary 3.4 gives polynomial-time distinguishers for proper GRS subcodes in regimes not covered by the direct Schur-power test or its standard shortening variant.

Distinguishers for broader parameter regimes. More recent work replaces a single Schur-power dimension defect by richer polynomial relations, allowing distinguishers to reach constant-rate regimes relevant to Classic McEliece. At a high level, these attacks mainly study algebraic relations among the columns of a parity-check representation, and their complexity analyses in the Classic McEliece regime rely on heuristic predictions about the corresponding algebraic systems. We first summarize these attacks and then compare them with our primal-code hold-out test.

Couvreur, Mora, and Tillich [CMT23] study the quadratic equations satisfied by the columns of the dual public code. For alternant and Goppa codes, these equations contain matrices of unexpectedly low rank, which can distinguish the structured code from a random code and, in some regimes, reveal its hidden algebraic structure. Their approach applies to a much broader parameter range than the ordinary square-code distinguisher, although the cost of finding the required low-rank matrices ranges from polynomial to exponential depending on the parameters.

Randriambololona's syzygy distinguisher [Ran25] instead studies higher-order algebraic dependencies associated with a shortening of the dual code, summarized by its so-called graded Betti numbers. Under a heuristic prediction for these invariants on random codes, it gives the first subexponential distinguisher in the asymptotic Classic Goppa–McEliece regime. For a fixed base field and constant code rate, its running time is

$$2^{O\left(\frac{n(\log\log n)^3}{(\log n)^2}\right)}.$$

Hemmert and Wiemers [HW25] and Hemmert [Hem26] study homogeneous polynomials that vanish to high order on the columns of a shortened parity-check matrix. For their chosen parameters, the corresponding space is predicted to be trivial for a random code, whereas explicit polynomials survive for a Goppa code. Their analysis gives the strongest previously published concrete distinguisher estimates for the Classic McEliece parameter sets among the structural approaches considered here. The work presents parameter-dependent estimates rather than a separate asymptotic running-time theorem under the Classic McEliece scaling. We therefore use it for the concrete comparison in Table 1 without assigning it an asymptotic complexity class. This technique was extended to support key recovery at a comparable but higher cost [Hem26]; see below.

Briaud, Lemoine, Randriambololona, and Tillich [BLRT26] return to the quadratic-relation framework of [CMT23], introducing a bilinear parametrization tailored to finding rank-two matrices. Under heuristic assumptions on the ranks of related matrices, they obtain a subexponential-time distinguisher in the constant-rate regime. An advantage over [Ran25] is that the associated rank-two matrices also support key recovery, as discussed below.

Comparison with our distinguisher. Although our attack also uses polynomial vanishing, it organizes the structural information differently. The quadratic-relation and higher-order-vanishing attacks above work on the dual side, studying polynomial relations among parity-check columns.

Our attack instead works directly with a generator matrix of the primal code and applies even when the public code is only a proper subcode of an ambient GRS code. It holds out one column and asks whether multiplicity-s constraints at the remaining columns force evaluation at the held-out point. For a structured code, this implication follows from Hermite interpolation after restricting the test polynomial to the hidden univariate GRS parametrization. For a random code, soundness follows unconditionally from Schwartz–Zippel or, in the binary case, from distance and low-weight bounds for Reed–Muller codes. Thus, unlike the recent Classic-regime distinguishers surveyed above, our distinguishing analysis requires no heuristic assumptions. The resulting algorithm runs in quasipolynomial time in the asymptotic Classic McEliece regime and, after shortening, applies to every concrete parameter set considered in the NIST process.

The attack was developed independently of the higher-order-vanishing approach and arose from a failed attempt to construct doubly efficient PIR from algebraic locally decodable codes. We view the resulting PIR and interpolation interpretation as an additional conceptual distinction.

From distinguishing to key recovery. Turning a distinguisher into a key-recovery algorithm, which strictly implies a decryption algorithm, generally requires additional ideas. Bardet, Mora, and Tillich [BMT24] gave a polynomial-time key-recovery attack for alternant codes sampled generically over $\mathbb{F}_2$ or $\mathbb{F}_3$, throughout the very-high-rate regime in which the square-code distinguisher applies. Their algorithm repeatedly uses shortening and componentwise products to derive alternant codes of decreasing degree while preserving the hidden support. Once the degree has been reduced to three, a Gröbner-basis computation recovers the support and multipliers. This degree-reduction mechanism does not extend to the Goppa subclass.

Couvreur, Mora, and Tillich [CMT23] showed that low-rank quadratic relations can also reveal the hidden support and multipliers. For alternant codes satisfying the genericity conditions of their analysis, their direct recovery method applies when the alternant degree r satisfies $r < q + 1$. For Goppa codes, it requires the stronger condition $r < q - 1$, where q is the size of the base field and r is the alternant or Goppa degree. In particular, this condition excludes every nontrivial binary Goppa code and therefore does not cover either CFS or Classic McEliece.

Later work extended the quadratic-relation approach to restricted binary-Goppa regimes, including square-free degree-two codes, the very-high-rate parameters of the Courtois–Finiasz–Sendrier (CFS) signature scheme, and several public McEliece key-recovery challenges released by the Technology Innovation Institute (TII) [Mor25, BLT26, BLRT26]. The bilinear attack of [BLRT26], for example, recovers keys for several TII challenge instances and for a CFS instance, but its experiments do not reach the Classic McEliece parameter sets.

Hemmert [Hem26] subsequently showed how the structure of higher-order-vanishing polynomials can likewise be used for key recovery, demonstrating the attack on several previously unbroken TII challenges. He conjectures that the method extends to the Classic McEliece parameter sets with roughly the complexity of the underlying higher-order-vanishing distinguisher, although the resulting estimates remain well above the targeted security levels.

Related polynomial-method techniques. The argument for our distinguisher analysis follows a common "polynomial-method" paradigm in algebraic coding theory and theoretical computer science: construct a low-degree polynomial satisfying many local constraints, compose it with a structured low-degree parametrization, and use a degree-versus-zeros argument to force the result-

ing univariate polynomial to be identically zero. Guruswami and Sudan used this paradigm, with multiplicity constraints, in their list-decoding algorithm for Reed–Solomon and algebraic-geometry codes [GS99]. For decoding related algebraic codes, Coppersmith and Sudan [CS03] and Parvaresh and Vardy [PV05] similarly interpolated multivariate relations and forced them to vanish identically after substituting suitably correlated message polynomials. Guruswami and Rudra [GR08] instantiated this approach for folded Reed–Solomon codes, and Guruswami and Wang [GW13] and Kopparty [Kop15] later adapted it to univariate multiplicity codes. Dvir, Kopparty, Saraf, and Sudan [DKSS13] further abstracted and strengthened the multiplicity aspect of this method, propagating high-multiplicity vanishing along lines and developing multiplicity versions of Schwartz–Zippel. Kopparty, Saraf, and Yekhanin [KSY14] used the same restriction-to-lines viewpoint to construct multiplicity codes, whose symbols consist of Hasse jets and whose local correction reduces multivariate recovery to univariate multiplicity decoding.

While our algorithms follow a similar high-level approach, our use of this paradigm differs in two ways. First, the univariate coordinates and the low-degree curve are hidden from the algorithm. Still, the existence of this hidden parametrization forces a publicly testable linear relation among Hasse-jet functionals. Second, in our setting the forced polynomial identity is used as a hold-out distinguishing test, rather than as an equation for recovering a nearby codeword.

Connection with private information retrieval. As discussed above, our attacks originated from a failed attempt to construct a doubly efficient PIR scheme. Here we summarize the connection. Span-based attacks impose a standard obstacle for linear PIR: if the server's answers do not span the linear functional corresponding to a database position, the server can rule out that position as the client's target. This was already observed in the works of Boyle, Ishai, Pass, and Wootters [BIPW17] and Canetti, Holmgren, and Richelson [CHR17], and further generalized by Applebaum, Ishai, and Shechter [AIS26]. We repurpose this obstacle as a cryptanalytic tool, replacing the span test by a hold-out test that enables distinguishing the permuted PIR queries from random. Candidate PIR queries obtained from algebraic local decoders have exactly the form of permuted subcodes of evaluation codes used in McEliece-type public keys; the PIR hold-out test then becomes a hold-out test for the corresponding McEliece distribution. Reed–Muller encoding recovers the familiar Schur-product attack, while multiplicity-code encoding yields the stronger Hermite-interpolation attack used in this work.

2 Preliminaries

Notation. Let $\mathbb{N} = \{1, 2, \dots\}$ be the natural numbers, and define $[a, b] := \{a, a+1, \dots, b\}$, $[n] := [1, n]$. For a prime power q, we write $\mathbb{F}_q$ to denote the finite field with q elements, and $\mathbb{F}_q^\times$ to denote its nonzero elements. We write vectors and matrices in boldface, e.g. $\boldsymbol{v} \in \mathbb{F}_q^k$ and $\boldsymbol{M} \in \mathbb{F}_q^{n\times k}$. For $\boldsymbol{a} \in \mathbb{Z}_{\geqslant 0}^k$, we write $|\boldsymbol{a}| := \sum_{i=1}^k a_i$ for its ℓ_1-norm. For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{F}_q^k$, let $d_{\mathrm{H}}(\boldsymbol{x}, \boldsymbol{y}) := |\mathrm{supp}(\boldsymbol{x} - \boldsymbol{y})|$ denote their Hamming distance. Unless stated otherwise, the degree of a univariate polynomial means its usual degree, and the degree of a multivariate polynomial means total degree. We use "degree at most d" for polynomials in bounded-degree spaces such as $\mathcal{P}_{\leqslant d}$, and "degree exactly d" when exact degree is intended. In particular, a homogeneous degree-d polynomial has all nonzero monomials of total degree exactly d. A nonzero polynomial $f(x) \in \mathbb{F}_q[x]$ is called *square-free* if there is no non-constant polynomial $g(x) \in \mathbb{F}_q[x]$ such that $g(x)^2 \mid f(x)$. Equivalently, every irreducible

factor of $f(x)$ appears with multiplicity one. For a polynomial $P(U)$, we write $[U^x]P(U)$ for the coefficient of U^x in $P(U)$.

Computational indistinguishability. PPT denotes probabilistic polynomial time. We consider non-uniform algorithms, or equivalently circuit families, with a specified time or size bound T. When no explicit bound is given, T is taken to be polynomial by default. All asymptotic parameters and distributions are indexed by a security parameter κ, which is omitted for brevity when unambiguous. A distribution ensemble $\mathcal{D} = \{\mathcal{D}_\kappa\}_{\kappa\in\mathbb{N}}$ is called an efficient distribution ensemble if there is a PPT sampler Samp such that $\mathsf{Samp}(1^\kappa)$ outputs an element distributed according to $\mathcal{D}_\kappa$. For two distribution ensembles $\mathcal{D} = \{\mathcal{D}_\kappa\}_{\kappa\in\mathbb{N}}$ and $\mathcal{D}' = \{\mathcal{D}'_\kappa\}_{\kappa\in\mathbb{N}}$ over the same sample space, the distinguishing advantage of an adversary $\mathcal{A}$ is defined as

$$\mathsf{Adv}^{\mathsf{dist}}_{\mathcal{D},\mathcal{D}'}(\mathcal{A},\kappa) := \left| \mathop{\mathbf{Pr}}_{x\leftarrow\mathcal{D}_\kappa}[\mathcal{A}(1^\kappa,x)=1] - \mathop{\mathbf{Pr}}_{x\leftarrow\mathcal{D}'_\kappa}[\mathcal{A}(1^\kappa,x)=1] \right|.$$

For functions $\mathsf{T} = \mathsf{T}(\kappa)$ and $\delta = \delta(\kappa)$, we say that $\mathcal{D}$ and $\mathcal{D}'$ are (T,δ)-*computationally indistinguishable* if every non-uniform adversary $\mathcal{A}$ running in time at most $\mathsf{T}(\kappa)$ satisfies $\mathsf{Adv}^{\mathsf{dist}}_{\mathcal{D},\mathcal{D}'}(\mathcal{A},\kappa) \leqslant \delta(\kappa)$ for all sufficiently large κ.

2.1 (Generalized) Reed–Solomon Codes

Definition 2.1 (Reed–Solomon and Generalized Reed–Solomon Codes [RS60, MS77])**.** *Let q be a prime power, and let $n, K \in \mathbb{N}$ with $K \leqslant n \leqslant q$. Let $\boldsymbol{\alpha} = (\alpha_1,\dots,\alpha_n) \in \mathbb{F}_q^n$ be a vector of distinct elements, and let $\boldsymbol{\lambda} = (\lambda_1,\dots,\lambda_n) \in (\mathbb{F}_q^\times)^n$ be a vector of nonzero multipliers. The generalized Reed–Solomon code of dimension K with support $\boldsymbol{\alpha}$ and multipliers $\boldsymbol{\lambda}$ is the code*

$$\mathsf{GRS}_K(\boldsymbol{\alpha},\boldsymbol{\lambda}) = \{(\lambda_1 f(\alpha_1),\dots,\lambda_n f(\alpha_n)) : f \in \mathbb{F}_q[X],\ \deg(f) < K\} \subseteq \mathbb{F}_q^n.$$

Equivalently, for each polynomial $f \in \mathbb{F}_q[X]$ of degree less than K, the corresponding codeword is

$$\mathsf{Encode}_{\boldsymbol{\alpha},\boldsymbol{\lambda},K}(f) = (\lambda_i f(\alpha_i))_{i\in[n]}.$$

The Reed–Solomon code with support $\boldsymbol{\alpha}$ is the special case where all multipliers are equal to 1:

$$\mathsf{RS}_K(\boldsymbol{\alpha}) := \mathsf{GRS}_K(\boldsymbol{\alpha},(1,\dots,1)).$$

2.2 Binary Goppa Codes

Definition 2.2 (Binary Goppa Code [Gop70, CMT23])**.** *Let $m, n, t \in \mathbb{N}$. Let $\boldsymbol{\alpha} = (\alpha_1,\dots,\alpha_n) \in \mathbb{F}_{2^m}^n$ be a vector of distinct elements, and let $\Gamma(Z) \in \mathbb{F}_{2^m}[Z]$ be a polynomial of degree exactly t such that $\Gamma(\alpha_i) \neq 0$ for every $i \in [n]$. Define the dual multiplier vector*

$$\boldsymbol{\mu} = \left(\Gamma(\alpha_1)^{-1},\dots,\Gamma(\alpha_n)^{-1}\right) \in (\mathbb{F}_{2^m}^\times)^n.$$

The binary Goppa code associated with support $\boldsymbol{\alpha}$ and Goppa polynomial Γ is the binary subfield subcode

$$\mathsf{Goppa}(\boldsymbol{\alpha},\Gamma) := \mathsf{GRS}_t(\boldsymbol{\alpha},\boldsymbol{\mu})^\perp \cap \mathbb{F}_2^n \subseteq \mathbb{F}_2^n,$$

where the dual is taken over $\mathbb{F}_{2^m}$, and where we view $\mathbb{F}_2^n$ as a subset of $\mathbb{F}_{2^m}^n$.

Fact 2.3 (Goppa codes as subcodes of GRS codes [Del75, MS77])**.** *Let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n) \in (\mathbb{F}_{2^m})^n$ be a support of distinct elements, and let $\Gamma \in \mathbb{F}_{2^m}[Z]$ be a degree-t polynomial such that $\Gamma(\alpha_i) \neq 0$ for every $i \in [n]$. Then the binary Goppa code $\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma)$ is a binary subfield subcode of a length-n GRS code of dimension $n - t$ over $\mathbb{F}_{2^m}$. Equivalently, there exists a multiplier vector $\boldsymbol{\lambda} \in (\mathbb{F}_{2^m}^{\times})^n$ such that*

$$\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma) \subseteq \mathsf{GRS}_{n-t}(\boldsymbol{\alpha}, \boldsymbol{\lambda}) \subseteq \mathbb{F}_{2^m}^n.$$

Moreover, if Γ is square-free and $2t < n$, then $\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma) = \mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma^2)$. Consequently, there exists $\boldsymbol{\lambda}' \in (\mathbb{F}_{2^m}^{\times})^n$ such that

$$\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma) \subseteq \mathsf{GRS}_{n-2t}(\boldsymbol{\alpha}, \boldsymbol{\lambda}') \subseteq \mathbb{F}_{2^m}^n.$$

Writing $\Pi_{\boldsymbol{\alpha}}(Z) := \prod_{j=1}^{n}(Z - \alpha_j)$, the multiplier vectors may be chosen, up to a common nonzero scalar, as

$$\lambda_i = \frac{\Gamma(\alpha_i)}{\Pi'_{\boldsymbol{\alpha}}(\alpha_i)} \qquad \text{and} \qquad \lambda'_i = \frac{\Gamma(\alpha_i)^2}{\Pi'_{\boldsymbol{\alpha}}(\alpha_i)},$$

respectively.

We also consider Alternant Codes, Wild Goppa Codes, and $q$ary Goppa codes. We recall their definitions in Appendix B.

2.3 McEliece Assumptions

In the McEliece cryptosystem, the secret key consists of a structured generator matrix $\boldsymbol{G} \leftarrow \mathcal{G}_\kappa$ together with the hiding transformations $\boldsymbol{S}$ and $\boldsymbol{P}$, while the public key is the disguised matrix $\boldsymbol{Y} = \boldsymbol{SGP}$. To encrypt a message vector $\boldsymbol{u} \in \mathbb{F}_q^k$, the sender samples a correctable error vector $\boldsymbol{e} \leftarrow \mathcal{E}_\kappa$ and outputs the noisy codeword $\boldsymbol{c} = \boldsymbol{uY} + \boldsymbol{e}$. The secret-key holder uses $\boldsymbol{P}$ to undo the coordinate permutation, decodes using the hidden code generated by $\boldsymbol{G}$, and then uses $\boldsymbol{S}$ to recover $\boldsymbol{u}$.

If the public key $\boldsymbol{Y}$ is indistinguishable from random, which is a common conservative security requirement for McEliece, then recovering $\boldsymbol{u}$ from $\boldsymbol{c}$ is hard based on the standard Learning Parity with Noise (LPN) assumption [BFKL94]. We now formally define both the conservative indistinguishability assumption and the minimal "one-way security" assumption for McEliece.

Definition 2.4 (McEliece public-key pseudorandomness [McE78, DMQN09])**.** *Let κ denote a security parameter. Let $q = q(\kappa)$ be a prime power, and let $k = k(\kappa)$, $K = K(\kappa)$, and $n = n(\kappa)$ be code parameters with $k \leqslant K \leqslant n$. Let $\mathcal{G} = \{\mathcal{G}_\kappa\}_\kappa$ be an efficient distribution ensemble over full-rank generator matrices $\boldsymbol{G} \in \mathbb{F}_q^{K \times n}$.*

Define the McEliece public-key distribution associated with $\mathcal{G}_\kappa$ as

$$\mathcal{D}_{\mathsf{McE},\kappa} = \left\{ \boldsymbol{SGP} : \begin{array}{l} \boldsymbol{G} \leftarrow \mathcal{G}_\kappa, \\ \boldsymbol{S} \xleftarrow{\$} \{\textit{full-rank matrices in } \mathbb{F}_q^{k \times K}\}, \\ \boldsymbol{P} \xleftarrow{\$} \{\textit{permutation matrices in } \mathbb{F}_q^{n \times n}\} \end{array} \right\}.$$

Let $\mathcal{U}_{k,n,q}$ denote the uniform distribution over all matrices in $\mathbb{F}_q^{k \times n}$. We say that $\{\mathcal{G}_\kappa\}_{\kappa \in \mathbb{N}}$ satisfies (T, δ)-McEliece public-key pseudorandomness if $\mathcal{D}_{\mathsf{McE},\kappa}$ and $\mathcal{U}_{k,n,q}$ are (T, δ)-computationally indistinguishable.

Remark 2.5 (On code distributions)**.** *When we refer to the distribution of the public key, we always refer to a randomized representation of the corresponding (permuted) code. A more natural choice for the random-code distribution is to sample uniformly from full-rank matrices in $\mathbb{F}_q^{k\times n}$. We omit this conditioning throughout the paper: when $n-k$ is large, a uniformly random matrix in $\mathbb{F}_q^{k\times n}$ is full rank except with probability at most $q^{-(n-k)}/(q-1)$, so conditioning on full rank changes the distribution by only negligible statistical distance in our parameter regimes.*

Definition 2.6 (McEliece one-way CPA security [McE78, BCL+17])**.** *Let κ denote a security parameter. Let $q, k, K, n, \mathcal{G}_\kappa$ and $\mathcal{D}_{\mathsf{McE},\kappa}$ be as in Definition 2.4, and let $\mathcal{E}_\kappa$ be an efficiently samplable distribution over error vectors in $\mathbb{F}_q^n$ such that every error in the support of $\mathcal{E}_\kappa$ is decodable for any code in the family.*

Define the McEliece OW-CPA experiment associated with $(\mathcal{G}_\kappa, \mathcal{E}_\kappa)$ and an adversary $\mathcal{A}$ as follows:

$$\mathsf{Exp}^{\mathsf{ow\text{-}cpa}}_{\mathsf{McE},\mathcal{A}}(\kappa) = \left[\begin{array}{l} \boldsymbol{Y} \xleftarrow{\$} \mathcal{D}_{\mathsf{McE},\kappa}, \\ \boldsymbol{u} \xleftarrow{\$} \mathbb{F}_q^k, \quad \boldsymbol{e} \leftarrow \mathcal{E}_\kappa, \\ \boldsymbol{c} := \boldsymbol{u}\boldsymbol{Y} + \boldsymbol{e}, \\ \widehat{\boldsymbol{u}} \leftarrow \mathcal{A}(1^\kappa, \boldsymbol{Y}, \boldsymbol{c}) \end{array} : \begin{array}{l} \\ \\ \\ \textit{Output 1 if } \widehat{\boldsymbol{u}} = \boldsymbol{u} \\ \textit{and 0 otherwise.} \end{array}\right].$$

The OW-CPA success probability of $\mathcal{A}$ is

$$\mathsf{Adv}^{\mathsf{ow\text{-}cpa}}_{\mathsf{McE},\mathcal{A}}(\kappa) := \mathbf{Pr}\left[\mathsf{Exp}^{\mathsf{ow\text{-}cpa}}_{\mathsf{McE},\mathcal{A}}(\kappa) = 1\right].$$

We say that $(\mathcal{G}_\kappa, \mathcal{E}_\kappa)_{\kappa\in\mathbb{N}}$ satisfies (T, δ)-McEliece one-way CPA security if every non-uniform adversary $\mathcal{A}$ running in time at most $\mathsf{T}(\kappa)$ satisfies

$$\mathsf{Adv}^{\mathsf{ow\text{-}cpa}}_{\mathsf{McE},\mathcal{A}}(\kappa) \leqslant \delta(\kappa)$$

for all sufficiently large κ.

Classic McEliece: Asymptotic parameters. Unless stated otherwise, when we refer to the McEliece public-key pseudorandomness or one-way CPA security definitions above, we have in mind the following binary Goppa instantiation, which is meant to capture the asymptotic parameter scaling of Classic McEliece [ABC+22, BCL+17].[6] We take $q = 2$ and choose parameter functions $m = m(\kappa)$, $n = n(\kappa)$, and $t = t(\kappa)$ with $n \leqslant 2^m$, $m = \Theta(\log n)$, and $t = \Theta(n/\log n)$ so that $K := n - mt = \Theta(n)$. The distribution $\mathcal{G}_\kappa$ is obtained by sampling a support $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$ of distinct elements from $\mathbb{F}_{2^m}$ and an irreducible Goppa polynomial $\Gamma \in \mathbb{F}_{2^m}[Z]$ of degree exactly t satisfying $\Gamma(\alpha_i) \neq 0$ for every $i \in [n]$, and then outputting a full-rank generator matrix of a uniformly random K-dimensional subcode of the associated binary Goppa code $\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma)$. The error distribution $\mathcal{E}_\kappa$ samples uniformly from $\{\boldsymbol{e} \in \mathbb{F}_2^n : |\mathrm{supp}(\boldsymbol{e})| = t\}$. In the full designed-dimension setting corresponding to the usual Classic McEliece scaling, the public row dimension is $k = K = n - mt = \Theta(n)$. Concrete Classic McEliece parameter sets and the corresponding parameters used in our attacks are discussed in Section 5.

[6] To simplify the presentation, we consider here a "primal" variant with Classic parameter scaling. The actual Classic McEliece KEM uses the dual Niederreiter syndrome representation. Given a parity-check matrix $\boldsymbol{H}$ in this representation, one can apply the primal attack to a randomly permuted random basis of the code checked by $\boldsymbol{H}$.

2.4 Hasse Derivatives and Multiplicity Codes

We first introduce some notation before defining multiplicity codes.

We define $\mathcal{A}_{w,k}$ to be the set of all vectors of non-negative integers of length k and ℓ_1-norm exactly w:

$$\mathcal{A}_{w,k} = \left\{ \boldsymbol{a} \in \mathbb{Z}_{\geqslant 0}^k : \ |\boldsymbol{a}| = w \right\}.$$

Let $\mathcal{A}_{<w,k} := \mathcal{A}_{0,k} \cup \mathcal{A}_{1,k} \cup \ldots \cup \mathcal{A}_{w-1,k}$.

Henceforth, given a vector $\boldsymbol{X} := (X_1, \ldots, X_k)$ of variables and a vector $\boldsymbol{a} := (a_1, \ldots, a_k)$ of exponents, we use the following vector exponentiation notation:

$$\boldsymbol{X}^{\boldsymbol{a}} := \prod_{i=1}^{k} X_i^{a_i}.$$

Definition 2.7 (Hasse derivatives)**.** *For a k-variate polynomial $F \in \mathbb{F}[X_1, \ldots, X_k]$ over a field $\mathbb{F}$, the Hasse derivative of F with respect to $\boldsymbol{a} = (a_1, \ldots, a_k) \in \mathbb{Z}_{\geqslant 0}^k$ is defined as*

$$\partial^{[\boldsymbol{a}]} F = \sum_{\substack{\boldsymbol{b}=(b_1,\ldots,b_k)\in\mathbb{Z}_{\geqslant 0}^k \\ \boldsymbol{b}\geqslant\boldsymbol{a}}} \prod_{i=1}^{k} \binom{b_i}{a_i} \cdot \mathsf{Coeff}_{X_1^{b_1}\ldots X_k^{b_k}}(F) \boldsymbol{X}^{\boldsymbol{b}-\boldsymbol{a}},$$

where $\mathsf{Coeff}_{X_1^{b_1}\ldots X_k^{b_k}}(F)$ denotes the coefficient of $X_1^{b_1} \ldots X_k^{b_k}$ in F.

For a univariate degree-at-most-d polynomial $f(Z) = \sum_{j=0}^{d} c_j \cdot Z^j \in \mathbb{F}[Z]$, we omit the vector notation and denote its r-th Hasse derivative by $\partial^{[r]} f(Z) = \sum_{j=r}^{d} c_j \cdot \binom{j}{r} Z^{j-r}$. If the field $\mathbb{F}$ has characteristic 0, then $\partial^{[r]} f(Z) = \frac{1}{r!} f^{(r)}(Z)$.

Hermite interpolation. Over a field of characteristic 0, sufficiently many evaluations of a polynomial f and its higher-order ordinary derivatives determine f. This is known as Hermite interpolation. The analogous statement for Hasse derivatives holds over fields of characteristic zero and over finite fields:

Lemma 2.8 (Hermite interpolation of Hasse derivatives [Has36, BGKM22])**.** *Let f be an univariate polynomial of degree at most d over a finite field $\mathbb{F}$, let h be a positive integer, and let $\mu_1, \ldots, \mu_h$ be positive integers such that $\mu_1 + \cdots + \mu_h > d$. Let $\alpha_1, \ldots, \alpha_h$ be distinct elements of $\mathbb{F}$. For all $i \in [h]$ and $j \in [\mu_i]$, let $(\partial^{[j-1]} f)(\alpha_i) = y_{i,j}$. Then the coefficients of f can be recovered from $\{(\alpha_i, j, y_{i,j})\}_{i\in[h], j\in[\mu_i]}$ in time $\mathrm{poly}(d, h, \log|\mathbb{F}|)$.*

To jointly consider the evaluation of F and its bounded-order Hasse derivatives at a given point, we will use the following notion of a Hasse jet.

Definition 2.9 (Hasse jet)**.** *For $\boldsymbol{x} \in \mathbb{F}_q^k$ and $s \geqslant 1$, define the* Hasse jet of order $< s$ *of F at $\boldsymbol{x}$ (also referred to as the* order-s evaluation of F at $\boldsymbol{x}$*) by*

$$J_{\boldsymbol{x}}^{<s}(F) := \left((\partial^{[\boldsymbol{a}]} F)(\boldsymbol{x}) \right)_{\boldsymbol{a}\in\mathcal{A}_{<s,k}}.$$

We say that F vanishes with multiplicity (at least) s *at $\boldsymbol{x}$ if $J_{\boldsymbol{x}}^{<s}(F) = \boldsymbol{0}$.*

Multiplicity codes. We now define multiplicity codes. We use the definition from [KSY14].

Definition 2.10 (Multiplicity Code [KSY14])**.** *Let s, d, k be non-negative integers and let q be a prime power. Let $\Sigma = \mathbb{F}_q^{\binom{k+s-1}{k}} = \mathbb{F}_q^{\{\boldsymbol{a} \in \mathcal{A}_{<s,k}\}}$. The multiplicity code of order-s evaluations of polynomials of total degree at most d in k variables over $\mathbb{F}_q$ is defined as follows. The code is over the alphabet Σ and has length q^k, where the coordinates are indexed by elements of $\mathbb{F}_q^k$. For each polynomial $F(\boldsymbol{X}) \in \mathbb{F}_q[X_1, \ldots, X_k]$ of (total) degree at most d, there is a codeword given by*

$$\mathsf{Encode}_{s,d,k,q}(F) = \left(J_{\boldsymbol{x}}^{<s}(F)\right)_{\boldsymbol{x} \in \mathbb{F}_q^k} .$$

Binary Hasse derivatives and Hamming balls. Our main algorithms for binary Goppa–McEliece will use Hasse derivatives of multilinear polynomials $F \in \mathbb{F}_2[X_1, \ldots, X_k]$. In this case, if $\boldsymbol{a} \in \mathbb{Z}_{\geqslant 0}^k$ has some coordinate $a_i \geqslant 2$, then $\partial^{[\boldsymbol{a}]} F \equiv 0$. Thus, for the potentially nonzero derivatives, we identify a subset $S \subseteq [k]$ with its indicator vector $\mathbf{1}_S$, and write

$$\partial^{[S]} F := \partial^{[\mathbf{1}_S]} F.$$

For every $S, T \subseteq [k]$ and $\boldsymbol{x} \in \mathbb{F}_2^k$, Boolean Möbius inversion (cf. [Lai94, Proposition 3]) gives

$$\partial^{[S]} F(\boldsymbol{x}) = \sum_{T \subseteq S} F(\boldsymbol{x} + \mathbf{1}_T), \qquad F(\boldsymbol{x} + \mathbf{1}_T) = \sum_{S \subseteq T} \partial^{[S]} F(\boldsymbol{x}),$$

where all sums are in $\mathbb{F}_2$. For $\boldsymbol{x} \in \mathbb{F}_2^k$ and $0 \leqslant r \leqslant k$, define the radius-$r$ Hamming ball by

$$B_H(\boldsymbol{x}, r) := \{\boldsymbol{y} \in \mathbb{F}_2^k : d_{\mathrm{H}}(\boldsymbol{x}, \boldsymbol{y}) \leqslant r\}.$$

Consequently,

$$\partial^{[S]} F(\boldsymbol{x}) = 0 \quad \text{for every } |S| \leqslant r$$

if and only if

$$F(\boldsymbol{y}) = 0 \quad \text{for every } \boldsymbol{y} \in B_H(\boldsymbol{x}, r).$$

Equivalently, writing $B_{<s}(\boldsymbol{x}) := B_H(\boldsymbol{x}, s-1)$, the Hasse jet $J_{\boldsymbol{x}}^{<s}(F)$ vanishes if and only if F vanishes on $B_{<s}(\boldsymbol{x})$. A constructive variant of this identity was recently used in the PIR context to recover Hasse derivatives from nearby evaluations [HR26].

3 Distinguishing Subcodes of GRS Codes

We begin with a distinguisher for public matrices generated by arbitrary subcodes of generalized Reed–Solomon codes. Fix a parent GRS dimension K and write $D := K-1$. If $\boldsymbol{Y} = \boldsymbol{S}\boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha}, \boldsymbol{\lambda})\boldsymbol{P}$, then, after absorbing the permutation into the ordering of the support and multipliers, there exist polynomials $f_1, \ldots, f_k \in \mathbb{F}_q[Z]$ of degree at most D and a multiplier vector $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n) \in (\mathbb{F}_q^\times)^n$ such that each public column has the form $\boldsymbol{y}_j = \lambda_j \cdot (f_1(\alpha_j), \ldots, f_k(\alpha_j))$. Thus, the public columns are independently rescaled points on a hidden low-degree curve. The RS–McEliece distribution is the special case in which all the multipliers are equal to one.

Unlike the span test from Fig. 2, here we will use an equivalent dual formulation using annihilating polynomials, which will facilitate the analysis. Our distinguisher holds out one column and computes the space of homogeneous degree-d polynomials whose order-$< s$ Hasse jets, namely

the derivatives of all orders below s, vanish on every non-held-out column. It accepts if every polynomial in this space also vanishes at the held-out column. Homogeneity makes these vanishing conditions invariant under the unknown GRS multipliers, while composition with the hidden curve reduces the structured-case analysis to univariate Hermite interpolation. In particular, the condition $s(n-1) > dD$ forces every kernel polynomial to vanish at the held-out GRS column. On the other hand, if $(n-1)\binom{k+s-1}{s-1} < \binom{k+d-1}{d}$, the derivative constraints leave a nonzero homogeneous kernel polynomial, which rejects a uniformly random held-out column with probability at least $1 - d/q$.

The distinguishing task. Fix a parent GRS dimension K and set $D := K - 1$. We define the following distributions over matrices in $\mathbb{F}_q^{k\times n}$. The GRS–McEliece distribution is

$$\mathcal{D}_{\mathsf{GRS}} := \left\{ \boldsymbol{S}\boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha},\boldsymbol{\lambda})\boldsymbol{P} : \begin{array}{l} \boldsymbol{\alpha} = (\alpha_1,\dots,\alpha_n) \xleftarrow{\$} \{\boldsymbol{a} \in \mathbb{F}_q^n : a_i \neq a_j \text{ for all } i \neq j\}, \\ \boldsymbol{\lambda} = (\lambda_1,\dots,\lambda_n) \xleftarrow{\$} (\mathbb{F}_q^\times)^n, \\ \boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha},\boldsymbol{\lambda}) = \left(\lambda_j \alpha_j^r\right)_{r\in[0,D],j\in[n]} \in \mathbb{F}_q^{K\times n}, \\ \boldsymbol{S} \xleftarrow{\$} \{\text{full-rank matrices in } \mathbb{F}_q^{k\times K}\}, \\ \boldsymbol{P} \xleftarrow{\$} \{\text{permutation matrices in } \mathbb{F}_q^{n\times n}\} \end{array} \right\}.$$

The RS–McEliece distribution $\mathcal{D}_{\mathsf{RS}}$ is the special case in which the multiplier vector is fixed to $\boldsymbol{\lambda} = (1,\dots,1)$. The random distribution is $\mathcal{U}_{k,n,q}$, the uniform distribution over matrices in $\mathbb{F}_q^{k\times n}$. Our goal is to distinguish $\mathcal{D}_{\mathsf{GRS}}$ from $\mathcal{U}_{k,n,q}$; the same distinguisher therefore also applies to $\mathcal{D}_{\mathsf{RS}}$. Throughout this section, we assume $k \leqslant K$, so that full-rank scramblers in $\mathbb{F}_q^{k\times K}$ exist.

3.1 Distinguishing Attack on GRS–McEliece

The unknown column multipliers prevent the public columns themselves from lying on a common affine low-degree curve. We remove these multipliers by restricting the polynomial test to homogeneous polynomials, following a standard technique for handling GRS subcodes [Wie10].

Lemma 3.1 (Homogeneous scaling)**.** *Let $H \in \mathbb{F}_q[X_1,\dots,X_k]$ be homogeneous of degree d. Then, for every $\lambda \in \mathbb{F}_q^\times$, every $\boldsymbol{x} \in \mathbb{F}_q^k$, and every $\boldsymbol{a} \in \mathbb{Z}_{\geqslant 0}^k$ with $|\boldsymbol{a}| \leqslant d$,*

$$(\partial^{[\boldsymbol{a}]}H)(\lambda\boldsymbol{x}) = \lambda^{d-|\boldsymbol{a}|} \cdot (\partial^{[\boldsymbol{a}]}H)(\boldsymbol{x}).$$

In particular, for every $s \in \mathbb{N}$,

$$J_{\lambda\boldsymbol{x}}^{<s}(H) = \boldsymbol{0} \iff J_{\boldsymbol{x}}^{<s}(H) = \boldsymbol{0}.$$

Proof. For homogeneous H of degree d, the Hasse derivative $\partial^{[\boldsymbol{a}]}H$ is homogeneous of degree $d-|\boldsymbol{a}|$ when $|\boldsymbol{a}| \leqslant d$, and it is identically zero when $|\boldsymbol{a}| > d$. Since $\lambda \neq 0$, multiplication by $\lambda^{d-|\boldsymbol{a}|}$ does not affect whether the derivative vanishes. □

We now instantiate the polynomial-kernel test for GRS–McEliece.

Algorithm 1 (The homogeneous hold-out polynomial distinguisher over $\mathbb{F}_q$)**.** *Let d and s be attack parameters. Define*

$$\mathcal{P}_{=d}^{\mathrm{hom}} := \{H \in \mathbb{F}_q[X_1,\dots,X_k] : H \text{ is homogeneous of degree } d\}.$$

On input $\boldsymbol{Y} \in \mathbb{F}_q^{k\times n}$, the distinguisher proceeds as follows.

1. *Choose a hold-out column $\tau \in [n]$. By symmetry, one may fix $\tau = n$.*
2. *Compute the linear space*

$$\mathcal{K}_{\tau}^{\mathrm{hom}}(\boldsymbol{Y}) := \left\{ H \in \mathcal{P}_{=d}^{\mathrm{hom}} : (\partial^{[\boldsymbol{a}]} H)(\boldsymbol{y}_j) = 0 \text{ for every } j \in [n] \setminus \{\tau\} \text{ and every } \boldsymbol{a} \in \mathcal{A}_{<s,k} \right\}.$$

3. *Compute a basis $H_1, \ldots, H_{h_\tau}$ of $\mathcal{K}_{\tau}^{\mathrm{hom}}(\boldsymbol{Y})$.*
4. *If $H_i(\boldsymbol{y}_\tau) = 0$ for every $i \in [h_\tau]$, output* GRS*. Otherwise, output* random*.*

Theorem 3.2 (GRS–McEliece distinguisher)**.** *Let k, K, d, s, n, q be parameters with $n \leqslant q$, $k \leqslant K$, and $d < q$, and set $D := K - 1$. Suppose that*

$$s(n-1) > d \cdot D, \tag{1}$$

$$(n-1) \cdot \binom{k+s-1}{s-1} < \binom{k+d-1}{d}. \tag{2}$$

Then Algorithm 1 runs in time $\mathrm{poly}\left(\binom{k+d-1}{d}, n, \log q\right)$ *and distinguishes $\mathcal{D}_{\mathsf{GRS}}$ from $\mathcal{U}_{k,n,q}$. More precisely, if $\boldsymbol{Y} \leftarrow \mathcal{D}_{\mathsf{GRS}}$, then the distinguisher outputs* GRS *with probability* 1*, while if $\boldsymbol{Y}$ is sampled uniformly from $\mathbb{F}_q^{k \times n}$, then the distinguisher outputs* GRS *with probability at most $\frac{d}{q}$. Thus, the acceptance-probability gap is at least $1 - \frac{d}{q}$.*

Remark 3.3. *Condition (1) is the reconstruction condition: the composition of a homogeneous degree-d polynomial with the hidden degree-D curve has degree at most dD, while the $n-1$ non-hold-out columns provide zeros of multiplicity s. Condition (2) is the kernel condition: the number of multiplicity constraints is smaller than the dimension of the homogeneous degree-d polynomial space, so a nonzero kernel polynomial exists. Since RS–McEliece is the special case in which all GRS multipliers are one, Theorem 3.2 also gives a distinguisher for RS–McEliece. If one is interested only in the RS case, homogeneity can be dropped and the right-hand side of Condition (2) can be enlarged from $\binom{k+d-1}{d}$ to $\binom{k+d}{d}$.*

Proof. The main computation of the distinguisher is Gaussian elimination for the kernel basis, which can be performed in time $\mathrm{poly}\left(\binom{k+d-1}{d}, n, \log q\right)$.

For the advantage analysis, we first consider the case $\boldsymbol{Y} \leftarrow \mathcal{D}_{\mathsf{GRS}}$. Write

$$\boldsymbol{Y} = \boldsymbol{S}\boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha}, \boldsymbol{\lambda})\boldsymbol{P}.$$

Absorb the hidden permutation into the indexing of the support and multiplier vectors. Viewing the rows of $\boldsymbol{S}$ as coefficient vectors, there exist polynomials $f_1, \ldots, f_k \in \mathbb{F}_q[Z]$ of degree at most D and a multiplier vector $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n) \in (\mathbb{F}_q^{\times})^n$ such that every column of $\boldsymbol{Y}$ has the form

$$\boldsymbol{y}_j = \lambda_j \cdot \boldsymbol{p}_j, \qquad \boldsymbol{p}_j := (f_1(\alpha_j), \ldots, f_k(\alpha_j)).$$

Fix any polynomial $H \in \mathcal{K}_{\tau}^{\mathrm{hom}}(\boldsymbol{Y})$. Since H is homogeneous and $\lambda_j \neq 0$, Lemma 3.1 gives

$$(\partial^{[\boldsymbol{a}]} H)(\boldsymbol{p}_j) = 0 \qquad \text{for every } j \neq \tau \text{ and every } \boldsymbol{a} \in \mathcal{A}_{<s,k}.$$

Define the composed univariate polynomial

$$\phi(Z) := H(f_1(Z), \ldots, f_k(Z)).$$

Since H has degree d and $\deg(f_i) \leqslant D$ for every $i \in [k]$, we have $\deg(\phi) \leqslant d \cdot D$.

For every non-held-out column $j \neq \tau$ and every $0 \leqslant x < s$, the x-th Hasse derivative of ϕ at α_j can be written as

$$(\partial^{[x]}\phi)(\alpha_j) = \sum_{\boldsymbol{a} \in \mathbb{Z}^k_{\geqslant 0}} (\partial^{[\boldsymbol{a}]}H)(\boldsymbol{p}_j) \cdot [U^x] \prod_{i \in [k]} \left(\sum_{r \geqslant 1} (\partial^{[r]}f_i)(\alpha_j) U^r \right)^{a_i}.$$

In the coefficient of U^x, only terms with $|\boldsymbol{a}| \leqslant x$ can contribute. Since $x < s$ and $H \in \mathcal{K}^{\mathrm{hom}}_\tau(\boldsymbol{Y})$, it follows that

$$(\partial^{[x]}\phi)(\alpha_j) = 0 \qquad \text{for every } j \neq \tau \text{ and every } 0 \leqslant x < s.$$

Thus, ϕ has a zero of multiplicity at least s at each of the $n-1$ distinct points $\{\alpha_j : j \neq \tau\}$. By Condition (1), these zeros have total multiplicity greater than $\deg(\phi)$, so Hermite interpolation forces ϕ to be identically zero. In particular,

$$H(\boldsymbol{y}_\tau) = H(\lambda_\tau \boldsymbol{p}_\tau) = \lambda_\tau^d H(\boldsymbol{p}_\tau) = \lambda_\tau^d \phi(\alpha_\tau) = 0.$$

This holds for every $H \in \mathcal{K}^{\mathrm{hom}}_\tau(\boldsymbol{Y})$, so the distinguisher outputs GRS with probability 1 when $\boldsymbol{Y} \leftarrow \mathcal{D}_{\mathsf{GRS}}$.

It remains to analyze the random case. We first show that $\mathcal{K}^{\mathrm{hom}}_\tau(\boldsymbol{Y}) \neq \{0\}$. The space $\mathcal{K}^{\mathrm{hom}}_\tau(\boldsymbol{Y})$ is the solution space of a homogeneous linear system whose unknowns are the coefficients of a polynomial in $\mathcal{P}^{\mathrm{hom}}_{=d}$. Hence, the number of unknowns is

$$\dim \mathcal{P}^{\mathrm{hom}}_{=d} = \binom{k+d-1}{d}.$$

Each non-held-out column imposes one linear constraint for every derivative index in $\mathcal{A}_{<s,k}$, so the total number of constraints is at most

$$(n-1) \cdot |\mathcal{A}_{<s,k}| = (n-1) \cdot \binom{k+s-1}{s-1}.$$

By Condition (2), the number of unknowns is larger than the number of constraints. Therefore, $\mathcal{K}^{\mathrm{hom}}_\tau(\boldsymbol{Y})$ contains a nonzero polynomial for every input matrix $\boldsymbol{Y}$.

Now fix all non-held-out columns $\{\boldsymbol{y}_j : j \neq \tau\}$, which fixes the nonzero space $\mathcal{K}^{\mathrm{hom}}_\tau(\boldsymbol{Y})$, while the held-out column $\boldsymbol{y}_\tau$ remains uniformly random and independent in $\mathbb{F}_q^k$. Choose a deterministic nonzero polynomial $H^\star \in \mathcal{K}^{\mathrm{hom}}_\tau(\boldsymbol{Y})$, for example the first nonzero element in a fixed basis of $\mathcal{K}^{\mathrm{hom}}_\tau(\boldsymbol{Y})$. Since $H^\star$ is a nonzero polynomial of total degree d, the Schwartz–Zippel lemma [Sch80] gives

$$\Pr_{\boldsymbol{y}_\tau \xleftarrow{\$} \mathbb{F}_q^k} [H^\star(\boldsymbol{y}_\tau) = 0] \leqslant \frac{d}{q}.$$

If the all-kernel test accepts, then in particular $H^\star(\boldsymbol{y}_\tau) = 0$. Averaging over the non-held-out columns, the distinguisher outputs GRS with probability at most d/q on a uniformly random matrix. □

Corollary 3.4 (Polynomial-time attack regime)**.** *Let $0 < \gamma < 1$, $\rho > 1$, and $C \geqslant \rho$ be constants. Consider a parameter sequence with $D := K - 1$ satisfying*

$$n \leqslant q, \qquad k \leqslant K, \qquad k \geqslant D^{\gamma}, \qquad \text{and} \qquad \rho D \leqslant n \leqslant CD.$$

Let r be any positive integer satisfying $\gamma(r+1) > 1$, and let s be any positive integer satisfying $s(\rho - 1) > r$. Set

$$d := s + r.$$

Then, for all sufficiently large D, the conditions of Theorem 3.2 hold. In particular, the homogeneous hold-out polynomial distinguisher runs in time $\mathrm{poly}(k, n, \log q)$ and distinguishes $\mathcal{D}_{\mathsf{GRS}}$ from $\mathcal{U}_{k,n,q}$, with random-case acceptance probability at most $\frac{d}{q}$.

Proof. Since $n \geqslant \rho D$ and $d = s + r$, the condition $s(\rho - 1) > r$ implies that, for all sufficiently large D,

$$s(n-1) > d \cdot D.$$

It remains to verify the kernel condition. Since s, d are constants and $n \leqslant CD$, we have

$$(n-1)\binom{k+s-1}{s-1} = O(D \cdot k^{s-1}).$$

On the other hand,

$$\binom{k+d-1}{d} = \Omega(k^d) = \Omega(k^{s+r}).$$

Hence, the ratio between the left-hand side and the right-hand side is at most

$$O\left(\frac{D}{k^{r+1}}\right) \leqslant O\left(D^{1-\gamma(r+1)}\right),$$

which tends to zero by the choice of r. Therefore, for all sufficiently large D,

$$(n-1)\binom{k+s-1}{s-1} < \binom{k+d-1}{d}.$$

Since $q \geqslant n \geqslant \rho D$ and d is constant, we also have $d < q$ for all sufficiently large D. The claim follows from Theorem 3.2. Since d and s are constants, $\binom{k+d-1}{d} = \mathrm{poly}(k)$, so the distinguisher runs in time $\mathrm{poly}(k, n, \log q)$. □

Remark 3.5 (A regime beyond prior attacks)**.** *As a concrete instance of Corollary 3.4, take*

$$K = D + 1, \qquad k = \lfloor 0.1D \rfloor, \qquad n = \lfloor 1.1D \rfloor, \qquad s = 11, \qquad d = 12,$$

and assume $q \geqslant n$. For all sufficiently large D, the conditions of Theorem 3.2 hold, so our attack runs in polynomial time. This regime is not covered by full-GRS key-recovery attacks, since the public code has dimension $k < K$ and is only a subcode of the hidden GRS code. It is also beyond the standard shortening-plus-Schur-square attack: the square of the parent GRS code is already full because $n < 2D + 1$, while exposing a Schur-square defect after shortening ℓ positions requires $\ell > 2D + 1 - n \approx 0.9D$. In contrast, a generic k-dimensional public subcode can only be usefully shortened by $\ell < k \approx 0.1D$. Thus, the homogeneous multiplicity distinguisher applies in a regime beyond the reach of shortening-plus-Schur.

4 Distinguishing Goppa–McEliece

We next extend the attack to McEliece instantiated with binary Goppa codes. The reason this is possible is that a binary Goppa code is not an arbitrary binary linear code: it is a binary subfield subcode of an ambient GRS code. Consequently, any public subcode generated from a Goppa code also lies inside a suitable GRS code after embedding the binary coordinates into the extension field. The structured-side part of the GRS argument carries over after embedding binary coordinates into $\mathbb{F}_{2^m}$. However, the random-side Schwartz–Zippel analysis over $\mathbb{F}_{2^m}^k$ does not directly apply to a uniformly random binary matrix, whose columns lie in $\mathbb{F}_2^k$. For the random case, we therefore use a multilinear homogeneous variant and the minimum-distance bound for binary Reed–Muller codes.

The distinguishing task. Let $m, n, t, k \in \mathbb{N}$. For a support $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n) \in (\mathbb{F}_{2^m})^n$, let

$$\mathcal{P}_{t,\boldsymbol{\alpha}}^{\mathsf{Goppa}} := \left\{ \Gamma \in \mathbb{F}_{2^m}[Z] : \begin{array}{l} \Gamma \text{ is square-free of degree } t, \\ \Gamma(\alpha_i) \neq 0 \text{ for every } i \in [n] \end{array} \right\}.$$

For $\Gamma \in \mathcal{P}_{t,\boldsymbol{\alpha}}^{\mathsf{Goppa}}$, let

$$K(\boldsymbol{\alpha}, \Gamma) := \dim_{\mathbb{F}_2} \mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma).$$

We restrict to parameters satisfying $k \leqslant n - mt$.

Since a degree-t binary Goppa code satisfies $K(\boldsymbol{\alpha}, \Gamma) \geqslant n - mt$, this guarantees $k \leqslant K(\boldsymbol{\alpha}, \Gamma)$ for every sampled support and Goppa polynomial. Without this condition, the public matrix has rank at most $K(\boldsymbol{\alpha}, \Gamma)$, which may give a trivial rank distinguisher. Recall the McEliece public-key distribution in Definition 2.4. When instantiated with a binary Goppa code, the distribution over matrices in $\mathbb{F}_2^{k \times n}$ becomes

$$\mathcal{D}_{\mathsf{Goppa}} := \left\{ \boldsymbol{S}\boldsymbol{G}_{\mathsf{Goppa}}(\boldsymbol{\alpha}, \Gamma)\boldsymbol{P} : \begin{array}{l} \boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n) \xleftarrow{\$} \{\boldsymbol{a} \in (\mathbb{F}_{2^m})^n : a_i \neq a_j \text{ for all } i \neq j\}, \\ \Gamma \xleftarrow{\$} \mathcal{P}_{t,\boldsymbol{\alpha}}^{\mathsf{Goppa}}, \\ \boldsymbol{G}_{\mathsf{Goppa}}(\boldsymbol{\alpha}, \Gamma) \in \mathbb{F}_2^{K(\boldsymbol{\alpha},\Gamma) \times n} \text{ generates } \mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma), \\ \boldsymbol{S} \xleftarrow{\$} \{\text{full-rank matrices in } \mathbb{F}_2^{k \times K(\boldsymbol{\alpha},\Gamma)}\}, \\ \boldsymbol{P} \xleftarrow{\$} \{\text{permutation matrices in } \mathbb{F}_2^{n \times n}\} \end{array} \right\}.$$

The random distribution is $\mathcal{U}_{k,n,2}$, the uniform distribution over matrices in $\mathbb{F}_2^{k \times n}$. The distinguishing task is to distinguish $\mathcal{D}_{\mathsf{Goppa}}$ from $\mathcal{U}_{k,n,2}$.[7]

The all-kernel distinguisher for Goppa–McEliece is quite similar to the homogeneous distinguisher for GRS–McEliece, except that all computations are now over $\mathbb{F}_2$.

Algorithm 2 (Binary homogeneous all-kernel hold-out distinguisher)**.** *Let d and s be attack parameters, and define $\mathcal{P}_{=d,2}^{\mathrm{ml}}$ to be the space of multilinear homogeneous degree-d polynomials over $\mathbb{F}_2$:*

$$\mathcal{P}_{=d,2}^{\mathrm{ml}} := \left\{ h \in \mathbb{F}_2[X_1, \ldots, X_k] : \begin{array}{l} h \textit{ is homogeneous of degree } d, \\ \textit{every monomial of } h \textit{ is multilinear} \end{array} \right\}.$$

[7] Classic McEliece represents the public code by a systematic parity-check matrix, whereas our distribution uses a randomly scrambled and permuted generator matrix. From the public matrix, one can compute a generator matrix $\boldsymbol{G}$, sample a random invertible matrix $\boldsymbol{S}$ and a random permutation matrix $\boldsymbol{P}$, and run the distinguisher on $\boldsymbol{SGP}$; conditioned on the underlying code, this yields exactly the same public key distribution as ours.

On input $\boldsymbol{Y} \in \mathbb{F}_2^{k\times n}$ with columns $\boldsymbol{y}_1, \dots, \boldsymbol{y}_n \in \mathbb{F}_2^k$, the distinguisher proceeds as follows.

1. *Choose a hold-out column $\tau \in [n]$. By symmetry, one may fix $\tau = n$.*
2. *Compute the linear space*

$$\mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y}) := \left\{ h \in \mathcal{P}_{=d,2}^{\mathrm{ml}} : (\partial^{[\boldsymbol{a}]} h)(\boldsymbol{y}_j) = 0 \text{ for every } j \in [n] \setminus \{\tau\} \text{ and every } \boldsymbol{a} \in \mathcal{A}_{<s,k} \right\}.$$

3. *Set $h_\tau := \dim \mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y})$ and compute a basis $h_1, \dots, h_{h_\tau}$ of this space.*
4. *If $h_i(\boldsymbol{y}_\tau) = 0$ for every $i \in [h_\tau]$, output* Goppa*. Otherwise, output* random*.*

Because evaluation is linear, testing a basis is equivalent to testing $h(\boldsymbol{y}_\tau) = 0$ for every $h \in \mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y})$; hence the outcome does not depend on the chosen basis.

Theorem 4.1 (Goppa–McEliece distinguisher)**.** *Let m, k, d, t, s, n be parameters with $n \leqslant 2^m$, $k \leqslant n - mt$, $d \leqslant k$, and $n - 2t \geqslant 1$. Set*

$$D := n - 2t - 1.$$

Suppose that

$$s(n-1) > d \cdot D, \tag{3}$$

$$(n-1) \cdot \binom{k+s-1}{s-1} < \binom{k}{d}. \tag{4}$$

Then Algorithm 2 runs in time $\mathrm{poly}\left(\binom{k}{d}, n\right)$ and distinguishes $\mathcal{D}_{\mathsf{Goppa}}$ from $\mathcal{U}_{k,n,2}$. More precisely, if $\boldsymbol{Y} \leftarrow \mathcal{D}_{\mathsf{Goppa}}$, then the distinguisher outputs Goppa *with probability 1, while if $\boldsymbol{Y}$ is sampled uniformly from $\mathbb{F}_2^{k\times n}$, then the distinguisher outputs* Goppa *with probability at most $1 - 2^{-d}$.*

Moreover, if $d < k/4$, then for any fixed constant $2 \leqslant \Delta \leqslant d-1$, there exist constants $0 < \varepsilon_\Delta < 1/2$ and $C_\Delta > 0$ such that, if

$$(n-1) \cdot \binom{k+s-1}{s-1} < \binom{k}{d} - C_\Delta \cdot \binom{k}{d-\Delta},$$

then, on a uniformly random input $\boldsymbol{Y} \leftarrow \mathbb{F}_2^{k\times n}$, the distinguisher outputs Goppa *with probability at most $1 - \varepsilon_\Delta$.*

Proof. We first show that the kernel is nonzero. The distinguisher works over the multilinear homogeneous degree-d space

$$\mathcal{P}_{=d,2}^{\mathrm{ml}} = \mathrm{span}_{\mathbb{F}_2} \left\{ \prod_{i\in I} X_i : I \subseteq [k],\ |I| = d \right\},$$

whose dimension is $\binom{k}{d}$. Each non-held-out column imposes one linear constraint for every derivative index in $\mathcal{A}_{<s,k}$, so the total number of constraints is at most

$$(n-1) \cdot |\mathcal{A}_{<s,k}| = (n-1) \cdot \binom{k+s-1}{s-1}.$$

By Condition (4), the number of unknowns is larger than the number of constraints. Hence $\mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y})$ contains a nonzero polynomial for every input matrix $\boldsymbol{Y}$.

The same constraint matrix yields a basis of the kernel by Gaussian elimination. Condition (4) bounds both its number of rows and its number of columns by $\binom{k}{d}$. The basis contains at most $\binom{k}{d}$ polynomials, each with at most $\binom{k}{d}$ coefficients, so evaluating the entire basis at $\boldsymbol{y}_\tau$ preserves the claimed $\mathrm{poly}(\binom{k}{d}, n)$ running time.

We next analyze the case $\boldsymbol{Y} \leftarrow \mathcal{D}_{\mathsf{Goppa}}$. By Fact 2.3, after embedding $\mathbb{F}_2$ into $\mathbb{F}_{2^m}$, every row of the public matrix lies in an ambient GRS code of dimension $n - 2t$. Therefore there exist polynomials

$$f_1, \ldots, f_k \in \mathbb{F}_{2^m}[Z]$$

of degree at most

$$D := n - 2t - 1,$$

a permutation π of $[n]$, and a multiplier vector $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n) \in (\mathbb{F}_{2^m}^\times)^n$ such that, after viewing the binary columns $\boldsymbol{y}_j \in \mathbb{F}_2^k$ as vectors over $\mathbb{F}_{2^m}$,

$$\boldsymbol{y}_j = \lambda_{\pi(j)} \boldsymbol{p}_j, \qquad \boldsymbol{p}_j := \big(f_1(\alpha_{\pi(j)}), \ldots, f_k(\alpha_{\pi(j)})\big) \qquad \text{for every } j \in [n].$$

To prove that the all-kernel test accepts, it suffices to consider an arbitrary nonzero polynomial $h \in \mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y})$, since the zero polynomial vanishes trivially. Let $H \in \mathbb{F}_{2^m}[X_1, \ldots, X_k]$ be the same polynomial after embedding its coefficients from $\mathbb{F}_2$ into $\mathbb{F}_{2^m}$. Since $h \in \mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y})$, for every $j \neq \tau$ and every $\boldsymbol{a} \in \mathcal{A}_{<s,k}$ we have

$$(\partial^{[\boldsymbol{a}]} h)(\boldsymbol{y}_j) = 0 \qquad \text{over } \mathbb{F}_2.$$

The Hasse derivatives of H are obtained by the same formal expressions, now viewed over $\mathbb{F}_{2^m}$. Since $\boldsymbol{y}_j \in \mathbb{F}_2^k$, the above equalities lift to

$$(\partial^{[\boldsymbol{a}]} H)(\boldsymbol{y}_j) = 0 \qquad \text{over } \mathbb{F}_{2^m}$$

for every $j \neq \tau$ and every $\boldsymbol{a} \in \mathcal{A}_{<s,k}$.

Since H is homogeneous of degree d and $\boldsymbol{y}_j = \lambda_{\pi(j)} \boldsymbol{p}_j$ with $\lambda_{\pi(j)} \neq 0$, Lemma 3.1 gives

$$(\partial^{[\boldsymbol{a}]} H)(\boldsymbol{p}_j) = 0 \qquad \text{for every } j \neq \tau \text{ and every } \boldsymbol{a} \in \mathcal{A}_{<s,k}.$$

Now define the composed univariate polynomial

$$\phi(Z) := H(f_1(Z), \ldots, f_k(Z)) \in \mathbb{F}_{2^m}[Z].$$

Since $\deg(H) = d$ and $\deg(f_i) \leqslant D$ for every $i \in [k]$, we have

$$\deg(\phi) \leqslant d \cdot D.$$

For every $j \neq \tau$ and every $0 \leqslant x < s$, the Hasse chain rule gives

$$(\partial^{[x]} \phi)(\alpha_{\pi(j)}) = \sum_{\boldsymbol{a} \in \mathbb{Z}_{\geqslant 0}^k} (\partial^{[\boldsymbol{a}]} H)(\boldsymbol{p}_j) \cdot [U^x] \prod_{i \in [k]} \left(\sum_{r \geqslant 1} (\partial^{[r]} f_i)(\alpha_{\pi(j)}) U^r \right)^{a_i}.$$

Only terms with $|\boldsymbol{a}| \leqslant x$ can contribute to the coefficient of U^x. Since $x < s$ and

$$(\partial^{[\boldsymbol{a}]} H)(\boldsymbol{p}_j) = 0 \qquad \text{for every } j \neq \tau \text{ and every } \boldsymbol{a} \in \mathcal{A}_{<s,k},$$

we get

$$(\partial^{[x]} \phi)(\alpha_{\pi(j)}) = 0 \qquad \text{for every } j \neq \tau \text{ and every } 0 \leqslant x < s.$$

Thus ϕ has a zero of multiplicity at least s at each of the $n-1$ distinct points $\alpha_{\pi(j)}$ for $j \neq \tau$. By Condition (3), we have $s(n-1) > d \cdot D \geqslant \deg(\phi)$. Hence, by Hermite interpolation, ϕ is the zero polynomial. In particular,

$$H(\boldsymbol{p}_\tau) = \phi(\alpha_{\pi(\tau)}) = 0.$$

Since H is homogeneous and $\boldsymbol{y}_\tau = \lambda_{\pi(\tau)} \boldsymbol{p}_\tau$, we get

$$h(\boldsymbol{y}_\tau) = H(\boldsymbol{y}_\tau) = H(\lambda_{\pi(\tau)} \boldsymbol{p}_\tau) = \lambda_{\pi(\tau)}^d H(\boldsymbol{p}_\tau) = 0.$$

Since h was arbitrary, every polynomial in $\mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y})$ vanishes at $\boldsymbol{y}_\tau$. Therefore the distinguisher outputs Goppa with probability 1 on $\mathcal{D}_{\mathsf{Goppa}}$.

It remains to analyze the random case. Fix the non-held-out columns. This fixes the nonzero space $\mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y})$, while the held-out column $\boldsymbol{y}_\tau$ remains uniform and independent in $\mathbb{F}_2^k$. Choose any nonzero $h^\star \in \mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y})$ as a function of the fixed non-held-out columns. If the all-kernel test accepts, then in particular $h^\star(\boldsymbol{y}_\tau) = 0$. Since $h^\star$ is a nonzero multilinear polynomial of degree at most d, the minimum-distance bound for binary Reed–Muller codes [Ree53, Mul54] implies

$$\mathop{\mathbf{Pr}}_{\boldsymbol{y}_\tau \xleftarrow{\$} \mathbb{F}_2^k} [h^\star(\boldsymbol{y}_\tau) \neq 0] \geqslant 2^{-d}.$$

Hence, conditioned on the non-held-out columns,

$$\mathbf{Pr}[\text{the all-kernel test accepts}] \leqslant \mathbf{Pr}[h^\star(\boldsymbol{y}_\tau) = 0] \leqslant 1 - 2^{-d}.$$

Averaging over the non-held-out columns gives the claimed false-acceptance bound.

For the stronger random-case bound, fix a constant $\Delta \geqslant 2$ and set

$$\eta_\Delta := 2^{-(\Delta+1)} \qquad \text{and} \qquad \varepsilon_\Delta := \eta_\Delta.$$

By Theorem 1.5 of [ASW15], assuming $d < k/4$, the number of degree-at-most-d Boolean polynomials whose evaluations are nonzero on at most an η_Δ fraction of $\mathbb{F}_2^k$ is bounded by

$$2^{C_\Delta \binom{k}{d-\Delta}}$$

for some constant $C_\Delta > 0$ depending only on Δ. In particular, the same upper bound applies to polynomials in $\mathcal{P}_{=d,2}^{\mathrm{ml}}$. Under the stronger kernel condition, after fixing the non-held-out columns we have

$$\dim \mathcal{K}_\tau^{\mathrm{ml},2}(\boldsymbol{Y}) > C_\Delta \binom{k}{d-\Delta}.$$

The kernel therefore contains more polynomials than the total number of degree-at-most-d polynomials of relative weight at most η_Δ. Hence it contains a polynomial $h^\star$ such that

$$\Pr_{\boldsymbol{y}_\tau \xleftarrow{\$} \mathbb{F}_2^k} [h^\star(\boldsymbol{y}_\tau) \neq 0] > \eta_\Delta.$$

Acceptance of the all-kernel test implies $h^\star(\boldsymbol{y}_\tau) = 0$, so, conditioned on the non-held-out columns, its acceptance probability is less than $1 - \eta_\Delta = 1 - \varepsilon_\Delta$. Averaging over the non-held-out columns proves the stronger random-case bound. □

Corollary 4.2 (Quasipolynomial distinguisher for Classic McEliece)**.** *Consider the asymptotic Classic McEliece parameter regime discussed in Section 2:*

$$k = \Theta(n), \qquad m = \Theta(\log n), \qquad t = \Theta\left(\frac{n}{\log n}\right),$$

with $n \leqslant 2^m$, $k \leqslant n - mt$, and $n - 2t \geqslant 1$ for all sufficiently large n. In particular, the constants hidden in $m = \Theta(\log n)$ and $t = \Theta(n/\log n)$ are chosen so that $n - mt = \Theta(n)$, leaving room for $k = \Theta(n)$. Then one can choose

$$s = \Theta(\log n), \qquad d = s + 1 = \Theta(\log n),$$

so that the reconstruction condition, the kernel condition, and the stronger kernel condition in the moreover part of Theorem 4.1 all hold for any fixed constant $\Delta \geqslant 2$. Consequently, the Goppa–McEliece distinguisher runs in quasipolynomial time $n^{O(\log n)}$ and has constant distinguishing advantage. Moreover, the distinguishing advantage can be amplified to $1 - o(1)$ by picking multiple hold-out columns.

Proof. Write $t \geqslant c_t n/\log n$ and $k \geqslant c_k n$ for positive constants c_t, c_k. Choose a constant $c_s > 1/(2c_t)$ and set

$$s := \lceil c_s \log n \rceil, \qquad d := s + 1.$$

Since $D = n - 2t - 1$, the reconstruction condition becomes

$$s(n-1) > d \cdot D = (s+1)(n - 2t - 1).$$

Equivalently, it suffices that

$$2(s+1)t > n - 1,$$

which holds for all sufficiently large n by the choices of s and t.

It remains to verify the stronger kernel condition. Fix any constant $\Delta \geqslant 2$, and let C_Δ be the corresponding constant from the moreover part of Theorem 4.1. Since $d = s+1 = o(k)$ and $d < k/4$ for all sufficiently large n, it suffices to show

$$(n-1)\binom{k+s-1}{s-1} + C_\Delta \cdot \binom{k}{s+1-\Delta} < \binom{k}{s+1}.$$

Since $\Delta \geqslant 2$ is constant and $n = \Theta(k)$, the left-hand side is at most

$$O(n) \cdot \binom{k+s-1}{s-1}.$$

We compare the binomial coefficients on both sides of the inequality directly:

$$\frac{\binom{k+s-1}{s-1}}{\binom{k}{s+1}} = \frac{s(s+1)}{k(k-s)} \prod_{i=1}^{s-1} \frac{k+i}{k-i}.$$

Since $s = O(\log n)$ and $k = \Theta(n)$, we have $s = o(k)$, and hence

$$\prod_{i=1}^{s-1} \frac{k+i}{k-i} = \exp\left(O\left(\frac{s^2}{k}\right)\right) = 1 + o(1).$$

Therefore

$$O(n) \cdot \frac{\binom{k+s-1}{s-1}}{\binom{k}{s+1}} = O(n) \cdot O\left(\frac{s^2}{k^2}\right) = O\left(\frac{s^2}{n}\right) = o(1),$$

so the stronger kernel condition holds for all sufficiently large n.

Finally,

$$\binom{k}{d} \leqslant k^d = n^{O(\log n)}.$$

Therefore the running time is quasipolynomial. By Theorem 4.1, the random-case acceptance probability is at most $1 - \varepsilon_\Delta$ for some constant $\varepsilon_\Delta > 0$, while the structured-case acceptance probability is 1. Hence the acceptance-probability gap is constant.

To further amplify the advantage, we hold out a set $R \subseteq [n]$ of r columns, compute the kernel using only the $n - r$ remaining columns, and accept only if every polynomial in this kernel vanishes at every held-out column. Using the same choice of s and d, the reconstruction condition would become

$$s(n-r) > d \cdot D \iff 2(s+1)t - s(r-1) > n - 1,$$

which holds for sufficiently large n when $r \ll t$.

For a random matrix, conditioned on the non-held-out columns, the columns in R are independent and uniform, so false acceptance occurs with probability at most $(1 - \varepsilon_\Delta)^r$. Thus, taking $r = \omega(1)$ with $r \ll t$ increases the advantage to $1 - o(1)$. □

Remark 4.3 (On the square-free requirement)**.** *Even if the binary Goppa code generator matrix is defined using a Goppa polynomial that is not square-free, our distinguishing attack still applies in quasipolynomial time for the Classic McEliece parameter regime, because we can still set $D := n-t-1$ in Theorem 4.1 according to the (simpler) first part of Fact 2.3. When the Goppa polynomial is square-free, which is anyway needed for decoding with a high error rate, the attack yields a sharper concrete bound, as shown in Section 5.*

Remark 4.4 (Subfield subcodes of GRS)**.** *This section focuses on binary Goppa codes. More generally, our attack applies to any code contained in an ambient GRS code. This includes alternant codes and other codes in the Goppa family.*

5 Concrete Distinguishing Estimates

In this section we estimate the concrete efficiency of our quasipolynomial distinguishing attack using standard optimization techniques from the literature.

Concrete cost model. For the estimates below, we count *binary operations* in the Boolean-circuit cost model [BC24]. Here a binary operation is the evaluation of an arbitrary two-input Boolean function $\{0,1\}^2 \to \{0,1\}$, such as AND, OR, or XOR, and has unit cost. A NOT operation also has unit cost, while constants and copying, and hence fan-out, are free. The resulting count measures Boolean circuit size and ignores circuit depth, physical layout, and communication costs, so it should be interpreted as an optimistic operation count rather than a full implementation cost.

In particular, our cost estimates do not exploit Hellman-style time-space tradeoffs [Hel80]: we count the direct per-instance running time of the attack without parameter-dependent preprocessing or stored tables.

5.1 Sharpened Kernel Condition from Systematic Form

Fix a held-out coordinate τ of a binary matrix $\boldsymbol{Y} \in \mathbb{F}_2^{k\times n}$, and suppose that the remaining columns have rank k. Choose a set $I \subseteq [n] \setminus \{\tau\}$ of k linearly independent columns and replace $\boldsymbol{Y}$ by

$$\boldsymbol{Y}' := (\boldsymbol{Y}_I)^{-1}\boldsymbol{Y}.$$

This invertible row operation preserves the public code and puts the columns indexed by I in the form $\boldsymbol{u}_1, \dots, \boldsymbol{u}_k$, where $\boldsymbol{u}_i$ is the i-th standard basis vector of $\mathbb{F}_2^k$. In the random case, conditioned on the non-held-out columns, the row operation is fixed and the transformed held-out column remains uniform. Computing this systematic form costs $O(k^2 n)$ binary operations, which is negligible compared with the whole distinguisher algorithm.

These k coordinate columns impose no nontrivial constraints on the polynomial space used by our distinguisher. Indeed, for a degree-d multilinear monomial X_A and a square-free derivative index $S \subseteq [k]$,

$$(\partial^{[S]} X_A)(\boldsymbol{u}_i) \neq 0 \quad \Longrightarrow \quad A \setminus S \subseteq \{i\} \quad \Longrightarrow \quad |S| \geqslant d-1.$$

Hence, when $d > s$, every order-$<s$ Hasse derivative of every polynomial in $\mathcal{P}^{\mathrm{ml}}_{=d,2}$ vanishes at every $\boldsymbol{u}_i$. At a general point, the number of potentially nonzero derivative constraints is

$$\sum_{r=0}^{s-1} \binom{k}{r}.$$

The kernel condition can therefore be sharpened to

$$(n-k-1)\sum_{r=0}^{s-1}\binom{k}{r} < \binom{k}{d}, \qquad d > s. \tag{5}$$

The reconstruction condition is unchanged: although the jet constraints at the systematic columns need not be stored, they still vanish identically and therefore still contribute multiplicity-s zeros in the structured-case interpolation argument.

5.2 Shortening Optimization

Shortening is a standard way to reduce both the length and the dimension of a public code before applying a structural distinguisher [CGG+14].

Let $\boldsymbol{Y} \in \mathbb{F}_q^{k\times n}$ be a public generator matrix and let $\mathcal{I} \subseteq [n]$ be a set of ℓ coordinates. Let $\boldsymbol{B}_{\mathcal{I}}$ be a full-rank matrix whose rows span the left kernel of the restricted matrix $\boldsymbol{Y}_{\mathcal{I}}$, namely the space

$$\{\boldsymbol{v} \in \mathbb{F}_q^{1\times k} : \boldsymbol{v}\boldsymbol{Y}_{\mathcal{I}} = 0\}.$$

The shortened public matrix is

$$\mathsf{Short}_{\mathcal{I}}(\boldsymbol{Y}) := \boldsymbol{B}_{\mathcal{I}}\boldsymbol{Y}_{[n]\setminus\mathcal{I}}.$$

When $\mathsf{rank}(\boldsymbol{Y}_{\mathcal{I}}) = \ell$, the shortened matrix has size $(k-\ell)\times(n-\ell)$. If the code generated by $\boldsymbol{Y}$ is a subcode of a GRS code with degree D and length n, then every polynomial representing a codeword in the shortened code contains the factors corresponding to the ℓ shortened coordinates. Factoring them out reduces the polynomial degree by ℓ, while the resulting nonzero evaluation factors are absorbed into the GRS multipliers. Thus, the shortened matrix generates a subcode of a GRS code with degree $D-\ell$ and length $n-\ell$.

The hold-out polynomial distinguishers above can be applied directly to $\mathsf{Short}_{\mathcal{I}}(\boldsymbol{Y})$. On the structured side, shortening a subcode of an RS, GRS, or ambient GRS code gives a subcode of the correspondingly shortened ambient code. On the random side, conditioned on $\mathsf{rank}(\boldsymbol{Y}_{\mathcal{I}}) = \ell$, the matrix $\mathsf{Short}_{\mathcal{I}}(\boldsymbol{Y})$ is distributed as a uniform random matrix of size $(k-\ell)\times(n-\ell)$, because each remaining random column is independently mapped by the full-rank linear map $\boldsymbol{B}_{\mathcal{I}}$.

For the Classic McEliece parameters, consider a full public key of a degree-t binary Goppa code over $\mathbb{F}_{2^m}$. We take

$$k = n - mt.$$

After shortening by ℓ coordinates, set

$$n_\ell := n-\ell, \qquad k_\ell := k-\ell, \qquad D_{\mathrm{amb},\ell} := n_\ell - 2t - 1.$$

The shortened Goppa distinguisher uses the multilinear homogeneous space

$$\mathcal{P}^{\mathrm{ml}}_{=d,2}(k_\ell) := \mathrm{span}_{\mathbb{F}_2}\{X_I : I \subseteq [k_\ell],\ |I| = d\},$$

where X_I denotes $\prod_{i\in I} X_i$. The reconstruction condition becomes

$$s(n_\ell - 1) > d \cdot D_{\mathrm{amb},\ell}. \tag{6}$$

Applying (5) after shortening gives the kernel condition

$$(n_\ell - k_\ell - 1)\sum_{r=0}^{s-1}\binom{k_\ell}{r} < \binom{k_\ell}{d}. \tag{7}$$

5.3 Optimized Implementation Estimates

For a fixed hold-out column τ, our first step is to put the shortened matrix in the systematic form of Section 5.1. We reuse the notation and denote the resulting matrix by $\boldsymbol{Y}$ with columns denoted as $\boldsymbol{y}_j$. The all-kernel test then amounts to checking the consistency of the following linear system:

$$\boldsymbol{M}_\tau \boldsymbol{x} = \boldsymbol{b}_\tau.$$

The vector $\boldsymbol{b}_\tau$ above corresponds to the evaluation vector at $\boldsymbol{y}_\tau$, and $\boldsymbol{M}_\tau$ is a matrix of size

$$\binom{k_\ell}{d} \times \left((n_\ell - k_\ell - 1) \sum_{r=0}^{s-1} \binom{k_\ell}{r} \right).$$

The rows are indexed by the degree-d multilinear monomials. In each row, the corresponding entry of $\boldsymbol{b}_\tau$ is the value of that monomial at the input $\boldsymbol{y}_\tau$. Each column of $\boldsymbol{M}_\tau$ records the Hasse derivative, evaluated at one non-held-out point, across all the monomials specified by the row index. Thus, the columns associated with a fixed point form its order-$< s$ Hasse jet. Moreover, the matrix $\boldsymbol{M}_\tau$ is sparse, with at most

$$(n_\ell - k_\ell - 1) \binom{k_\ell}{d} \sum_{r=0}^{s-1} \binom{d}{r}$$

nonzero entries.

Which linear system solver to use. Since our matrix is sparse, it is reasonable to use faster sparse linear system solvers as opposed to Gaussian elimination or Strassen-type methods. In particular, we rely on a block Krylov method. This method not only yields smaller running time but also reduces the memory cost as one does not need to store dense matrices. Below, we estimate concrete costs using the reliable small-field block Lanczos algorithm of [Ebe10].

To solve a linear system of the form $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, this algorithm requires an upper bound for $\mu(\boldsymbol{A})$, where $\mu(\boldsymbol{A}) := \dim\big(\mathrm{im}(\boldsymbol{A}) \cap \ker(\boldsymbol{A})\big)$ is the dimension of the intersection of the image and kernel of $\boldsymbol{A}$. As specified in [Ebe04], to obtain such a bound for our matrix, we sample $\boldsymbol{P}_\tau \in \mathbb{F}_2^{\widetilde{N} \times N}$ and $\boldsymbol{Q}_\tau \in \mathbb{F}_2^{J \times \widetilde{N}}$ from the sparse-preconditioner distribution of [Ebe04], where

$$N := \binom{k_\ell}{d}, \qquad J := (n_\ell - k_\ell - 1) \sum_{j=0}^{s-1} \binom{k_\ell}{j}, \qquad \widetilde{N} := J + \lceil 2 \log_2 N \rceil,$$

and set

$$\widetilde{\boldsymbol{M}}_\tau := \boldsymbol{P}_\tau \boldsymbol{M}_\tau \boldsymbol{Q}_\tau, \qquad \widetilde{\boldsymbol{b}}_\tau := \boldsymbol{P}_\tau \boldsymbol{b}_\tau.$$

By [Ebe04, Lemma 5],

$$\mathbf{Pr}\Big[\mathsf{rank}(\widetilde{\boldsymbol{M}}_\tau) \neq \mathsf{rank}(\boldsymbol{M}_\tau)\Big] \leqslant \frac{1}{N}, \qquad \mathbb{E}[\mu(\widetilde{\boldsymbol{M}}_\tau)] < 5.$$

When $\mathsf{rank}(\widetilde{\boldsymbol{M}}_\tau) = \mathsf{rank}(\boldsymbol{M}_\tau)$, the consistency of the original linear system implies the consistency of the new one, and a final test can always reject inconsistent cases. Therefore, we can work with the linear system $\widetilde{\boldsymbol{M}}_\tau \boldsymbol{x} = \widetilde{\boldsymbol{b}}_\tau$. For the algorithm of [Ebe10], we set the upper bound for μ to

$\mu_0 = 572$.[8] We set the reliability parameters to $\Delta := 16$ and $\delta := 18$, and define the right and left block sizes by

$$b_{\mathrm{R}} := \mu_0 + \Delta + 1, \qquad b_{\mathrm{L}} := b_{\mathrm{R}} + 2(\left\lceil \log_2 \widetilde{N} \right\rceil + \delta).$$

We use the truncated version of the algorithm, which aborts if its exceptional elimination space has dimension greater than δ. Combining [Ebe10, Theorem 1 and Lemma 10] with the preceding bounds, a consistent system is accepted with probability at least 99% in our parameter regime, whereas an inconsistent system is never accepted.

The algorithm proceeds by computing many matrix-vector multiplications with the matrix $\widetilde{\boldsymbol{M}}_\tau$. The product bounds in [Ebe10, App. A.6] show that the algorithm requires at most

$$K_{\mathrm{lin}} := 5\widetilde{N} + 3b_{\mathrm{R}} + 4b_{\mathrm{L}}$$

matrix-vector multiplications.

We denote the number of operations required for one such multiplication by T_{mv}. One can inspect and directly count the remaining steps taken by the algorithm. This gives the following concrete bound on the total number of operations:

$$T_{\mathrm{lin}} \leqslant K_{\mathrm{lin}} T_{\mathrm{mv}} + (72b_{\mathrm{L}} + 8)\widetilde{N}^2 + (112b_{\mathrm{L}}^2 + 5184)\widetilde{N}.$$

As shown in [Ebe04], the expected total number of nonzero entries in $\boldsymbol{P}_\tau$ and $\boldsymbol{Q}_\tau$ is

$$\mathbb{E}[Z_{\boldsymbol{P}_\tau, \boldsymbol{Q}_\tau}] = (N + J)\left(\sum_{i=1}^{J} \min\left\{ \frac{1}{2}, \frac{3\log_2 N}{i} \right\} + \frac{\lceil 2\log_2 N \rceil}{2} \right).$$

We estimate T_{mv} as twice the number of nonzero entries, accounting for binary multiplication and addition:

$$T_{\mathrm{mv}} := 2\left((n_\ell - k_\ell) N \sum_{r=0}^{s-1} \binom{d}{r} + \mathbb{E}[Z_{\boldsymbol{P}_\tau, \boldsymbol{Q}_\tau}] \right).$$

Together with the preceding bound on T_{lin}, this gives a concrete cost estimate for our algorithm.

Finally, when $\boldsymbol{P}_\tau$ and $\boldsymbol{Q}_\tau$ contain a total of Z nonzero entries, the implementation uses at most

$$S_{\mathrm{lin}}(Z) := (24b_{\mathrm{L}} + 4)\widetilde{N} + 8b_{\mathrm{L}}^2 + 24b_{\mathrm{L}} \left\lceil \log_2 \widetilde{N} \right\rceil + n_\ell k_\ell + 2\lceil \log_2 \widetilde{N} \rceil Z + (n_\ell - k_\ell) N \sum_{r=0}^{s-1} \binom{d}{r}$$

bits of storage. In the estimates below, we substitute $\mathbb{E}[Z_{\boldsymbol{P}_\tau, \boldsymbol{Q}_\tau}]$ for Z. We have not attempted to optimize the storage cost, so the reported values are only indicative. Lower storage may be obtained by generating $\boldsymbol{M}_\tau, \boldsymbol{P}_\tau, \boldsymbol{Q}_\tau$ on demand, using a smaller block size, or making other time–storage tradeoffs.

We now discuss the shortening conditions from Section 5.2 for Classic McEliece parameters. For the public-key distinguisher, we search over integer choices of (ℓ, d, s) with $d > s$ satisfying the reconstruction condition in (6) and the shortened kernel condition in (7). Among all feasible choices, we choose the one minimizing the dimension of the multilinear polynomial space, namely $\dim \mathcal{P}^{\mathrm{ml}}_{=d,2}(k - \ell) = \binom{k-\ell}{d}$.

[8] We choose this large upper bound because our estimate uses only Markov's inequality and aims only to give a lower bound on the costs of our distinguisher. A smaller value of μ_0 could conjecturally work and would give a better estimate.

parameter set	m	n	t	k	ℓ	d	s	Dim	$\log_2 T_{\mathrm{atk}}$	$\log_2 S_{\mathrm{atk}}$	advantage
`mceliece348864/f`	12	3488	64	2720	2503	8	7	$\binom{217}{8} \approx 2^{46.60}$	114.16	66.45	> 0.42
`mceliece460896/f`	13	4608	96	3360	3086	8	7	$\binom{274}{8} \approx 2^{49.34}$	120.29	69.54	> 0.42
`mceliece6688128/f`	13	6688	128	5024	4709	8	7	$\binom{315}{8} \approx 2^{50.97}$	123.95	71.37	> 0.45
`mceliece6960119/f`	13	6960	119	5413	5109	8	7	$\binom{304}{8} \approx 2^{50.55}$	123.02	70.91	> 0.45
`mceliece8192128/f`	13	8192	128	6528	6213	8	7	$\binom{315}{8} \approx 2^{50.97}$	123.95	71.37	> 0.45

Table 2: Optimized parameters for the Goppa–McEliece distinguisher on Classic McEliece parameter sets. Here $\mathsf{Dim} = \dim \mathcal{P}^{\mathrm{ml}}_{=d,2}(k-\ell)$. The columns headed $\log_2 T_{\mathrm{atk}}$ and $\log_2 S_{\mathrm{atk}}$ report, respectively, the estimated number of binary operations and the estimated storage in bits for one execution of the distinguisher under the preceding cost model. The final column reports the improved concrete lower bound on the distinguishing advantage obtained from the all-kernel count, after the one-percent completeness loss of the randomized linear solver. The time and storage exponents are rounded, while the advantage bounds are truncated downward. See Remark 5.1 for further optimizations that are not reflected in the table.

We give the resulting estimates in Table 2, where each row lists our choice of (ℓ, d, s) and the corresponding dimension N of the polynomial space. We set $T_{\mathrm{atk}} := T_{\mathrm{lin}}$ and $S_{\mathrm{atk}} := S_{\mathrm{lin}}(\mathbb{E}[Z_{\boldsymbol{P}_\tau, \boldsymbol{Q}_\tau}])$. The minimum-distance argument used in Theorem 4.1 gives us an advantage of only 2^{-d}. The stronger asymptotic statement in that theorem does not make its constants concrete. A more careful count using all polynomials in the kernel and generalized Hamming weights [Wei91, HP98] gives the higher concrete lower bounds on the advantage reported in the table. This advantage can be amplified by holding out more columns; however, this will require a reparameterization of the concrete attack above.

Remark 5.1 (Further concrete optimizations)**.** Weis [Wei26b] pointed out to us two further optimizations that are not reflected in Table 2. First, the nonzero count used above is only an upper bound: for the actual public points, many of the potentially nonzero Hasse-derivative entries vanish, making the linear system considerably sparser. Second, the sufficient kernel condition (7) can be sharpened by requiring the actual rank of the jet matrix to be smaller than the polynomial-space dimension. As in the heuristic kernel analysis of Appendix C.2, one can estimate this rank from the predicted nullity instead of upper-bounding it by the number of jet columns. Reoptimizing the parameters using these two improvements reduces the estimated time exponent by roughly 20 bits. These are the high-level improvement ideas; several lower-level optimizations and their detailed analysis are expected in a forthcoming report by Weis.

6 Heuristic Decryption Algorithm

Although our distinguishing attack does not recover the hidden support, multipliers, or scrambler of the GRS public key, it suggests a heuristic ciphertext-decryption attack. We prove only the one-sided completeness statement that "clean" (noiseless) ciphertext coordinates pass the test; rejection of "dirty" (noisy) coordinates is justified by a heuristic argument, so the attack is not proven to work in general. The main observation is that a ciphertext can be appended as an additional row to the public code: clean coordinates remain consistent with the ambient GRS structure, while dirty coordinates contain an additional error term. This approach applies to both GRS–McEliece and Goppa–McEliece; we describe both formulations below.

The ciphertext-decryption task for GRS–McEliece. Let

$$\boldsymbol{Y} := \boldsymbol{S}\boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha}, \boldsymbol{\lambda})\boldsymbol{P} \in \mathbb{F}_q^{k\times n}$$

be the public key, where $\boldsymbol{G} = \boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha}, \boldsymbol{\lambda})$ is a dimension-K, degree $D = K - 1$ GRS generator matrix defined in the GRS–McEliece distribution above. Above $\boldsymbol{S} \in \mathbb{F}_q^{k\times K}$ is the full-rank hidden row scrambler, and $\boldsymbol{P}$ is the hidden permutation.

A McEliece ciphertext consists of an LPN sample of the form

$$\boldsymbol{c} = \boldsymbol{u}\boldsymbol{Y} + \boldsymbol{e} = \boldsymbol{u}\boldsymbol{S}\boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha}, \boldsymbol{\lambda})\boldsymbol{P} + \boldsymbol{e},$$

where $\boldsymbol{u} \in \mathbb{F}_q^{1\times k}$ is a random vector and $\boldsymbol{e}$ is sampled uniformly at random from the set

$$\left\{\boldsymbol{e} \in \mathbb{F}_q^n : |\mathrm{supp}(\boldsymbol{e})| = \left\lfloor \frac{n-K}{2} \right\rfloor \right\}.$$

The decryption task asks to recover the secret $\boldsymbol{u}$ given $\boldsymbol{Y}$ and $\boldsymbol{c}$. Here and below, "ciphertext" refers to the noisy codeword $\boldsymbol{u}\boldsymbol{Y} + \boldsymbol{e}$. Note that the actual Classic McEliece KEM uses the dual Niederreiter syndrome representation, but it is equivalent to the decoding view; see Remark 6.1.

Goppa–McEliece ciphertexts. A Goppa–McEliece public-key/ciphertext pair has essentially the same form as in the GRS case, except that the public code is a structured binary subcode of an ambient GRS code. Namely, the public key is of the form $\boldsymbol{Y} = \boldsymbol{S}\boldsymbol{G}_{\mathsf{Goppa}}(\boldsymbol{\alpha}, \Gamma)\boldsymbol{P} \in \mathbb{F}_2^{k\times n}$, where $\boldsymbol{G}_{\mathsf{Goppa}}(\boldsymbol{\alpha}, \Gamma)$ generates the binary Goppa code $\mathsf{Goppa}(\boldsymbol{\alpha}, \Gamma)$, $\boldsymbol{S}$ has full row rank, and $\boldsymbol{P}$ is a permutation matrix. By Fact 2.3, after extension of scalars to $\mathbb{F}_q = \mathbb{F}_{2^m}$, this code is contained in an ambient GRS code of dimension $n - 2t$. Let $\boldsymbol{G}_{\mathrm{amb}}$ be the generator matrix of that ambient code. Then there is a matrix $\boldsymbol{T}_{\mathrm{amb}} \in \mathbb{F}_q^{K(\boldsymbol{\alpha},\Gamma)\times(n-2t)}$ such that

$$\boldsymbol{G}_{\mathsf{Goppa}}(\boldsymbol{\alpha}, \Gamma) = \boldsymbol{T}_{\mathrm{amb}}\boldsymbol{G}_{\mathrm{amb}}.$$

Thus $\boldsymbol{Y} = \widetilde{\boldsymbol{S}}\boldsymbol{G}_{\mathrm{amb}}\boldsymbol{P}$ over $\mathbb{F}_q$, where $\widetilde{\boldsymbol{S}} := \boldsymbol{S}\boldsymbol{T}_{\mathrm{amb}}$ has full row rank because $\boldsymbol{Y}$ has full row rank.

The ciphertext is of the form $\boldsymbol{c} = \boldsymbol{u}\boldsymbol{Y} + \boldsymbol{e}$, where $\boldsymbol{u} \in \mathbb{F}_2^{1\times k}$ and $\boldsymbol{e}$ is sampled uniformly at random from $\{\boldsymbol{e} \in \mathbb{F}_2^n : |\mathrm{supp}(\boldsymbol{e})| = t\}$. After embedding $\mathbb{F}_2$ in $\mathbb{F}_q$, this is a GRS-type ciphertext whose scrambler, message, and error vector follow the binary distributions induced by the Goppa–McEliece instance. For the attack description below, we always treat the public key and ciphertext as a GRS-style instance, with $D = K - 1$ in the GRS case and $D = n - 2t - 1$ in the binary Goppa case.

The heuristic ciphertext-decryption attack below applies to McEliece instantiations based on both GRS codes and binary Goppa codes. For both the GRS and binary Goppa versions, the attack uses the kernel test with homogeneous polynomials of degree exactly d over the extension field $\mathbb{F}_q$, rather than the binary multilinear test used by the Goppa distinguisher. This choice is justified by our heuristics in Appendix A.

Algorithm 3 (Heuristic decryption attack for GRS–McEliece and Goppa–McEliece)**.** *The attack proceeds as follows.*

1. *Attach the ciphertext to the public key to form*

$$\boldsymbol{A} := \begin{pmatrix} \boldsymbol{Y} \\ \boldsymbol{c} \end{pmatrix} \in \mathbb{F}_q^{(k+1)\times n}.$$

2. *For every $\tau \in [n]$, compute the homogeneous hold-out kernel*

$$\mathcal{K}_{\tau}^{\mathrm{hom}}(\boldsymbol{A}) := \left\{ H \in \mathcal{P}_{=d}^{\mathrm{hom}} : (\partial^{[\boldsymbol{a}]} H)(\boldsymbol{A}_j) = 0 \ \forall j \in [n] \setminus \{\tau\}, \forall \boldsymbol{a} \in \mathcal{A}_{<s,k+1} \right\},$$

where $\mathcal{P}_{=d}^{\mathrm{hom}}$ denotes the $k+1$-variate homogeneous polynomials of degree exactly d and $\boldsymbol{A}_j$ denotes the j-th column of $\boldsymbol{A}$. Mark τ as clean if $H(\boldsymbol{A}_\tau) = 0$ for every $H \in \mathcal{K}_{\tau}^{\mathrm{hom}}(\boldsymbol{A})$, equivalently, if the homogeneous hold-out distinguisher shown in Algorithm 1 accepts τ. This produces the candidate clean set

$$\widehat{\mathcal{I}}_{\mathrm{cl}} := \{\tau \in [n] : \textit{the homogeneous hold-out GRS distinguisher accepts } \tau\}.$$

3. *Solve the following linear system to recover $\boldsymbol{u}$:*

$$\boldsymbol{u}\boldsymbol{Y}_{\widehat{\mathcal{I}}_{\mathrm{cl}}} = \boldsymbol{c}_{\widehat{\mathcal{I}}_{\mathrm{cl}}},$$

where $\boldsymbol{Y}_{\widehat{\mathcal{I}}_{\mathrm{cl}}}$ and $\boldsymbol{c}_{\widehat{\mathcal{I}}_{\mathrm{cl}}}$ denote the restrictions to the columns in $\widehat{\mathcal{I}}_{\mathrm{cl}}$. If the linear system is inconsistent, abort.

Remark 6.1 (Reduction from syndrome to primal decryption)**.** *In Niederreiter-form McEliece cryptosystems such as Classic McEliece, the code-based component of a ciphertext is a syndrome $\boldsymbol{z} = \boldsymbol{H}\boldsymbol{e}^\top$, where $\boldsymbol{H} \in \mathbb{F}_2^{(n-k)\times n}$ is the public parity-check matrix and $\boldsymbol{e} \in \mathbb{F}_2^n$ is the error vector. Our primal decryption procedure also applies to this syndrome representation. Let $\boldsymbol{Y} \in \mathbb{F}_2^{k\times n}$ generate $\ker(\boldsymbol{H})$. Choose any $\boldsymbol{x} \in \mathbb{F}_2^n$ satisfying $\boldsymbol{H}\boldsymbol{x}^\top = \boldsymbol{z}$, sample $\boldsymbol{r} \xleftarrow{\$} \mathbb{F}_2^{1\times k}$, and form $\boldsymbol{c} = \boldsymbol{x} + \boldsymbol{r}\boldsymbol{Y}$. Since $\boldsymbol{H}(\boldsymbol{x}-\boldsymbol{e})^\top = 0$, we have $\boldsymbol{x} - \boldsymbol{e} \in \ker(\boldsymbol{H})$, and hence there is a unique $\boldsymbol{v} \in \mathbb{F}_2^{1\times k}$ such that $\boldsymbol{x} = \boldsymbol{v}\boldsymbol{Y} + \boldsymbol{e}$. Therefore, $\boldsymbol{c} = (\boldsymbol{v}+\boldsymbol{r})\boldsymbol{Y} + \boldsymbol{e}$ is a primal ciphertext with uniformly random message $\boldsymbol{u} := \boldsymbol{v} + \boldsymbol{r}$. Recovering $\boldsymbol{u}$ using the primal procedure then recovers the error vector as $\boldsymbol{e} = \boldsymbol{c} - \boldsymbol{u}\boldsymbol{Y}$.*

Claim 6.2 (Completeness on clean ciphertext coordinates)**.** *Let $\mathcal{I}_{\mathrm{cl}} := [n] \setminus \mathrm{supp}(\boldsymbol{e})$ be the set of clean ciphertext coordinates. Suppose that*

$$s(|\mathcal{I}_{\mathrm{cl}}| - 1) > d \cdot D. \tag{8}$$

Then, for every $\tau \in \mathcal{I}_{\mathrm{cl}}$, the GRS distinguisher marks τ as clean.

Proof. Fix a clean coordinate $\tau \in \mathcal{I}_{\mathrm{cl}}$. It remains to show that every polynomial in this kernel vanishes at $\boldsymbol{A}_\tau$. Define

$$\boldsymbol{S}_{\boldsymbol{u}} := \begin{pmatrix} \boldsymbol{S} \\ \boldsymbol{u}\boldsymbol{S} \end{pmatrix}.$$

Since $\boldsymbol{Y} = \boldsymbol{S}\boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha}, \boldsymbol{\lambda})\boldsymbol{P}$, for every clean coordinate $j \in \mathcal{I}_{\mathrm{cl}}$ the j-th column of $\boldsymbol{A}$ is a clean GRS–McEliece column of $\boldsymbol{S}_{\boldsymbol{u}}\boldsymbol{G}_{\mathsf{GRS}}(\boldsymbol{\alpha}, \boldsymbol{\lambda})\boldsymbol{P}$. Hence there exist polynomials $f_1, \ldots, f_k \in \mathbb{F}_q[Z]$ of degree at most D and a permutation π of $[n]$ such that, for every $j \in \mathcal{I}_{\mathrm{cl}}$,

$$\boldsymbol{A}_j = \lambda_{\pi(j)} \cdot \left(f_1(\alpha_{\pi(j)}), \ldots, f_k(\alpha_{\pi(j)}), \sum_{r=1}^{k} u_r f_r(\alpha_{\pi(j)}) \right)^\top.$$

We use f' to denote $\sum_{r=1}^{k} u_r \cdot f_r$. For any $H \in \mathcal{K}_\tau^{\hom}(\boldsymbol{A})$, for every $j \in \mathcal{I}_{\mathrm{cl}} \setminus \{\tau\}$ and every $\boldsymbol{a} \in \mathcal{A}_{<s,k+1}$, the definition of $\mathcal{K}_\tau^{\hom}(\boldsymbol{A})$ gives

$$(\partial^{[\boldsymbol{a}]} H)(\boldsymbol{A}_j) = 0.$$

Since H is homogeneous, Lemma 3.1 implies

$$(\partial^{[\boldsymbol{a}]} H)\left(f_1(\alpha_{\pi(j)}), \ldots, f_k(\alpha_{\pi(j)}), f'(\alpha_{\pi(j)})\right) = 0$$

for every $j \in \mathcal{I}_{\mathrm{cl}} \setminus \{\tau\}$ and every $\boldsymbol{a} \in \mathcal{A}_{<s,k+1}$. Define the composed univariate polynomial

$$\phi(Z) := H(f_1(Z), \ldots, f_k(Z), f'(Z)).$$

As in the proof of Theorem 3.2, the above Hasse-derivative vanishing implies that ϕ has a zero of multiplicity at least s at every point $\alpha_{\pi(j)}$ with $j \in \mathcal{I}_{\mathrm{cl}} \setminus \{\tau\}$. Moreover, $\deg(\phi) \leqslant d \cdot D$. By (8), Hermite interpolation forces ϕ to be identically zero. Therefore

$$H(\boldsymbol{A}_\tau) = H\left(\lambda_{\pi(\tau)} \cdot \left(f_1(\alpha_{\pi(\tau)}), \ldots, f_k(\alpha_{\pi(\tau)}), f'(\alpha_{\pi(\tau)})\right)\right) = \lambda_{\pi(\tau)}^d \phi(\alpha_{\pi(\tau)}) = 0.$$

Thus the all-kernel test marks τ as clean. □

To make the linear system consistent, we need $\widehat{\mathcal{I}}_{\mathrm{cl}}$ to only contain the coordinates with $e_\tau = 0$. We currently do not have a proof to lower bound the probability of this event. Nevertheless, we provide a heuristic analysis in Appendix A, which suggests that, in the asymptotic Classic McEliece parameter regime, one can choose $d, s = \Theta(\log n)$, yielding a quasipolynomial-time attack.

Remark 6.3 (One-sided filtering for binary Goppa decryption)**.** *For binary Goppa codes, we can avoid the requirement to reject every dirty coordinate by modifying Step 2 of Algorithm 3. Let* $\boldsymbol{\xi} := (0, \ldots, 0, 1)^\top \in \mathbb{F}_2^{k+1}$. *Instead of marking* τ *as clean when* $H(\boldsymbol{A}_\tau) = 0$ *for every* $H \in \mathcal{K}_\tau^{\hom}(\boldsymbol{A})$, *we retain* τ *when* $H(\boldsymbol{A}_\tau + \boldsymbol{\xi}) \neq 0$ *for some* $H \in \mathcal{K}_\tau^{\hom}(\boldsymbol{A})$, *and denote the resulting set by* $\widehat{\mathcal{I}}'_{\mathrm{cl}}$.

If τ *is dirty, then* $\boldsymbol{A}_\tau + \boldsymbol{\xi}$ *is the corresponding clean point on the hidden GRS curve. By the same interpolation argument as in Claim 6.2, every* $H \in \mathcal{K}_\tau^{\hom}(\boldsymbol{A})$ *vanishes at this point, and hence* $\tau \notin \widehat{\mathcal{I}}'_{\mathrm{cl}}$. *Therefore,* $\widehat{\mathcal{I}}'_{\mathrm{cl}}$ *contains no dirty coordinates. If* τ *is clean, the shifted point instead behaves as a dirty point, and the same heuristics in Appendix A suggest that some kernel polynomial is nonzero at the shifted point. It therefore suffices that enough clean coordinates survive so that* $\mathsf{rank}(\boldsymbol{Y}_{\widehat{\mathcal{I}}'_{\mathrm{cl}}}) = k$. *The message* $\boldsymbol{u}$ *is then uniquely recovered from the linear system* $\boldsymbol{u}\boldsymbol{Y}_{\widehat{\mathcal{I}}'_{\mathrm{cl}}} = \boldsymbol{c}_{\widehat{\mathcal{I}}'_{\mathrm{cl}}}$.

7 Heuristic Key-Recovery Algorithm

While the heuristic decryption algorithm from Section 6 uses our distinguisher for message recovery, it does not recover a decryption key. It also introduces a large concrete overhead. In this section, we instead use the coefficients of the interpolation relations obtained by the distinguisher to recover an ambient GRS representation of the public code. This representation can be used for a fast decryption of any future ciphertext with respect to the same public key. The concrete cost of this heuristic key-recovery algorithm is higher than that of our distinguisher, but much closer to distinguishing than our decryption attack.

We start in Section 7.1 with an overview of the main idea behind this key-recovery technique, demonstrated by applying it to *generic* GRS subcodes and alternant codes (with random evaluation

points and multipliers). In Section 7.2, we give the formal description of the algorithm for (generic) GRS and alternant codes. Finally, in Section 7.3, we describe our recovery algorithm for Goppa codes. Although a binary Goppa code is contained in an ambient GRS code, its induced polynomial space has additional structure that requires changing the interpolation parameters.

7.1 Technical Overview

Recall that by $\mathsf{GRS}_{D+1}(\boldsymbol{\alpha}, \boldsymbol{\lambda})$ we denote a GRS code consisting of evaluations of polynomials of degree at most D over $\mathbb{F}_q$ on the support vector $\boldsymbol{\alpha} \in \mathbb{F}_q^n$, where the evaluation at α_i is multiplied by the scalar λ_i. The starting point of our key-recovery algorithm is the Sidelnikov–Shestakov algorithm [SS92] for recovering the hidden structure of a GRS code. Their main observation is that the Lagrange interpolation coefficients retain enough structure to reveal the support vector $\boldsymbol{\alpha}$, namely the set of evaluation points. Once the evaluation points are known, recovering a compatible multiplier vector becomes much simpler by just solving a linear system related to the dual code. We therefore focus here on the core challenge of recovering the support set.

Recalling the Sidelnikov–Shestakov attack. The Sidelnikov–Shestakov attack [SS92] starts from a (scrambled) generator matrix of a full GRS code of degree D and selects $D+1$ of its columns, say the first $D+1$ columns. Since values at the corresponding $D+1$ distinct support points determine every polynomial of degree at most D, any remaining column τ has a unique Lagrange interpolation representation in terms of the selected columns. Taking the GRS multipliers into account, one can show that the interpolation coefficients reconstructing the τ-th held-out column carry enough structure to recover the support points up to "projective equivalence" defined by a map of the form $\phi(X) = \frac{aX+b}{cX+d}$.

Concretely, letting $C_{\tau,i}$ denote the coefficient of column $i \in [D+1]$ when expressing the τ-th held-out column as a linear combination of the selected columns, the coefficients satisfy the Cauchy-type identity $C_{\tau,i} = A_\tau B_i/(\alpha_\tau - \alpha_i)$, where A_τ depends only on τ and B_i only on i. Consequently, ratios of these coefficients cancel the unknown factors and reveal cross-ratios of the support points: for held-out columns $\tau \neq \tau'$ and selected indices $i \neq j$,

$$\frac{C_{\tau,i}\, C_{\tau',j}}{C_{\tau,j}\, C_{\tau',i}} \;=\; \frac{(\alpha_\tau - \alpha_j)\,(\alpha_{\tau'} - \alpha_i)}{(\alpha_\tau - \alpha_i)\,(\alpha_{\tau'} - \alpha_j)},$$

referred to as the *cross-ratio* of the four support points $\alpha_\tau, \alpha_{\tau'}, \alpha_j, \alpha_i$. Applying a common fractional-linear map $\phi(X) = \frac{aX+b}{cX+d}$ (with $ad - bc \neq 0$) to the support leaves the GRS code unchanged once the multipliers are rescaled: for any $\boldsymbol{\alpha}$ with $c\alpha_i + d \neq 0$,

$$\mathsf{GRS}_{D+1}(\boldsymbol{\alpha}, \boldsymbol{\lambda}) = \mathsf{GRS}_{D+1}\big(\phi(\boldsymbol{\alpha}), \boldsymbol{\lambda}'\big), \qquad \lambda'_i = \lambda_i\,(c\alpha_i + d)^D.$$

Hence recovering $\boldsymbol{\alpha}$ up to such a map suffices.

The argument above relies on the uniqueness of interpolation in the full GRS code. For a proper public subcode it does not necessarily apply: the public matrix is rank-deficient relative to the ambient code, so there may be additional relations that hold only on the subcode and one may not isolate the ambient interpolation coefficients. Standard Schur-product attacks can sometimes overcome this problem by first expanding the subcode into a full GRS code and then applying Sidelnikov–Shestakov [Wie10, FGO+13, CGG+14]. This route fails in our setting: we deal with

subcodes of GRS codes whose degree is very large compared with the rank of the public matrix, so the relevant Schur products saturate before exposing a full GRS code. Our distinguisher dealt with this difficulty by leveraging Hermite interpolation, and our recovery follows the same route: the algorithm is a natural extension of the Sidelnikov–Shestakov attack [SS92] to the setting of interpolation with multiplicities.

Hermite interpolation. We now recall the Hermite interpolation identity, which underlies our recovery algorithm, in terms of Hasse jets. In a univariate Hermite interpolation, a polynomial of degree at most D has $D+1$ degrees of freedom in the coefficients. The (univariate) Hermite interpolation theorem proves that the polynomial can be uniquely determined by its order-$< s$ Hasse jet $\big(\phi(\alpha_j), (\partial^{[1]}\phi)(\alpha_j), \ldots, (\partial^{[s-1]}\phi)(\alpha_j)\big)$ at N distinct points $\alpha_1, \ldots, \alpha_N$ as soon as $Ns > D$, since these jets impose Ns linear conditions on its $D+1$ coefficients. At the "no-slack" point $Ns = D+1$, the value $\phi(\alpha_\tau)$ at any further point is therefore a fixed $\mathbb{F}_q$-linear combination of the jets. For $s = 1$ this is exactly the ordinary Lagrange interpolation.

Similarly to Lagrange interpolation, one can show (we omit the details for this overview) that, after absorbing the GRS multipliers, the coefficients of the top-order derivative (of order $s-1$) take the Cauchy form $C_{\tau,j,s-1} = A_\tau B_{j,s-1}/(\alpha_\tau - \alpha_j)$. Thus, if these coefficients were available, the same strategy as in the Sidelnikov–Shestakov attack would recover the support up to projective equivalence.

Accessing univariate coefficients through multivariate Hasse reconstruction. The reason the subcode does not hand us the canonical relation is that we never observe the univariate derivatives $(\partial^{[u]}\phi_H)(\alpha_j)$ directly. To address this, we examine the *multivariate* Hasse evaluations $(\partial^{[\boldsymbol{a}]}H)(\boldsymbol{y}_j)$ of the homogeneous test polynomial H at the columns of the public key matrix, and the univariate derivatives are recovered from them only through the Hasse chain rule (27), which writes $(\partial^{[u]}\phi_H)(\alpha_j)$ as a fixed combination of the $(\partial^{[\boldsymbol{a}]}H)(\boldsymbol{p}_j)$ with $|\boldsymbol{a}| \leqslant u$. One can show that the resulting order-$(s-1)$ derivative coefficients have a Cauchy structure of the form $C_{\tau,j,\boldsymbol{a}} = A_\tau B_{j,\boldsymbol{a}}/(\alpha_\tau - \alpha_j)$, where $|\boldsymbol{a}| = s-1$ and $\boldsymbol{a}$ is the multi-index of the multivariate Hasse derivative. We call such coefficients *canonical solutions*. Assuming we can get hold of such a canonical solution, we would be able to run the same ideas as before to learn the support up to projective equivalence.

Issue: Spurious kernel relations. The idea above is natural, but it runs into an issue the recovery must address: the public jet matrix does not expose only a single canonical Hermite relation. Its kernel also contains additional spurious relations between multivariate Hasse jets, which occur even for generic public points and carry no information about the hidden support. These irrelevant relations effectively introduce noise that must be filtered out. Let $\boldsymbol{E}$ be the jet-evaluation matrix whose column indexed by $(j, \boldsymbol{a})$, $|\boldsymbol{a}| < s$, encodes the functional $H \mapsto (\partial^{[\boldsymbol{a}]}H)(\boldsymbol{y}_j)$ on homogeneous degree-d polynomials. A coefficient vector $\boldsymbol{c} = (c_{j,\boldsymbol{a}})$ lies in $\ker(\boldsymbol{E})$ precisely when

$$\sum_{j=1}^{n} \sum_{|\boldsymbol{a}|<s} c_{j,\boldsymbol{a}} \, (\partial^{[\boldsymbol{a}]}H)(\boldsymbol{y}_j) = 0 \qquad \text{for every homogeneous } H \text{ of degree } d. \tag{$\star$}$$

The kernel contains several well-understood spurious families. The first are the *one-block projective relations*: since H is homogeneous of degree d, the Euler–Hasse identity gives, for every multi-index

$\boldsymbol{a}$,

$$\sum_{i=1}^{k}(a_i+1)\,X_i\,(\partial^{[\boldsymbol{a}+\boldsymbol{e}_i]}H) = (d-|\boldsymbol{a}|)\,(\partial^{[\boldsymbol{a}]}H). \qquad (\star\star)$$

Evaluated at $\boldsymbol{y}_j$, each instance with $|\boldsymbol{a}| < s-1$ is a relation of the form $(\star)$ supported on the j-th block. There are also relations between every collection of few blocks. We collect both in a local space $\mathcal{K}_{\mathrm{loc}}$. For a generic GRS subcode the heuristic decomposition relevant to recovery is

$$\ker(\boldsymbol{E}) \;\approx\; \mathcal{K}_{\mathrm{loc}} \;\oplus\; \mathcal{H}_{\boldsymbol{\alpha}},$$

where $\mathcal{H}_{\boldsymbol{\alpha}}$ is the space of global Hermite relations induced by the hidden GRS parametrization. Only this global component carries the support.

Toward a Candidate Algorithm. To address this issue of detecting a canonical solution in a space masked by spurious relations, we motivate a slight change in the equations we examine. This change will convert the Cauchy structure of the canonical solutions into a low-rank structure that is easier to extract by linear algebra. Namely, instead of working with $N = n-h$ columns and using them to expand on h remaining hold-out columns and hoping to extract Cauchy type canonical solutions in the space of solutions, we consider n columns that has h special columns (say the first h). For those special hold-out columns we will express a degree d polynomial $H(\boldsymbol{y}_\tau)$ as a linear combination of the order-$< s$ Hasse jets at the remaining columns,

$$H(\boldsymbol{y}_\tau) = \sum_{j\neq\tau}\sum_{|\boldsymbol{a}|<s} \widetilde{\eta}_{\tau,(j,\boldsymbol{a})}\,\big(\partial^{[\boldsymbol{a}]}H\big)(\boldsymbol{y}_j).$$

The main point is that working with the no-slack condition of $(n-1)\cdot s = dD+1$ (to enable unique interpolation) the univariate interpolation coefficients are of the form:

$$\eta_{\tau,j,s-1} = A_\tau\,(\alpha_\tau-\alpha_j)^{s-1}\,B_{j,s-1}.$$

Thus, their multivariate projection for every $\boldsymbol{a}$ with $|\boldsymbol{a}| = s-1$ looks like

$$\widetilde{\eta}_{\tau,(j,\boldsymbol{a})} = A_\tau\,(\alpha_\tau-\alpha_j)^{s-1}\,\widetilde{B}_{j,\boldsymbol{a}},$$

where $\widetilde{B}_{j,\boldsymbol{a}}$ depends only on $(j,\boldsymbol{a})$ (it collects the $B_{j,u}$ through the Hasse chain rule and the multiplier scaling). If $s-1$ is a power of two, $s-1=2^e$, and the GRS field has characteristic two, then $(\alpha_\tau-\alpha_j)^{s-1} = (\alpha_\tau-\alpha_j)^{2^e} = \alpha_\tau^{2^e}+\alpha_j^{2^e} = \alpha_\tau^{s-1}+\alpha_j^{s-1}$, so this splits as

$$\widetilde{\eta}_{\tau,(j,\boldsymbol{a})} = \big(A_\tau\,\alpha_\tau^{s-1}\big)\,\widetilde{B}_{j,\boldsymbol{a}} \;+\; \big(A_\tau\big)\,\big(\alpha_j^{s-1}\,\widetilde{B}_{j,\boldsymbol{a}}\big).$$

As a consequence of this (and as formally argued in Section **??**) we can argue that the canonical solutions are generated by a rank-two matrix product of the form $\boldsymbol{U}\cdot\boldsymbol{V}$, where the rows of $\boldsymbol{U}$ are indexed by the h held-out columns and the columns of $\boldsymbol{V}$ by the jet coordinates $(j,\boldsymbol{a})$. Here $\boldsymbol{U}$ and $\boldsymbol{V}$ have the following structure for every $\mathbf{a}$ with $1\leqslant|\mathbf{a}|\leqslant s-1$:

$$\boldsymbol{U}_\tau = \begin{pmatrix} A_\tau\,\alpha_\tau^{s-1} & A_\tau\end{pmatrix} \in \mathbb{F}_q^{1\times 2}, \qquad \boldsymbol{V}_{(j,\boldsymbol{a})} = \begin{pmatrix}\widetilde{B}_{j,\boldsymbol{a}}\\ \alpha_j^{s-1}\,\widetilde{B}_{j,\boldsymbol{a}}\end{pmatrix} \in \mathbb{F}_q^{2\times 1},$$

so that $\boldsymbol{U}\in\mathbb{F}_q^{h\times 2}$, $\boldsymbol{V}\in\mathbb{F}_q^{2\times M}$ with $M = n\binom{k+s-1}{s-1}$, and $\widetilde{\eta}_{\tau,(j,\boldsymbol{a})} = (\boldsymbol{U}\boldsymbol{V})_{\tau,(j,\boldsymbol{a})}$.

Summarizing the Recovery Algorithm for GRS. Summarizing the above, our algorithm starts with a few requirements:

- No-slack condition: $(n-1)\cdot s = dD+1$ to ensure uniqueness of the univariate canonical solutions.
- Kernel condition: the full jet matrix has $M = n\binom{k+s-1}{s-1}$ columns and $N_d = \binom{k+d-1}{d}$ rows. A condition $M < N_d$ as in our distinguisher is required heuristically to ensure that the number of spurious relations is small and that the rank one structure is hopefully extractable. Our algorithm uses a slightly sharper condition than this that yields better concrete parameters.
- Finally, $s-1$ should be a power of two. For concrete parameter sets this does not cause an issue because of an optimization we discuss later.

Once the canonical interpolation coefficients have been observed to be of the rank-two form above, the recovery proceeds as follows. The algorithm sets aside $h = 3$ hold-out columns $T = \{\tau_1, \tau_2, \tau_3\}$. These three points fix the fractional-linear normalization of the support, so there is no remaining projective freedom to search over. It then guesses the corresponding factor $\boldsymbol{U} \in \mathbb{F}_q^{3\times 2}$ whose τ-th row is the intended solution of that hold-out,

$$\boldsymbol{U}_\tau = \begin{pmatrix} A_\tau\, \alpha_\tau^{s-1} & A_\tau \end{pmatrix} \;\propto\; \begin{pmatrix} \alpha_\tau^{s-1} & 1 \end{pmatrix},$$

that is, projectively the point $[\alpha_\tau^{s-1} : 1]$. When we discuss the concrete numbers for this algorithm, we will work by choosing these ratios for every row of $\boldsymbol{U}$ to be determined by arbitrary three distinct points (so that the guessing is not required). This can be done by exploiting projective freedom that comes with GRS support. Moreover, A_τ's need not be guessed either (and we will discuss this later when we discuss concrete run times for the attack).

With $\boldsymbol{U}$ fixed, the remaining unknowns are linear, and the algorithm solves for the affine space $\mathcal{V}_{\boldsymbol{U}}$ of second factors $\boldsymbol{V}$ for which the rows of $\boldsymbol{UV}$ lie in $\ker(\boldsymbol{E})$ and obey the hold-out normalization $c_{\tau,(\tau,\boldsymbol{0})} = 1$ and $c_{\tau,(\tau,\boldsymbol{a})} = 0$ for $0 < |\boldsymbol{a}| < s$. Parsing an affine space element by coordinate, $\boldsymbol{V} = [\,\boldsymbol{W}_1 \mid \cdots \mid \boldsymbol{W}_n\,]$ with $\boldsymbol{W}_j$ the block of columns indexed by $(j, \boldsymbol{a})$, each $\boldsymbol{W}_j$ is the canonical component of $\mathcal{C}_{\boldsymbol{\alpha}}$ at coordinate j superposed with whatever spurious relations of $\mathcal{K}_{\mathrm{loc}}$ that survive the normalization.

The support is then read off coordinate by coordinate from the top-derivative block $\boldsymbol{W}'_j$ of $\boldsymbol{W}_j$ – the sub-block on the exact order $|\boldsymbol{a}| = s-1$ [9], where the rank one- structure lives. For the canonical solution this block is rank one and has the form:

$$\begin{pmatrix} 1 \\ \alpha_j^{s-1} \end{pmatrix} \left(\widetilde{B}_{j,\boldsymbol{a}}\right)_{|\boldsymbol{a}|=s-1} \in \mathbb{F}_q^{2\times\binom{k+s-2}{s-1}},$$

its two rows proportional in the ratio α_j^{s-1}. The algorithm therefore sweeps the candidate locators $x \in \mathbb{F}_q$ and accepts x precisely when the two rows satisfy $V_{2,(j,\boldsymbol{a})} = x^{\,s-1} V_{1,(j,\boldsymbol{a})}$ for every $\boldsymbol{a}$ of size $s-1$ – equivalently when the left combination $(x^{\,s-1}, 1)\,\boldsymbol{W}'_j = \boldsymbol{0}$ has a valid solution in the affine space, i.e. when $\boldsymbol{W}'_j$ is rank one with ratio $x^{\,s-1}$. Since $z \mapsto z^{\,s-1} = z^{\,2^e}$ is an automorphism of $\mathbb{F}_{2^m}$, the accepted value pins down α_j^{s-1}, hence α_j.

[9] In principle, we can consider derivatives of all orders from 1 through $s-1$.

The case of alternant codes. An alternant code is defined over a base field $\mathbb{F}_{q_0} \subseteq \mathbb{F}_q$, so it is stable under the Frobenius automorphisms. After extension of scalars it is therefore contained not in a single ambient GRS code but in the $m = [\mathbb{F}_q : \mathbb{F}_{q_0}]$ Frobenius conjugates, whose raw supports are the entrywise powers $\boldsymbol{\alpha}^{q_0^a}$ for $0 \leqslant a < m$. Let $\mathcal{H}^{(a)}$ denote the canonical relation space contributed by the a-th conjugate ambient GRS representation. We then expect (and confirm experimentally in Appendix C.2) that

$$\ker(\boldsymbol{E}) \;\approx\; \mathcal{K}_{\mathrm{loc}} \oplus \bigoplus_{a=0}^{m-1} \mathcal{H}^{(a)}.$$

Consequently, the extraction mechanism above no longer works with $h = 3$ hold-outs. For each conjugate support, some fractional-linear map carries its three hold-out locators onto the guessed reference values, so all m branches survive the normalization. At coordinate j, the rank-one test may then accept the m values $\alpha_j^{(0)}, \ldots, \alpha_j^{(m-1)}$, one from each branch. Once these m values for every coordinate are learned, it may be possible to patch them into descriptions of m different GRS codes. For conceptual simplicity we take $h = 4$. For a generic support, the fourth hold-out selects a single branch.

Rigidity assumption. The correctness of this algorithm relies on the rank-one rigidity conjecture (Conjecture C.3). For the correct normalized matrix $\boldsymbol{U}^\star$, it predicts (with the convention $m = 1$ for a GRS code):

$$\mathcal{C}_j(\boldsymbol{U}^\star) = \{\alpha_j^{(a)} : 0 \leqslant a < m\} \quad \text{when } h = 3, \qquad \mathcal{C}_j(\boldsymbol{U}^\star) = \{\alpha_j\} \quad \text{when } h = 4.$$

Here $\mathcal{C}_j(\boldsymbol{U}^\star)$ is a set of locators that pass the rank one test in a given block j. Thus, the local relations create no additional admissible locator. Our conjecture requires some conditions to be satisfied. They should include the three conditions (no-slack, kernel condition and $s-1$ being a power of 2) mentioned above. Besides, we additionally require that $\dim(\pi_j(\ker(\boldsymbol{E}))) < |\mathcal{B}_j|$, so that the local kernel relations do not span the whole local block. In Appendix C.2 we study this assumption. Finally, once the support $\boldsymbol{\alpha}$ is recovered, a compatible multiplier vector is obtained by solving a linear system [SS92, Wie10], yielding an ambient GRS representation of the public code.

Optimization using a multiplicity profile. Our recovery algorithm needs a no-slack condition requiring $s(n-1) = dD + 1$. This can be quite restrictive, especially when we also require $s-1$ to be a power of two. To get more freedom with concrete parameters, we use a different multiplicity profile $\boldsymbol{s} = (s_1, \ldots, s_n)$. Unlike setting them to be all equal, we require $s_\tau = s$ for every hold-out $\tau \in [h]$, where $s-1$ is a nontrivial power of two, but may choose the remaining s_j freely. The no-slack condition becomes $\sum_{j \neq \tau} s_j = dD + 1$ for every hold-out $\tau \in [h]$. As shown in Appendix C.1, where we study the structure of the canonical coefficients and show that they satisfy the required properties. Besides the required rank two condition of being expressable as $\boldsymbol{U} \cdot \boldsymbol{V}$, the top derivative block of $\boldsymbol{W}_j$ (and in fact derivative orders from $s_j - s + 1$ to $s_j - 1$) all have the required rank-one structure with the column direction governed by the corresponding support point.

Additional optimizations. Several further optimizations improve the concrete running time. They concern eliminating guesses by fixing the projective freedom, solving the linear systems more efficiently, and sharpening the kernel condition. We defer them to Section 7.4.

7.2 Key Recovery for GRS and Alternant Codes

The key-recovery task. Let $\mathbb{F}_q$ be a field of characteristic two. In the GRS case, the public matrix $\boldsymbol{Y} = [\boldsymbol{y}_1|\cdots|\boldsymbol{y}_n]$ lies in $\mathbb{F}_q^{k\times n}$. In the alternant case, let $\mathbb{F}_{q_0} \subseteq \mathbb{F}_q$ be the base field, where $m = [\mathbb{F}_q : \mathbb{F}_{q_0}]$ and $q = q_0^m$, and view the public matrix $\boldsymbol{Y} \in \mathbb{F}_{q_0}^{k\times n}$ as a matrix over $\mathbb{F}_q$. In both cases, the row span of $\boldsymbol{Y}$ is contained in an unknown ambient GRS code of dimension $D+1$ over $\mathbb{F}_q$. Given $\boldsymbol{Y}$ and the public parameters, the goal is to output

$$\widehat{\boldsymbol{\alpha}} \in \mathbb{F}_q^n \qquad \text{and} \qquad \widehat{\boldsymbol{\lambda}} \in (\mathbb{F}_q^\times)^n$$

such that the entries of $\widehat{\boldsymbol{\alpha}}$ are distinct and

$$\operatorname{rowspan}_{\mathbb{F}_q}(\boldsymbol{Y}) \subseteq \mathsf{GRS}_{D+1}(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\lambda}}).$$

Thus, the algorithm need not recover the original support or multipliers; any ambient GRS representation containing the public code is a valid output.

Algorithm 4 (Key recovery for GRS and alternant codes)**.** *Let d, s, h and a multiplicity profile $\boldsymbol{s} = (s_1, \ldots, s_n) \in \mathbb{N}^n$ be attack parameters. Designate $[h]$ as the hold-out set, where $2 \leqslant h \leqslant n$, $s_\tau = s$ for every $\tau \in [h]$, $s-1$ is a power of two, and*

$$\sum_{j\neq\tau} s_j = dD + 1 \qquad \forall\, \tau \in [h]. \tag{9}$$

On input $\boldsymbol{Y} = [\boldsymbol{y}_1|\cdots|\boldsymbol{y}_n]$, proceed as follows.

1. *Set*

$$N_d := \binom{k+d-1}{d}, \qquad \mathcal{J}_{\boldsymbol{s}} := \{(j, \boldsymbol{a}) : j \in [n],\ \boldsymbol{a} \in \mathcal{A}_{<s_j,k}\},$$
$$M := |\mathcal{J}_{\boldsymbol{s}}| = \sum_{j=1}^{n} \binom{k+s_j-1}{s_j-1}.$$

   Define the homogeneous jet matrix

$$\boldsymbol{E} := \left[\left((\partial^{[\boldsymbol{a}]}\boldsymbol{X}^{\boldsymbol{b}})(\boldsymbol{y}_j)\right)_{\boldsymbol{b}\in\mathcal{A}_{d,k}}\right]_{(j,\boldsymbol{a})\in\mathcal{J}_{\boldsymbol{s}}} \in \mathbb{F}_q^{N_d\times M}.$$

   For every $j \in [n]$, define the locator block

$$\mathcal{B}_j := \left\{\boldsymbol{a} \in \mathcal{A}_{<s_j,k} : s_j - s + 1 \leqslant |\boldsymbol{a}| \leqslant s_j - 1\right\}.$$

2. *For every rank-two matrix $\boldsymbol{U} \in \mathbb{F}_q^{h\times 2}$, with rows $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_h$, compute the affine space $\mathcal{V}_{\boldsymbol{U}}$ of matrices*

$$\boldsymbol{V} = \begin{pmatrix} \boldsymbol{V}_1 \\ \boldsymbol{V}_2 \end{pmatrix} \in \mathbb{F}_q^{2\times M}$$

   satisfying

$$\boldsymbol{E}\boldsymbol{V}_1^\top = \boldsymbol{0}, \qquad \boldsymbol{E}\boldsymbol{V}_2^\top = \boldsymbol{0},$$
$$(\boldsymbol{u}_\tau \boldsymbol{V})_{(\tau,\boldsymbol{a})} = 0 \qquad \forall \tau \in [h], \quad \forall \boldsymbol{a} \in \mathcal{A}_{<s,k} \setminus \{\boldsymbol{0}\},$$

   and

$$(\boldsymbol{u}_1 \boldsymbol{V})_{(1,\boldsymbol{0})} = 1.$$

3. *For $\boldsymbol{V} \in \mathcal{V}_{\boldsymbol{U}}$ and $j \in [n]$, define its locator block by*

$$\boldsymbol{W}_j(\boldsymbol{V}) := \big(V_{\ell,(j,\boldsymbol{a})}\big)_{\substack{\ell\in[2]\\ \boldsymbol{a}\in\mathcal{B}_j}} \in \mathbb{F}_q^{2\times|\mathcal{B}_j|}.$$

Compute the candidate set

$$\mathcal{C}_j(\boldsymbol{U}) := \left\{ x \in \mathbb{F}_q : \begin{array}{l} \text{there exists } \boldsymbol{V} \in \mathcal{V}_{\boldsymbol{U}} \text{ such that} \\ V_{2,(j,\boldsymbol{a})} = x^{s-1} V_{1,(j,\boldsymbol{a})} \quad \forall \boldsymbol{a} \in \mathcal{B}_j \end{array} \right\}.$$

Thus, $x \in \mathcal{C}_j(\boldsymbol{U})$ precisely when the affine space contains a locator block whose second row is x^{s-1} times its first row; in particular, this block has rank at most one. Each membership test is a linear feasibility problem over $\mathbb{F}_q$. If some $\mathcal{C}_j(\boldsymbol{U})$ is not a singleton, move on to the next matrix $\boldsymbol{U}$.

4. *Write $\mathcal{C}_j(\boldsymbol{U}) = \{\widehat{\alpha}_j\}$ for every $j \in [n]$. If the entries of $\widehat{\boldsymbol{\alpha}}$ are not distinct, move on to the next matrix $\boldsymbol{U}$.*

5. *Recover a compatible multiplier vector $\widehat{\boldsymbol{\lambda}}$ by linear algebra as in [SS92, Wie10]. If*

$$\mathrm{rowspan}_{\mathbb{F}_q}(\boldsymbol{Y}) \subseteq \mathsf{GRS}_{D+1}(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\lambda}}),$$

output $(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\lambda}})$. If no rank-two matrix $\boldsymbol{U} \in \mathbb{F}_q^{h\times 2}$ yields a valid output, output $\perp$.

The canonical-solution analysis and the heuristic justification for the singleton locator test are deferred to Appendix C.

Throughout the section, for every $j \in [n]$, let $\pi_j : \mathbb{F}_q^M \to \mathbb{F}_q^{|\mathcal{B}_j|}$ denote the coordinate projection onto the locator block $\mathcal{B}_j$.

Theorem 7.1 (Conditional key recovery for GRS and alternant codes)**.** *Suppose that the parameters of Algorithm 4 satisfy the no-slack condition (9) and the kernel conditions*

$$\mathsf{rank}(\boldsymbol{E}) < N_d, \qquad \dim \pi_j\big(\ker(\boldsymbol{E})\big) < |\mathcal{B}_j| \quad \forall\, j \in [n].$$

Assume Conjecture C.3. When $h = 3$, the locator sets contain exactly the normalized support and its $m-1$ Frobenius conjugates, where $m := 1$ for a GRS code. Consequently, Algorithm 4 succeeds with probability $1 - o(1)$ for a GRS code, whereas for an alternant code with $m > 1$, the singleton test does not isolate one canonical branch. When $h = 4$, every locator set contains only the target normalized support point, and Algorithm 4 succeeds with probability $1 - o(1)$ for both GRS and alternant codes.

More precisely, set

$$L_s := \binom{k+s-1}{s-1}, \qquad B := \max_{j\in[n]} |\mathcal{B}_j|, \qquad \Lambda_{\mathrm{sys}} := 2M + 2N_d + hL_s + B.$$

The support-recovery stage solves at most $q^{2h}(nq+1)$ linear systems, each of dimension at most $O(\Lambda_{\mathrm{sys}})$, and hence runs in time

$$O\Big(nq^{2h+1} \cdot \mathrm{poly}(\Lambda_{\mathrm{sys}}, \log q)\Big).$$

When h is constant, this running time is quasipolynomial, namely $n^{O(\log n)}$, in the asymptotic Classic McEliece parameter regime described in Section 2.

Proof. The enumeration of rank-two matrices contains the canonical matrix $\boldsymbol{U}^\star$ induced by the hidden GRS representation. By Conjecture C.3, when $h = 3$, exactly the m normalized Frobenius-conjugate supports survive the locator test. For a GRS code, $m = 1$, so every candidate set is a singleton containing the target support point. For an alternant code with $m > 1$, all m conjugate branches survive, so the singleton test does not isolate the target branch. When $h = 4$, the conjecture leaves only the target branch, so every candidate set is a singleton containing the target support point. Multiplier recovery and the final containment test then produce a valid ambient GRS representation.

We next explain the size parameter Λ_{sys}. For fixed $\boldsymbol{U}$, the unknown matrix $\boldsymbol{V} \in \mathbb{F}_q^{2\times M}$ contributes $2M$ scalar unknowns. The two kernel equations $\boldsymbol{E}\boldsymbol{V}_1^\top = \boldsymbol{E}\boldsymbol{V}_2^\top = \boldsymbol{0}$ contribute $2N_d$ equations. Each of the h hold-out blocks has $L_s = |\mathcal{A}_{<s,k}|$ jet coordinates, so the hold-out normalization contributes at most hL_s further equations. Finally, for a fixed coordinate j and candidate $x \in \mathbb{F}_q$, the rank-one locator test adds $|\mathcal{B}_j| \leqslant B$ equations. More precisely, this feasibility system has $2M$ unknowns and at most $2N_d + h(L_s - 1) + 1 + |\mathcal{B}_j|$ equations. Thus, its total number of rows and columns is bounded by Λ_{sys}. Consequently, Λ_{sys} is a uniform bound, up to a constant factor, on the total row-and-column dimension of every linear system in the support-recovery stage; it is not an additional algebraic parameter.

For each $\boldsymbol{U}$, there are nq locator feasibility tests. One additional system constructs the common affine space. There are fewer than q^{2h} matrices in $\mathbb{F}_q^{h\times 2}$, so the support-recovery stage solves at most $q^{2h}(nq + 1)$ systems of size $O(\Lambda_{\mathrm{sys}})$. Constructing $\boldsymbol{E}$ and evaluating the recovered candidates also take $\mathrm{poly}(\Lambda_{\mathrm{sys}}, \log q)$ time. This proves the displayed running-time bound. In the Classic McEliece regime, $q = 2^m = n^{O(1)}$, and the relevant choices $d, \max_j s_j = O(\log n)$ give $M, N_d, B = n^{O(\log n)}$. Hence $\Lambda_{\mathrm{sys}} = n^{O(\log n)}$, and the running time is quasipolynomial when h is constant.

□

7.3 Key Recovery for Binary Goppa Codes

Let $q = 2^m$, and let $\boldsymbol{Y} \in \mathbb{F}_2^{k\times n}$ be a public matrix generated from a square-free binary Goppa code with Goppa polynomial Γ of degree t. Viewing $\boldsymbol{Y}$ over $\mathbb{F}_q$ and setting $D := n - 2t - 1$, the key-recovery task is the one defined in Section 7.2: output an ambient GRS representation of dimension $D + 1$ over $\mathbb{F}_q$ whose row span contains that of $\boldsymbol{Y}$.

A binary Goppa code is, in particular, an alternant code, so one might hope to run the recovery of the previous section verbatim. The alternant algorithm does not work unchanged, but a single parametric adjustment repairs it. We first explain the obstruction.

Reduced dimension of the underlying univariate polynomials. Recall that each column of $\boldsymbol{Y}$ is parameterized by a point $\big(f_1(\alpha_j), \dots, f_k(\alpha_j)\big)$ with $\deg f_i \leqslant D$, and that the recovery operates on the univariate polynomials $\phi_H(X) := H\big(f_1(X), \dots, f_k(X)\big)$, of degree at most dD, as H ranges over the homogeneous polynomials of degree d. For a generic alternant code these span the full space of polynomials of degree at most dD, of dimension $dD + 1$, and the no-slack condition of the alternant algorithm (Algorithm 4) is calibrated to this dimension. For a binary Goppa code the span is strictly smaller: empirically we find its dimension to be exactly $dD + 1 - t$, a deficit of t relative to the generic alternant expectation. This reduction reflects the Goppa-specific geometry. For example, for a square-free Goppa polynomial Γ of degree t, when the degree d is

even the derivative ϕ'_H of every such polynomial is divisible by the Goppa Γ; such constraints is what removes the missing t dimensions.

The fix is therefore to recalibrate the no-slack condition to this reduced dimension. We choose parameters satisfying

$$\sum_{j \neq \tau} s_j = dD + 1 - t \qquad \forall\, \tau \in [h]. \tag{10}$$

With this single substitution in place of the alternant condition $\sum_{j \neq \tau} s_j = dD + 1$, the recovery algorithm for binary Goppa codes is otherwise identical to the one in Section 7.2.

A reader may reasonably worry that this unexpected dimension drop destroys the canonical relations used by the locator test. We therefore conjecture directly that the required canonical structure is preserved even with the new condition. In Section C.1.1, we show that empirically we show that this turns out to be true. The intuition, experimental evidence, and the corresponding analysis are deferred to Section C.1.1.

Theorem 7.2 (Conditional key recovery for binary Goppa codes)**.** *Suppose that the attack parameters satisfy* (10) *and*

$$\mathsf{rank}(\boldsymbol{E}) < N_d, \qquad \dim \pi_j\big(\ker(\boldsymbol{E})\big) < |\mathcal{B}_j| \quad \forall\, j \in [n],$$

and assume Conjectures C.2 and C.3. When $h = 3$, the locator sets contain exactly the normalized support and its $m - 1$ Frobenius conjugates, so the singleton test does not isolate one branch when $m > 1$. When $h = 4$, every locator set contains only the target normalized support point, and the binary Goppa variant of Algorithm 4 succeeds with probability $1 - o(1)$.

Let Λ_{sys} be as in Theorem 7.1. The support-recovery stage solves at most $q^{2h}(nq + 1)$ linear systems, each of dimension at most $O(\Lambda_{\mathrm{sys}})$, and hence runs in time

$$O\Big(nq^{2h+1} \cdot \mathrm{poly}(\Lambda_{\mathrm{sys}}, \log q)\Big) = O\Big(n2^{(2h+1)m} \cdot \mathrm{poly}(\Lambda_{\mathrm{sys}}, m)\Big)\,.$$

For the asymptotic Classic McEliece parameter regime described in Section 2, when h is constant and $d, s_j = O(\log n)$ for every $j \in [n]$, this running time is quasipolynomial, namely $n^{O(\log n)}$.

Proof. By Conjecture C.2, there is a rank-two factorization $\boldsymbol{C}^\star = \boldsymbol{U}^\star \boldsymbol{V}^\star$ with the canonical left factor, so $\boldsymbol{U}^\star$ occurs in the enumeration. The row spaces of $\boldsymbol{C}^\star$ and $\boldsymbol{V}^\star$ coincide, so both rows of $\boldsymbol{V}^\star$ belong to $\ker(\boldsymbol{E})$. The diagonal convention gives the hold-out constraints and normalization, while (33) gives the required locator-block identity. Therefore, $\boldsymbol{V}^\star \in \mathcal{V}_{\boldsymbol{U}^\star}$ and $\alpha_j \in \mathcal{C}_j(\boldsymbol{U}^\star)$ for every coordinate j.

The rank-one rigidity conjecture now gives the two asserted cases. With three hold-outs, the locator sets also contain the normalized Frobenius conjugates, so they are not singletons when $m > 1$. With four hold-outs, $\mathcal{C}_j(\boldsymbol{U}^\star) = \{\alpha_j\}$ for every j. The candidate support is therefore recovered, and multiplier recovery followed by the public containment test returns a valid ambient GRS representation.

The binary Goppa modification changes only the interpolation threshold; it does not change the dimensions of $\boldsymbol{E}$, $\boldsymbol{U}$, $\boldsymbol{V}$, or the locator blocks. Thus, the count in the proof of Theorem 7.1 applies verbatim: fewer than q^{2h} choices of $\boldsymbol{U}$ are considered, and for each choice there are at most nq locator tests and one affine space, all of size $O(\Lambda_{\mathrm{sys}})$. Substituting $q = 2^m$ gives the displayed running time. In the asymptotic Classic McEliece parameter regime described in Section 2, we have $q = n^{O(1)}$. When $d, s_j = O(\log n)$ for every $j \in [n]$, the quantities M, N_d, and B, and hence Λ_{sys}, are $n^{O(\log n)}$. For constant h, the displayed running time is therefore $n^{O(\log n)}$. □

7.4 Baseline Concrete Recovery Estimates

We give a baseline estimate for the binary Goppa recovery algorithm on the Classic McEliece parameter sets. (Further optimizations will be discussed in Remark 7.3 below.) We use the binary-operation convention and the sparse block Lanczos cost model from Section 5. The conjectures and the nullity and local-projection predictions used below are developed in Appendix C; see in particular Appendix C.2 and Conjecture C.3.

Shortening. We apply the shortening operation defined in Section 5.2. It is necessary here to bring the recovery systems into the range reported below. For a Classic McEliece parameter set $(m, n, t, k = n - mt)$, choose a shortening set $\mathcal{I} \subseteq [n]$ of size ℓ, and write

$$n_\ell := n - \ell, \qquad k_\ell := k - \ell, \qquad D_\ell := n_\ell - 2t - 1, \qquad N_{d,\ell} := \binom{k_\ell + d - 1}{d}.$$

The shortened matrix is again a public subcode of the binary Goppa code on the surviving support coordinates, with ambient GRS degree D_ℓ, so the same recovery algorithm applies to it. Throughout this subsection, a subscript ℓ denotes the corresponding object for this shortened instance.

One shortened execution recovers only the coordinates outside $\mathcal{I}$. We therefore use four common hold-outs and choose surviving coordinate sets whose union is $[n]$. Three common coordinates align the recovered supports projectively, and the fourth aligns their Frobenius branches. It suffices to use $R_\ell := \lceil (n-4)/(n_\ell - 4) \rceil$ shortened executions.

Parameter selection. For a multiplicity profile $\boldsymbol{s} = (s_1, \ldots, s_{n_\ell})$, set

$$M_\ell := \sum_{j=1}^{n_\ell} \binom{k_\ell + s_j - 1}{s_j - 1}.$$

The recovery theorem requires

$$\mathsf{rank}(\boldsymbol{E}_\ell) < N_{d,\ell}, \qquad \dim \pi_j\big(\ker(\boldsymbol{E}_\ell)\big) < |\mathcal{B}_j| \quad \forall\, j \in [n_\ell].$$

Let $\kappa_{\mathrm{rnd},\ell}(\boldsymbol{s})$ denote the random-nullity prediction (34) for the shortened parameters and profile $\boldsymbol{s}$. The predicted rank is

$$\widehat{\kappa}_\ell := \kappa_{\mathrm{rnd},\ell}(\boldsymbol{s}) + ms + m - 1, \qquad \widehat{r}_\ell := M_\ell - \widehat{\kappa}_\ell.$$

For the projected kernel, let $\boldsymbol{s}^{(j)}$ replace s_j with $s_j - s + 1$, as in (41), and set

$$\widehat{r}^{\,\mathrm{loc}}_{j,\ell} := \kappa_{\mathrm{rnd},\ell}(\boldsymbol{s}) - \kappa_{\mathrm{rnd},\ell}(\boldsymbol{s}^{(j)}), \qquad \widehat{r}^{\,\mathrm{proj}}_{j,\ell} := \widehat{r}^{\,\mathrm{loc}}_{j,\ell} + ms + m - 1.$$

The second estimate conservatively allows the entire structured excess to survive the projection. Every row of Table 3 satisfies

$$N_{d,\ell} - \widehat{r}_\ell \geqslant 2^{30}, \qquad |\mathcal{B}_j| - \widehat{r}^{\,\mathrm{proj}}_{j,\ell} \geqslant 2^{23} \quad \forall\, j \in [n_\ell]. \tag{11}$$

For this baseline, we fix the hold-out multiplicity to $s = 5$, use the degree $d > s$ listed in each row, and distribute the non-hold-out multiplicities as evenly as possible subject to the binary Goppa no-slack condition

$$\sum_{j=1}^{n_\ell} s_j = dD_\ell + 1 - t + s. \tag{12}$$

This is (10) after adding back the multiplicity s of the selected hold-out.

Avoiding the enumeration of $\boldsymbol{U}$. The factor q^{2h} in Theorem 7.2 comes from enumerating all guesses of the $h \times 2$ matrix $\boldsymbol{U}$, but this enumeration is avoidable in the concrete implementation.

The rank-two factorization $\boldsymbol{W} = \boldsymbol{U}\boldsymbol{V}$ is unchanged under the change of basis $\boldsymbol{U} \mapsto \boldsymbol{U}\boldsymbol{g}$, $\boldsymbol{V} \mapsto \boldsymbol{g}^{-1}\boldsymbol{V}$ for any invertible $\boldsymbol{g} \in \mathbb{F}_q^{2\times 2}$. Because any three pairwise linearly independent rows of $\boldsymbol{U}$ can be carried by such a $\boldsymbol{g}$ to any three prescribed pairwise linearly independent vectors in $\mathbb{F}_q^2$ (up to the per-row scalings, which are free), we may fix $\boldsymbol{U}$ outright: we run the recovery system with $h = 3$ and a single fixed $\boldsymbol{U}$ whose three rows are prescribed. No enumeration over $\boldsymbol{U}$ is needed at this stage.

We then run the locator test at a fourth hold-out only. At a coordinate j, this test inspects the two-row top block $\boldsymbol{W}'_j$ and accepts a value $x \in \mathbb{F}_q$ when some $\boldsymbol{V} \in \mathcal{V}_{\boldsymbol{U}}$ satisfies $V_{2,(j,\boldsymbol{a})} = x^{s-1} V_{1,(j,\boldsymbol{a})}$ for every $\boldsymbol{a} \in \mathcal{B}_j$ — one linear feasibility system per candidate x. We augment it with a single *degenerate* system, which accepts when some $\boldsymbol{V} \in \mathcal{V}_{\boldsymbol{U}}$ has $V_{1,(j,\boldsymbol{a})} = 0$ for every $\boldsymbol{a} \in \mathcal{B}_j$, i.e. the first row of the block vanishes. Testing the q field values together with this one degenerate system is $q+1$ feasibility systems in all. By Conjecture C.3, the accepted set at the fourth hold-out is exactly the m Frobenius conjugates of its locator, each appearing either as a value in $\mathbb{F}_q$ or as the degenerate system.

Selecting any one of these conjugate solutions fixes the fourth row of $\boldsymbol{U}$: we append the row $(\alpha_4^{s-1}, 1)$ when the chosen solution is a field value $\alpha_4 \in \mathbb{F}_q$, or the row $(1, 0)$ when it is the degenerate solution. This produces the fixed $\boldsymbol{U}$ for the $h = 4$ run, and the same $(q+1)$-system locator test is applied at every remaining coordinate.

A coordinate may be assigned the degenerate solution rather than a field value. Since the recovered support has $n_\ell \leqslant n \leqslant q$ distinct entries while each coordinate has $q+1$ possible outcomes, we can re-prescribe the three fixed rows — equivalently, apply one further change of basis $\boldsymbol{g}$ — so that every coordinate is assigned a value in $\mathbb{F}_q$; the induced change of variables yields an equivalent, valid key with all support entries in $\mathbb{F}_q$. The selected conjugate branch likewise yields an equivalent valid key.

Thus one shortened execution uses $q+1$ feasibility systems to determine the fourth hold-out, followed by $n_\ell(q+1)$ systems for the full four-hold-out recovery; the preliminary three-hold-out systems are no larger than the four-hold-out ones. The total is therefore $(n_\ell + 1)(q+1)$ largest-system-equivalents, in place of an enumeration over $\boldsymbol{U}$.

Reliability of the locator Searches. To compute $\mathcal{C}_j(\boldsymbol{U})$, we use the same randomized linear-system solver as in the distinguisher, adapted to operate over $\mathbb{F}_q$; it may miss a consistent system with probability at most $\rho := 10^{-2}$, but every returned solution is checked exactly, so inconsistent locator systems cannot produce false candidates. For each coordinate we scan the q candidate values $x \in \mathbb{F}_q$ together with the additional degenerate system — the $q+1$ feasibility systems introduced above — and repeat the entire scan with fresh randomness only if the preceding scan found no verified candidate, stopping after three scans. Since a canonical candidate always exists, either as a value in $\mathbb{F}_q$ or as the degenerate system, the global solver-induced failure probability is at most $R_\ell(n_\ell+1)\rho^3 < 0.01$. Early stopping makes the expected number of scans $\eta_3 := 1 + \rho + \rho^2 = 1.0101$, which is included in the cost below.

parameter set	m	n	t	k	ℓ	d	profile	largest locator system	R_ℓ	$\log_2 T_{\mathrm{rec}}$
`mceliece348864/f`	12	3488	64	2720	2569	7	$5^{42}\,6^{877}$	$2^{39.57} \times 2^{40.21}$	4	130.38
`mceliece460896/f`	13	4608	96	3360	3159	8	$5^{4}\,6^{177}\,7^{1268}$	$2^{47.11} \times 2^{47.88}$	4	148.95
`mceliece6688128/f`	13	6688	128	5024	4802	7	$5^{35}\,6^{1851}$	$2^{43.40} \times 2^{44.02}$	4	141.28
`mceliece6960119/f`	13	6960	119	5413	5198	7	$5^{24}\,6^{1738}$	$2^{43.08} \times 2^{43.70}$	4	140.45
`mceliece8192128/f`	13	8192	128	6528	6306	7	$5^{35}\,6^{1851}$	$2^{43.40} \times 2^{44.02}$	5	141.60

Table 3: Baseline, unoptimized concrete recovery estimates for the Classic McEliece parameter sets using the binary-operation metric. Here $k = n - mt$, $n_\ell = n - \ell$, $k_\ell = k - \ell$, and $D_\ell = n_\ell - 2t - 1$. Every row uses hold-out multiplicity $s = 5$ and a final hold-out set of size $h = 4$. In the profile column, a^b means that b coordinates have multiplicity a, including the four multiplicity-5 hold-outs. The profiles satisfy the no-slack condition (12) and the predicted kernel margins (11). The largest-locator-system column gives equations by unknowns over $\mathbb{F}_{2^m}$. The value $R_\ell = \lceil (n-4)/(n_\ell - 4) \rceil$ is the number of shortened executions used to cover the full support. The final column is computed from (13), including the expected amplification factor η_3; all exponents are rounded to two decimal places. See Remark 7.3 for additional optimizations that are not reflected in the table.

Baseline cost. For the system-size count, set

$$L_{s,\ell} := \binom{k_\ell + s - 1}{s - 1}, \qquad B_\ell := \max_j \sum_{u=s_j-s+1}^{s_j-1} \binom{k_\ell + u - 1}{u}.$$

Thus, $L_{s,\ell}$ is the length of a hold-out jet block and B_ℓ is the largest locator-block length. The largest locator feasibility system has $2M_\ell$ unknowns and $2N_{d,\ell} + 4(L_{s,\ell} - 1) + 1 + B_\ell$ equations over $\mathbb{F}_q = \mathbb{F}_{2^m}$. Let $T_{\mathrm{loc}}^{(q)}$ denote its sparse block Lanczos cost in $\mathbb{F}_q$-operations. A degree-d monomial has at most $\binom{d}{r}$ potentially nonzero Hasse derivatives of order r, so the number of nonzero entries used in the sparse-solver estimate is bounded by

$$2N_{d,\ell} \sum_{j=1}^{n_\ell} \sum_{r=0}^{s_j-1} \binom{d}{r} + 2\big(4(L_{s,\ell} - 1) + 1 + B_\ell\big).$$

For simplicity, we compute $T_{\mathrm{loc}}^{(q)}$ using the same formula for T_{lin} from Section 5, with the two dimensions interchanged because the locator system is wider than tall: we set $N = 2M_\ell$ and $J = 2N_{d,\ell} + 4(L_{s,\ell} - 1) + 1 + B_\ell$, recompute the preconditioner sparsity for these dimensions, substitute the preceding nonzero bound for the original matrix, and count each field addition or multiplication as one $\mathbb{F}_q$-operation. The resulting total-cost estimate is

$$T_{\mathrm{rec}} := \eta_3 m^2 R_\ell (n_\ell + 1)(q + 1)\, T_{\mathrm{loc}}^{(q)}, \tag{13}$$

charging m^2 binary operations per extension-field operation.

Remark 7.3 (Further concrete optimizations and concurrent work)**.** Weis [Wei26b] (with the assistance of Anthropic's Claude) independently developed a key-recovery algorithm based on the distinguisher from [GIJS26]. This algorithm is very different from our Sidelnikov–Shestakov based approach and relies on analyzing tangent spaces.[10] Weis also proposed several optimizations that are likely to lead to a significant reduction in the concrete costs.

- First, the above baseline uses the nq-type approach, namely $(n_\ell + 1)(q + 1)$ separate locator systems per shortened execution. This requires us to solve many (concretely, $(n_\ell + 1)(q + 1)$) large linear systems. Instead, one can sample sufficiently many vectors from $\ker(\boldsymbol{E}_\ell)$ (where 200 seems to suffice for all cases). Then one can form a local basis for a much smaller space corresponding to the the local derivative block. This is done by augmenting the span of these vectors with an explicitly known description of kernel relations supported on one or two jet blocks. Then, one can hope to extract support by identifying rank-one matrices in the affine spaces for these blocks.

- A second suggestion is to use the sharper counting of the Kernel condition. This was already accounted for in our optimizations.

- Finally, because $\boldsymbol{E}_\ell$ is binary, its large kernel computation can be carried out over $\mathbb{F}_2$ and then extended to $\mathbb{F}_q$, leaving only the smaller normalization and locator calculations over the extension field.

- Another optimization augments the public matrix with the column $\boldsymbol{y}_\infty := \sum_{j\in[n]} \boldsymbol{y}_j$. This can be done because a binary Goppa code has additional structure. $\boldsymbol{y}_\infty$ corresponds to the top degree coefficient of the underlying GRS polynomials. Therefore $[\mathbf{Y} \mid \boldsymbol{y}_\infty]$ is a subcode of the extended GRS code of degree D on $\boldsymbol{\alpha} \cup \{\infty\}$. This gives an additional point and leads to better parameters.

Preliminary estimates incorporating these optimizations make heuristic key recovery more expensive but closer to distinguishing, requiring less than 2^{128} binary operations for every Classic McEliece parameter set in Table 3 [Wei26b].

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[10] This is conceptually related to Lemoine's tangent space attack [Lem25] and the key-recovery algorithms of Vedenev [Ved26] and Saarinen [Saa26], though the technical details are different.

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AI disclosure. Our main results emerged from the failed PIR attempt described above and were obtained through traditional human trial and error, without the use of AI. Still, AI was instrumental for many aspects of this work. OpenAI Codex (ChatGPT-5.6 Sol) was used to verify the results, analyze and improve concrete efficiency, draft some of the proofs, design and implement experiments, and help with literature review and other aspects of writing. Amazon Quick and Google AI harness were used for extra sanity checks.

AI played a more significant role in developing our key-recovery algorithm. The algorithm, inspired by the Sidelnikov–Shestakov algorithm, was proposed by the authors. ChatGPT-5.6 Sol then helped with the routine but tedious derivations of the structure of the interpolation coefficients. It also helped us quickly run the experiments measuring nullities, which were important for guiding the choice of parameters and gaining confidence in our conjectures.

Most importantly, an audit by ChatGPT identified a dimension gap that made our initial algorithm fail for binary Goppa codes and required changing the parameters. ChatGPT also helped run experiments to confirm the structure of the required interpolation coefficients for Goppa codes. We sent the resulting algorithm to Steve Weis from Anthropic, who was able (with the help of Claude) to successfully run our algorithm on small binary Goppa instances.

The authors verified the correctness and originality of all content.

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A Heuristic Analysis of Ciphertext Decryption

We start by noting that Algorithm 3 specifies the decryption attack for a polynomial degree d and a vanishing multiplicity s, but does not prescribe how to choose these parameters. This section develops heuristic dimension and genericity estimates that make this choice explicit. Appendix A.1 projects multivariate polynomials and their derivatives onto a natural space of bivariate polynomials to explain the interpolation behavior underlying the rejection of dirty coordinates in GRS instances. Appendix A.2 then considers the additional error structure of Goppa codes and develops heuristics specific to Classic McEliece. Finally, Appendix A.4 reports finite-parameter experiments designed to test our hypotheses.

Throughout this section, we absorb the hidden permutation π into the coordinate indexing. To analyze the soundness of Algorithm 3, recall that there are polynomials $f_1, \dots, f_k \in \mathbb{F}_q[Z]$, a polynomial $f' \in \operatorname{span}\{f_1, \dots, f_k\}$, and a vector of nonzero normalized column multipliers $\widetilde{\boldsymbol{\lambda}} := (\widetilde{\lambda}_1, \dots, \widetilde{\lambda}_n) \in (\mathbb{F}_q^\times)^n$ such that the j-th column of the augmented matrix $\boldsymbol{A}$ has the form

$$\boldsymbol{A}_j = \widetilde{\lambda}_j \big(f_1(\alpha_j), \dots, f_k(\alpha_j), f'(\alpha_j) + e_j/\widetilde{\lambda}_j\big).$$

For a held-out coordinate τ, the algorithm accepts $\boldsymbol{A}_\tau$ if and only if $H(\boldsymbol{A}_\tau) = 0$ for every $H \in \mathcal{K}_\tau^{\hom}(\boldsymbol{A})$, where $\mathcal{K}_\tau^{\hom}(\boldsymbol{A})$ is the space of homogeneous degree-d polynomials that vanish to multiplicity s at all columns $\boldsymbol{A}_j$ with $j \neq \tau$.

One can show that the acceptance of the above algorithm is invariant under invertible row transformations. Namely, applying invertible row transformations to $\boldsymbol{A}$ causes only a corresponding invertible change of polynomial variables. As a consequence, since f' is a linear combination of $f_1, \dots, f_k$, we may subtract f' from the last coordinate. Moreover, since the polynomials are homogeneous, the nonzero column multipliers $\widetilde{\lambda}_j$ do not affect their vanishing or multiplicity. We may therefore work with the normalized points:

$$\boldsymbol{p}_j := \big(f_1(\alpha_j), \dots, f_k(\alpha_j), \beta_j\big), \tag{14}$$

where $\beta_j = e_j/\widetilde{\lambda}_j$. Consequently, the algorithm accepts the held-out coordinate τ if and only if $H(\boldsymbol{p}_\tau) = 0$ for every $H \in \mathcal{K}^{\hom}\big(\boldsymbol{p}_j : j \neq \tau\big)$, where $\mathcal{K}^{\hom}(\boldsymbol{p}_j : j \neq \tau)$ denotes the space of

homogeneous degree-d polynomials vanishing to multiplicity s at all $\boldsymbol{p}_j$ with $j \neq \tau$. This view will be used to develop our first heuristic.

A.1 Bivariate Polynomial Projection Heuristic

The normalized representation in Eq. (14) allows us to extend the univariate Hermite interpolation viewpoint to account for errors. Similarly to the list decoding algorithm of Guruswami and Sudan [GS98], we consider bivariate polynomials to incorporate the error.

Let Z be the variable that tracks the support point α_j, and let W be the variable that tracks the normalized error value β_j: clean coordinates have $\beta_j = 0$, whereas dirty coordinates have $\beta_j \neq 0$. This motivates the following natural bivariate projection.

Define the polynomial map

$$\boldsymbol{p}(Z, W) := \big(f_1(Z), \ldots, f_k(Z), W\big),$$

that is supposed to capture the points $\{\boldsymbol{p}_j\}$ and let

$$\Phi_{\boldsymbol{p}}(H)(Z, W) := H\big(\boldsymbol{p}(Z, W)\big) = H\big(f_1(Z), \ldots, f_k(Z), W\big)$$

be the induced map from homogeneous degree-d polynomials to a bivariate polynomial in Z and W. By construction,

$$\Phi_{\boldsymbol{p}}(H)(\alpha_j, \beta_j) = H(\boldsymbol{p}_j).$$

Therefore, the algorithm rejects a dirty held-out coordinate τ if and only if

$$\exists\, G \in \Phi_{\boldsymbol{p}}\big(\mathcal{K}^{\mathrm{hom}}(\boldsymbol{p}_j : j \neq \tau)\big) \quad \text{such that} \quad G(\alpha_\tau, \beta_\tau) \neq 0.$$

For a GRS instance, when τ is dirty, the support point α_τ is independently uniform relative to the non-held-out coordinates, while $\beta_\tau = e_\tau / \widetilde{\lambda}_\tau$ is uniform in $\mathbb{F}_q^\times$. We therefore expect evaluation at $(\alpha_\tau, \beta_\tau)$ to behave generically on the projected space:[11]

Hypothesis A.1 (Bivariate genericity)**.** *Write*

$$r_\tau := \dim \Phi_{\boldsymbol{p}}\big(\mathcal{K}^{\mathrm{hom}}(\boldsymbol{p}_j : j \neq \tau)\big).$$

Consider an asymptotic family of GRS instances, and let

$$r_{\min} := \min_{\tau \in \mathrm{supp}(\boldsymbol{e})} r_\tau.$$

If $r_{\min} = \omega(1)$, then Algorithm 3 rejects all dirty coordinates with probability $1 - o(1)$, where the probability is over the random GRS public key and the random ciphertext. This hypothesis is examined in the GRS and alternant experiments.

[11] Unlike the distinguishing task, decryption must identify every erroneous coordinate. It would suffice for each dirty coordinate to be rejected with failure probability $o(1/w)$, where w is the error weight. A Schwartz–Zippel argument does not necessarily give such a bound here because the polynomial degree may exceed the field size; this motivates the genericity hypothesis.

It remains to estimate r_τ, which guides the choice of d and s. We separate this estimate into the dimension of the full projected space and the dimension lost after imposing the non-held-out constraints. More precisely, we define:

$$\begin{aligned} B_d &:= \dim \Phi_{\boldsymbol{p}}\big(\mathcal{P}^{\hom}_{=d}\big), \\ L_\tau &:= B_d - \dim \Phi_{\boldsymbol{p}}\big(\mathcal{K}^{\hom}(\boldsymbol{p}_j : j \neq \tau)\big). \end{aligned}$$

Then $r_\tau = B_d - L_\tau$.

To estimate B_d, observe that every $G \in \Phi_{\boldsymbol{p}}\big(\mathcal{P}^{\hom}_{=d}\big)$ can be written as:

$$G(Z, W) = \sum_{i=0}^{d} g_i(Z) W^{d-i}.$$

For each i, the coefficient $g_i(Z)$ lies in the span of the degree-i products of $f_1, \ldots, f_k$. This space is generated by $\binom{i+k-1}{i}$ polynomials, each of degree at most iD. This gives the maximal-rank estimate

$$\widehat{B}_d(k, D) := \sum_{i=0}^{d} \min\left\{ \binom{i+k-1}{i}, iD + 1 \right\}.$$

We define the induced-dimension defect by

$$\delta^{(\mathrm{ind})} := \widehat{B}_d(k, D) - B_d \geqslant 0,$$

and heuristically expect this defect to be small.

For the second term, define the space of projected polynomials satisfying the bivariate multiplicity constraints by:

$$\mathcal{Q}_\tau := \left\{ G \in \Phi_{\boldsymbol{p}}\big(\mathcal{P}^{\hom}_{=d}\big) : (\partial^{[a,b]} G)(\alpha_j, \beta_j) = 0, \quad \forall\ a + b < s,\ j \neq \tau \right\}.$$

Multiplicity s at each non-held-out pair (α_j, β_j) imposes at most $\binom{s+1}{2}$ bivariate constraints. Hence,

$$B_d - \dim(\mathcal{Q}_\tau) \leqslant (n-1)\binom{s+1}{2}.$$

Although every projected homogeneous-kernel polynomial belongs to $\mathcal{Q}_\tau$, the converse need not hold. We define the homogeneous-lifting defect by

$$\delta_\tau^{(\hom)} := \dim(\mathcal{Q}_\tau) - \dim \Phi_{\boldsymbol{p}}\big(\mathcal{K}^{\hom}(\boldsymbol{p}_j : j \neq \tau)\big) \geqslant 0.$$

Consequently,

$$\begin{aligned} L_\tau &= B_d - \dim(\mathcal{Q}_\tau) + \delta_\tau^{(\hom)} \\ &\leqslant (n-1)\binom{s+1}{2} + \delta_\tau^{(\hom)}. \end{aligned}$$

As in the previous heuristic, we expect the lifting defect to be small when the homogeneous kernel has a large positive dimension margin. Using the raw number of Hasse derivative indices in $k+1$ variables, this margin is

$$\binom{k+d}{d} - (n-1)\binom{k+s}{s-1}.$$

Combining the two estimates gives

$$r_\tau \geqslant \widehat{B}_d(k, D) - (n-1)\binom{s+1}{2} - \delta^{(\mathrm{ind})} - \delta_\tau^{(\mathrm{hom})}.$$

We summarize the two defect estimates in the following hypotheses.

Hypothesis A.2 (Induced-dimension defect)**.** *For an asymptotic family of random GRS public keys with dimension $k = \Omega(n^\gamma)$ for some constant $\gamma > 0$,*

$$\delta^{(\mathrm{ind})} = o(n)$$

with probability $1 - o(1)$ over key generation. For an asymptotic family of Classic McEliece public keys with Goppa degree t,

$$\delta^{(\mathrm{ind})} = (d-1)t + o(n)$$

with probability $1 - o(1)$ over key generation. This hypothesis is examined in the induced-dimension experiments.

Hypothesis A.3 (Homogeneous-lifting defect)**.** *If the homogeneous-kernel dimension margin is sufficiently large, then*

$$\max_{\tau \in \mathrm{supp}(\boldsymbol{e})} \delta_\tau^{(\mathrm{hom})} = o(n)$$

with probability $1 - o(1)$ over key generation and encryption randomness. This hypothesis is examined in the homogeneous-lifting experiments.

The hypotheses above give the following conditional guarantees for the decryption algorithm.

Theorem A.4 (Conditional decryption for GRS)**.** *Consider an asymptotic family of GRS instances with $w := |\mathrm{supp}(\boldsymbol{e})| < (n-D)/2$. Assume the GRS defect estimates in Hypotheses A.2 and A.3 and the bivariate-genericity hypothesis Hypothesis A.1. Suppose*

$$s(n-w-1) > dD, \tag{15}$$

$$\widehat{B}_d(k, D) - (n-1)\binom{s+1}{2} = \Omega(n), \tag{16}$$

$$\frac{\binom{k+d}{d}}{(n-1)\binom{k+s}{s-1}} = \omega(1). \tag{17}$$

Then Algorithm 3 recovers the message with probability $1 - o(1)$, where the probability is over key generation, encryption randomness, and any internal randomness of the algorithm.

Proof. The kernel-margin condition (17) satisfies the premise of Hypothesis A.3. By the two defect estimates and (16), uniformly over all dirty coordinates,

$$r_\tau \geqslant \widehat{B}_d(k, D) - (n-1)\binom{s+1}{2} - \delta^{(\mathrm{ind})} - \delta_\tau^{(\mathrm{hom})} = \Omega(n).$$

Thus, Hypothesis A.1 implies that all dirty coordinates are rejected with probability $1 - o(1)$.

Since the clean set has size $n - w$, (15) is precisely the reconstruction condition in Claim 6.2. Hence, every clean coordinate is accepted. Finally, the ambient GRS distance implies that the public generator matrix restricted to the clean coordinates has full row rank. So when all dirty coordinates are rejected, the final linear system uniquely recovers the message. □

A.2 Goppa-Specific Analysis

Although the algorithm views a binary Goppa code as a subcode of an ambient GRS code, its additional algebraic and binary structure may create new obstructions. We isolate three mechanisms that can cause a dirty coordinate to be accepted and argue heuristically that none occurs with noticeable probability in the asymptotic Classic McEliece regime. Throughout this subsection, we assume that $d > s \geqslant 1$.

Failures from Short Cycles. After the row transformation and projective rescaling used above, the normalized points may be represented by the binary augmented columns

$$\boldsymbol{y}'_i := (\boldsymbol{y}_i, e_i) \in \mathbb{F}_2^{k+1}.$$

Suppose that a dirty coordinate τ belongs to a short cycle I, namely:

$$\sum_{i \in I} \boldsymbol{y}'_i = \mathbf{0}, \qquad \tau \in I, \qquad |I| = r,$$

where r is small. This can cause a dirty held-out column to be accepted.

To see this, assume that I is inclusion-minimal and write $I = \{i_1, \ldots, i_{r-1}, \tau\}$. Then the vectors $\{\boldsymbol{y}'_{i_j}\}_{j \in [r-1]}$ are linearly independent. Choose an invertible binary change of variables that maps each $\boldsymbol{y}'_{i_j}$ to $\boldsymbol{y}''_{i_j}$, the j-th standard basis vector. The cycle relation then maps $\boldsymbol{y}'_\tau$ to $\boldsymbol{y}''_\tau = \sum_{j=1}^{r-1} \boldsymbol{y}''_{i_j}$.

To reject τ, we need a polynomial that vanishes on all $r-1$ points to multiplicity s, but does not vanish on τ. Since the vectors $\{\boldsymbol{y}''_i\}_{i \in I}$ are supported only on the first $r-1$ coordinates, we only need to consider degree-d monomials in $r-1$ variables denoted as:

$$X_1^{b_1} \cdots X_{r-1}^{b_{r-1}}, \qquad \sum_{i=1}^{r-1} b_i = d.$$

At $\boldsymbol{y}''_{i_j}$, its local degree in the variables vanishing at $\boldsymbol{y}''_{i_j}$ is $\sum_{a \neq j} b_a = d - b_j$. Therefore, vanishing to multiplicity s at $\boldsymbol{y}''_{i_j}$ requires $d - b_j \geqslant s$, or equivalently $b_j \leqslant d - s$. Hence, a degree-d homogeneous polynomial vanishing to multiplicity s at all $r-1$ coordinate points can contain only monomials satisfying:

$$b_i \leqslant d - s, \quad \forall i \in [r-1].$$

Consequently, if

$$d > (r-1)(d-s),$$

then there is no non-zero polynomial in these monomials on $r-1$ variables satisfying the vanishing conditions. As a consequence, the algorithm would necessarily accept τ.

The dangerous cycle sizes satisfy:

$$r \leqslant L_{\text{circ}} := \left\lfloor \frac{d}{d-s} \right\rfloor + 1.$$

Thus, only cycles of size at most L_{circ} create this particular obstruction. For uniformly random public columns $\boldsymbol{y}_i \in \mathbb{F}_2^k$ and any fixed error vector, if $L_{\mathrm{circ}} = O(\log n)$ and $k = \Theta(n)$, then a union bound gives

$$\mathbf{Pr}\left[\exists I \subseteq [n],\ 2 \leqslant |I| \leqslant L_{\mathrm{circ}} : \sum_{i \in I} \boldsymbol{y}'_i = \mathbf{0}\right] \leqslant 2^{-k} \sum_{r=2}^{L_{\mathrm{circ}}} \binom{n}{r} = 2^{-\Omega(n)}.$$

We heuristically expect the short-dependency profile of random binary Goppa public columns to behave similarly, and hence that no dangerous cycle of size $O(\log n)$ occurs. The experiments below audit this obstruction directly.

Nearby ambient codewords. Let

$$\mathcal{C}_{\mathrm{amb}} := \mathsf{GRS}_{D+1}(\boldsymbol{\alpha}, \boldsymbol{\lambda}'), \qquad D = n - 2t - 1.$$

Suppose that a nonzero word $\boldsymbol{v} \in \mathcal{C}_{\mathrm{amb}}$ has the form

$$v_i = \lambda'_i g(\alpha_i), \qquad \deg(g) \leqslant D.$$

At every coordinate where $v_i = e_i$, the normalized point lies on the alternative section $W = g(Z)$. Hence, if $v_\tau = e_\tau = 1$ and

$$s\big(n - d_{\mathrm{H}}(\boldsymbol{e}, \boldsymbol{v}) - 1\big) > dD,$$

then, for every $G \in \Phi_{\boldsymbol{p}}\big(\mathcal{K}^{\mathrm{hom}}(\boldsymbol{p}_j : j \neq \tau)\big)$, the polynomial $G(Z, g(Z))$ has degree at most dD and has roots of multiplicity s at $n - d_{\mathrm{H}}(\boldsymbol{e}, \boldsymbol{v}) - 1$ constrained points. It must therefore vanish identically, forcing the algorithm to accept τ.

Thus, a possible obstruction is a nonzero ambient codeword satisfying

$$d_{\mathrm{H}}(\boldsymbol{e}, \boldsymbol{v}) < R_{\mathrm{near}} := n - 1 - \frac{dD}{s}$$

and agreeing with $\boldsymbol{e}$ at a dirty coordinate. The zero word has distance w from an error vector of weight w, but it agrees with $\boldsymbol{e}$ at no dirty coordinate and is therefore harmless.

Consider a minimum-weight word $\boldsymbol{v} \in \mathcal{C}_{\mathrm{amb}}$ with $|\mathrm{supp}(\boldsymbol{v})| = 2t + 1$. If

$$d_{\mathrm{H}}(\boldsymbol{e}, \boldsymbol{v}) < R_{\mathrm{near}},$$

then $\boldsymbol{v}$ must coincide exactly with $\boldsymbol{e}$ on more than

$$2t + 1 - R_{\mathrm{near}} = 1 + \frac{(d - s)D}{s}$$

dirty coordinates. The number of minimum-weight words is

$$(q - 1)\binom{n}{2t + 1} = 2^{o(n)}$$

in the asymptotic Classic McEliece regime $t = \Theta(n/\log n)$, by the MDS weight distribution [MPS22], whereas the required extension-field coincidences have probability $q^{-\Theta(t)} = 2^{-\Theta(n)}$ when $\frac{(d-s)D}{s} = \Theta(t)$.[12]

[12] After removing the multipliers from $\boldsymbol{v}$ and replacing e_i by $\beta_i = e_i/\lambda'_i$, the statistical distance between $\boldsymbol{\beta}$ and a uniformly sampled t-sparse vector over $\mathbb{F}_q$ is at most $1 - 1/q$ when Γ is sampled uniformly from the irreducible polynomials of degree t.

More generally, if $|\mathrm{supp}(\boldsymbol{v})| = 2t+1+\eta$, then the number of candidates increases essentially by a factor q^{η}, while entering the radius R_{near} requires η additional exact coincidences, contributing a compensating factor $q^{-\eta}$. Thus, the minimum-weight estimate extends to all relevant weight layers, and we expect that no dangerous nearby ambient codeword exists.

Restriction to Goppa-Induced Error Section. Unlike in the GRS setting, the normalized error value at a dirty coordinate is not independently random. Since Γ is irreducible and therefore square-free, we may take the multiplier $\widetilde{\lambda}_i$ in the degree-$(n-2t)$ ambient GRS representation to be the multiplier λ'_i from Fact 2.3. Hence, for every dirty coordinate i,

$$\beta_i = \rho(\alpha_i), \qquad \rho(Z) := \frac{\Pi'_{\boldsymbol{\alpha}}(Z)}{\Gamma(Z)^2}.$$

Thus, the clean coordinates lie on the section $W = 0$, whereas the dirty coordinates lie on the error section $W = \rho(Z)$. We therefore restrict each $G(Z, W)$ to this section. Decryption may fail if every such restriction is zero or if all restrictions share many roots. Let $w := |\mathrm{supp}(\boldsymbol{e})|$, and let $\mathcal{R}^{\mathrm{cl}}$ be the image under $\Phi_{\boldsymbol{p}}$ of the homogeneous polynomials satisfying the multiplicity constraints at all clean coordinates. Write

$$G(Z, W) = \sum_{i=0}^{d} g_i(Z) W^i \in \mathcal{R}^{\mathrm{cl}}, \qquad \deg(g_i) \leqslant D(d-i).$$

The clean constraints imply that g_i vanishes to multiplicity at least $s-i$ at every clean support point for $i < s$. Hence, define

$$\ell := \left|\{i \in \{0, \ldots, s-1\} : (s-i)(n-w) > D(d-i)\}\right|.$$

Univariate interpolation then gives

$$W^{\ell} \mid G(Z, W) \qquad \text{for every } G \in \mathcal{R}^{\mathrm{cl}}.$$

Writing $G = W^{\ell} F_G$, we remove the resulting common factor and restrict the remaining polynomial to the error section:

$$P_G(Z) := \Gamma(Z)^{2(d-\ell)} F_G\big(Z, \rho(Z)\big).$$

Now fix a dirty held-out coordinate τ, and let $\mathcal{R}_\tau \subseteq \mathcal{R}^{\mathrm{cl}}$ be the image of the homogeneous hold-out kernel after imposing the constraints at every other dirty coordinate. Recall that $\mathcal{R}_\tau$ is exactly $\Phi_{\boldsymbol{p}}\big(\mathcal{K}^{\mathrm{hom}}(\boldsymbol{p}_j : j \neq \tau)\big)$ in Appendix A.1, and r_τ denotes its dimension. The remaining dirty constraints force the common factor

$$E_\tau(Z) := \prod_{j \in \mathrm{supp}(\boldsymbol{e}) \setminus \{\tau\}} (Z - \alpha_j)^s.$$

Define the residual section space

$$\mathcal{P}_\tau := \left\{ \frac{P_G}{E_\tau} : G \in \mathcal{R}_\tau \right\}, \qquad r_\tau^{\mathrm{sec}} := \dim(\mathcal{P}_\tau).$$

For the residual section space to distinguish the held-out point, it should remain large and should not have a common factor that vanishes at α_τ.

The degree calculation above gives $\deg(P_G) \leqslant (d-\ell)(n-1)$, while the common factor $E_\tau(Z)$ accounts for $s(w-1)$ forced roots. Set

$$T_\tau := (d-\ell)(n-1) + 1 - s(w-1).$$

We measure the failure of the restriction map to attain maximal rank by defining $\delta_\tau^{(\mathrm{sec})} \geqslant 0$ through

$$r_\tau^{\mathrm{sec}} = \min\{r_\tau,\, T_\tau\} - \delta_\tau^{(\mathrm{sec})},$$

where $\delta_\tau^{(\mathrm{sec})}$ is the additional dimension loss under restriction to the error section. Our parameter selection makes both terms inside the minimum large, and we heuristically expect this defect to be small. Thus, $\mathcal{P}_\tau$ should have large dimension.

Finally, define the monic residual GCD and its number of distinct field roots by

$$g_\tau^{\mathrm{res}}(Z) := \gcd\{P(Z) : 0 \neq P \in \mathcal{P}_\tau\}, \qquad \nu_\tau := \left|\{a \in \mathbb{F}_q : g_\tau^{\mathrm{res}}(a) = 0\}\right|.$$

The algorithm falsely accepts τ exactly when $g_\tau^{\mathrm{res}}(\alpha_\tau) = 0$. We heuristically expect that a large residual section space does not possess an adversarial common factor: g_τ^{res} should have few roots in $\mathbb{F}_q$, with no correlation between those roots and the held-out point α_τ.

Hypothesis A.5 (Goppa-section genericity)**.** *Consider an asymptotic family of binary Goppa instances satisfying*

$$L_{\mathrm{circ}} = \left\lfloor \frac{d}{d-s} \right\rfloor + 1 = O(\log n), \qquad 2t + 1 - R_{\mathrm{near}} = 1 + \frac{(d-s)D}{s} = \Theta(t),$$
$$\min_{\tau \in \mathrm{supp}(\boldsymbol{e})} r_\tau = \Omega(n), \qquad \min_{\tau \in \mathrm{supp}(\boldsymbol{e})} T_\tau = \Omega(n).$$

Then, with probability $1 - o(1)$,

$$\max_{\tau \in \mathrm{supp}(\boldsymbol{e})} \delta_\tau^{(\mathrm{sec})} = o(n),$$
$$\min_{\tau \in \mathrm{supp}(\boldsymbol{e})} r_\tau^{\mathrm{sec}} = \Omega(n),$$
$$\max_{\tau \in \mathrm{supp}(\boldsymbol{e})} \nu_\tau = o(n),$$
$$g_\tau^{\mathrm{res}}(\alpha_\tau) \neq 0 \qquad \text{for every } \tau \in \mathrm{supp}(\boldsymbol{e}).$$

In particular, neither a short cycle, a nearby ambient codeword, nor any other Goppa-specific structure creates a common root at a dirty held-out coordinate with noticeable probability. Consequently, Algorithm 3 rejects all dirty coordinates with probability $1 - o(1)$*. Here the probability is over key generation, encryption randomness, and any internal randomness of the attack, for any fixed plaintext.*

This hypothesis is examined in the genericity experiments.

Theorem A.6 (Conditional decryption for binary Goppa codes)**.** *Consider an asymptotic family of square free binary Goppa instances with Goppa degree* t*, ambient degree* $D = n - 2t - 1$*, and error*

weight $w := |\mathrm{supp}(\boldsymbol{e})| \leqslant t$. *Assume the binary Goppa defect estimates in Hypotheses A.2 and A.3 and the Goppa-section genericity hypothesis Hypothesis A.5. Suppose*

$$s(n-w-1) > dD, \tag{18}$$

$$\frac{\binom{k+d}{d}}{(n-1)\binom{k+s}{s-1}} = \omega(1), \tag{19}$$

$$\widehat{B}_d(k,D) - (n-1)\binom{s+1}{2} - (d-1)t = \Omega(n), \tag{20}$$

$$(d-\ell)(n-1) + 1 - s(w-1) = \Omega(n), \tag{21}$$

$$\left\lfloor \frac{d}{d-s} \right\rfloor + 1 = O(\log n), \tag{22}$$

$$1 + \frac{(d-s)D}{s} = \Theta(t). \tag{23}$$

Here, ℓ *is defined as above. Then Algorithm 3 recovers the message with probability* $1 - o(1)$*, where the probability is over key generation, encryption randomness, and any internal randomness of the algorithm.*

Proof. The kernel-margin condition (19) satisfies the premise of Hypothesis A.3. By the induced-dimension and homogeneous-lifting defect estimates and (20), uniformly over the dirty coordinates,

$$r_\tau \geqslant \widehat{B}_d(k,D) - (n-1)\binom{s+1}{2} - \delta^{(\mathrm{ind})} - \delta_\tau^{(\mathrm{hom})} = \Omega(n).$$

Eq. (21) gives $T_\tau = \Omega(n)$, while Eqs. (22) and (23) give the other two conditions of Hypothesis A.5. That hypothesis therefore implies that $r_\tau^{\mathrm{sec}} = \Omega(n)$ and $g_\tau^{\mathrm{res}}(\alpha_\tau) \neq 0$ for every dirty coordinate τ. Hence, all dirty coordinates are rejected with probability $1 - o(1)$.

Eq. (18) is the clean-coordinate reconstruction condition in Claim 6.2, so every clean coordinate is accepted. Finally, a binary Goppa code of degree t has minimum distance at least $2t + 1 > w$. Thus, the public generator matrix restricted to the clean coordinates has full row rank, and the final linear system uniquely recovers the message. □

A.3 Compatibility with the Asymptotic Classic McEliece Regime

We finally verify that the conditions of Theorem A.6 are simultaneously satisfiable in the asymptotic Classic McEliece parameter regime.

Claim A.7 (Asymptotic parameter compatibility)**.** *Suppose*

$$t = \Theta\left(\frac{n}{\log n}\right), \qquad k = \Theta(n), \qquad D = n - 2t - 1,$$

and let $1 \leqslant w \leqslant t$*. For a sufficiently large fixed constant* $h \in \mathbb{N}$*, set*

$$s := \left\lfloor \frac{hD}{t} \right\rfloor + 1, \qquad\qquad d := s + h.$$

Then $d, s = \Theta(\log n)$*, and all the deterministic conditions of Theorem A.6 hold uniformly over* $1 \leqslant w \leqslant t$*.*

Consequently, under Hypotheses A.2, A.3 and A.5, Algorithm 3 succeeds with probability $1 - o(1)$ *and runs in quasipolynomial time.*

Proof. By construction, $s, d = \Theta(D/t) = \Theta(\log n)$. Define $\zeta := st - hD$. The choice of s gives $0 < \zeta \leqslant t$. Since $w \leqslant t$,

$$\begin{aligned} s(n-w-1) - dD &\geqslant s(n-t-1) - (s+h)D \\ &= s(D+t) - (s+h)D \\ &= \zeta > 0, \end{aligned}$$

which proves (18).

For the homogeneous-kernel condition,

$$\begin{aligned} \frac{\binom{k+d}{d}}{\binom{k+s}{s-1}} &= \frac{k+1}{s} \prod_{j=s+1}^{d} \frac{k+j}{j} \\ &= \Omega\left(\left(\frac{n}{\log n}\right)^{h+1}\right). \end{aligned}$$

Dividing by $n-1$ proves (19).

Since $d - s = h$, we have $\lfloor d/(d-s) \rfloor + 1 = \lfloor (s+h)/h \rfloor + 1 = \Theta(\log n)$, which proves (22). Moreover, $1 + (d-s)D/s = 1 + hD/s = \Theta(t)$, proving (23).

Since $k, D = \Theta(n)$ and $d = O(\log n)$, we have $\widehat{B}_d(k, D) = D\binom{d+1}{2} + O(n)$. Using $n - 1 = D + 2t$, $d = s + h$, and $st = hD + \zeta$, we obtain

$$\begin{aligned} &\widehat{B}_d(k, D) - (n-1)\binom{s+1}{2} - (d-1)t \\ &\quad = \frac{h(h+1)}{2} D - s\zeta - 2st - (h-1)t + O(n). \end{aligned}$$

The positive term is $\Theta(h^2 n)$, whereas the remaining terms are $O(hn)$. Hence, for a sufficiently large fixed h, the expression is $\Omega(n)$, proving (20).

It remains to evaluate ℓ. Let $u := n - w - D = 2t + 1 - w$. Since $d = s + h$, the inequality defining the forced coefficients becomes $(s-i)u > hD$. As $i \in \{0, \ldots, s-1\}$, this gives $\ell = s - \lfloor hD/u \rfloor$, and therefore $d - \ell = h + \lfloor hD/u \rfloor$. Because $t + 1 \leqslant u \leqslant 2t$, we have $d - \ell = \Theta(D/t) = \Theta(\log n)$. Moreover, $s(w-1) \leqslant st = O(n)$. It follows that

$$(d-\ell)(n-1) + 1 - s(w-1) = \Theta(n \log n),$$

proving (21).

Thus, all the deterministic conditions of Theorem A.6 hold. Under the hypotheses stated in the claim, the conditional success probability follows from that theorem. Finally, since $d, s = O(\log n)$, all polynomial and Hasse-derivative spaces used by the algorithm have dimension $n^{O(\log n)}$, giving a quasipolynomial running time. □

A.4 Dimension and Genericity Experiments

Each paragraph below is organized around the hypothesis it audits. The first two address the induced-dimension and homogeneous-lifting defects, while the last audits the section dimension, residual GCD, and cycle conditions in Hypothesis A.5. Each underlying dimension is computed exactly over the relevant finite field; averages and ranges only summarize repeated samples.

Induced-dimension defect. This paragraph audits Hypothesis A.2 by comparing the exact induced dimension B_d with the maximal-rank benchmark $\widehat{B}_d(k, D)$.

For the GRS experiments, we sample k linearly independent polynomials of degree at most D. Each row of Table 4 contains five independent samples. In every sample, the projected space attains the maximal-rank benchmark.

q	(k, D)	$B_2/\widehat{B}_2$	$B_3/\widehat{B}_3$	$B_4/\widehat{B}_4$
32	$(14, 17)$	50/50	102/102	171/171
64	$(17, 25)$	69/69	145/145	246/246
64	$(19, 29)$	79/79	167/167	284/284

Table 4: Exact induced dimensions and maximal-rank benchmarks for the sampled generic GRS instances. Each entry gives $B_d/\widehat{B}_d(k, D)$; for each row and degree, this pair was identical across all five independent samples.

For binary Goppa instances, Table 5 reports the same comparison. Under the current projection, the appended ciphertext coordinate increases both B_d and $\widehat{B}_d(k, D)$ by one in all tested parameter sets. Consequently, the induced-dimension defect is unchanged. In every computed entry, we observe

$$\delta^{(\mathrm{ind})} = \widehat{B}_d(k, D) - B_d = (d-1)t.$$

(m, n, t)	(k, D)	$B_2/\widehat{B}_2$	$B_3/\widehat{B}_3$	$B_4/\widehat{B}_4$	$B_5/\widehat{B}_5$
$(4, 14, 1)$	$(10, 11)$	33/34	66/68	110/113	165/169
$(5, 28, 1)$	$(23, 25)$	74/75	149/151	249/252	374/378
$(5, 28, 2)$	$(18, 23)$	64/66	132/136	223/229	337/345
$(5, 32, 3)$	$(17, 25)$	66/69	139/145	237/246	360/372
$(6, 48, 1)$	$(42, 45)$	133/134	268/270	448/451	673/677
$(6, 48, 2)$	$(36, 43)$	122/124	250/254	421/427	635/643
$(6, 48, 3)$	$(30, 41)$	111/114	232/238	394/403	597/609
$(6, 56, 4)$	$(32, 47)$	124/128	262/270	447/459	679/695
$(6, 56, 5)$	$(26, 45)$	113/118	244/254	420/435	641/661
$(6, 56, 6)$	$(20, 43)$	102/108	226/238	393/411	603/627
$(6, 56, 7)$	$(14, 41)$	91/98	208/222	366/387	565/593
$(6, 64, 8)$	$(16, 47)$	104/112	238/254	419/443	647/679
$(7, 64, 2)$	$(50, 59)$	168/170	344/348	579/585	873/881
$(7, 80, 4)$	$(52, 71)$	192/196	402/410	683/695	1035/1051
$(7, 96, 6)$	$(54, 83)$	216/222	460/472	787/805	1197/1221

Table 5: Exact induced dimensions and maximal-rank benchmarks for the tested binary Goppa instances. Each entry gives $B_d/\widehat{B}_d(k, D)$.

Thus, the tested GRS instances have $\delta^{(\mathrm{ind})} = 0$, while the tested binary Goppa instances satisfy the stronger exact relation $\delta^{(\mathrm{ind})} = (d-1)t$.

Homogeneous-lifting defect. This paragraph audits Hypothesis A.3 by comparing the full bivariate constraint space $\mathcal{Q}_\tau$ with the image of the homogeneous kernel. It also records

$$r_\tau = \dim \Phi_{\boldsymbol{p}}\big(\mathcal{K}^{\mathrm{hom}}(\boldsymbol{p}_j : j \neq \tau)\big),$$

which will also be used in the genericity experiments below. Equivalently, we compute

$$\delta_\tau^{(\mathrm{hom})} = \dim(\mathcal{Q}_\tau) - r_\tau.$$

For every parameter row, we sample at least five independent instances.

Family	Base/Ambient	k	D	n	d	s	w	r_τ	$\delta_\tau^{(\mathrm{hom})}$
GRS*	$\mathbb{F}_{64}/\mathbb{F}_{64}$	19	29	56	3	2	3	20	0
GRS*	$\mathbb{F}_{67}/\mathbb{F}_{67}$	14	34	62	5	3	4	129	0
GRS	$\mathbb{F}_{32}/\mathbb{F}_{32}$	8	14	24	5	3	3	71	0
GRS	$\mathbb{F}_{32}/\mathbb{F}_{32}$	9	17	31	5	3	3	72	0
GRS	$\mathbb{F}_{32}/\mathbb{F}_{32}$	7	12	21	7	4	3	138	0
Alternant*	$\mathbb{F}_{16}/\mathbb{F}_{256}$	25	69	113	3	2	8	39	0
Alternant	$\mathbb{F}_{11}/\mathbb{F}_{121}$	14	52	92	5	3	4	201	0
Alternant	$\mathbb{F}_{16}/\mathbb{F}_{256}$	8	15	24	5	3	3	85	0
Alternant*	$\mathbb{F}_{16}/\mathbb{F}_{256}$	13	41	71	5	3	1	176	0
Goppa	$\mathbb{F}_2/\mathbb{F}_{32}$	12	25	32	4	2	1	138–139	0
Goppa	$\mathbb{F}_2/\mathbb{F}_{32}$	12	25	32	5	3	3	162–169	0–3

Table 6: Audits of the homogeneous-lifting defect at dirty held-out coordinates. The second column lists the base field followed by the ambient field; the GRS restriction is computed over the ambient field. For binary Goppa codes in this table, $t = 3$. The rows marked by $\star$ satisfy all deterministic parameter requirements of the decryption analysis. Ranges record variation across the sampled instances and dirty held-out coordinates.

The homogeneous-lifting defect is zero in every tested GRS and alternant instance. It is also zero in one of the two binary Goppa parameter rows; in the remaining row, it is at most 3, while $r_\tau \geqslant 162$. The starred GRS and alternant rows satisfy all deterministic parameter requirements of the decryption analysis. These experiments support Hypothesis A.3.

Every dirty coordinate in the sampled GRS and alternant instances was rejected, providing finite-parameter evidence for the bivariate-genericity hypothesis Hypothesis A.1.

Goppa-section genericity. This paragraph audits Hypothesis A.5. We first compare the error-section dimension r_τ^{sec} with its two upper bounds. Recall that

$$r_\tau := \dim(\mathcal{R}_\tau),$$
$$r_\tau^{\mathrm{sec}} := \dim(\mathcal{P}_\tau),$$

and that

$$T_\tau := (d-\ell)(n-1) + 1 - s(w-1)$$

is the dimension upper bound imposed by the degree of the restricted polynomial and the forced roots at the other dirty coordinates. We record the two gaps

$$\Delta_\tau^{(r)} := r_\tau - r_\tau^{\mathrm{sec}},$$
$$\Delta_\tau^{(T)} := T_\tau - r_\tau^{\mathrm{sec}}.$$

The measured section-dimension defect is therefore

$$\delta_\tau^{(\mathrm{sec})} = \min\left\{\Delta_\tau^{(r)}, \Delta_\tau^{(T)}\right\}.$$

The active upper bound is r_τ when $r_\tau \leqslant T_\tau$, and T_τ otherwise.

For each parameter row in Table 7, we sample 200 independent irreducible binary Goppa instances with error weight $w = t$ and audit every dirty coordinate as the held-out coordinate. Each dimension entry reports the average followed by the minimum and maximum in brackets. In each trial, we also test whether every dirty coordinate is rejected and, for every dirty hold-out, compute the residual GCD g_τ^{res} and count its distinct field roots outside the set of dirty evaluation points.

(m, n, t, D, k, d, s)	r_τ^{sec}	$\Delta_\tau^{(r)}$	$\Delta_\tau^{(T)}$	$\delta_\tau^{(\mathrm{sec})}$	roots outside dirty support	all dirty rejected
$(5, 32, 3, 25, 12, 4, 2)$	112.8 [112, 113]	25.8 [24, 26]	8.2 [8, 9]	8.2 [8, 9]	0.010 [0, 1]	200
$(5, 32, 2, 27, 20, 3, 2)$	61 [61, 61]	0 [0, 0]	31 [31, 31]	0 [0, 0]	0.005 [0, 2]	200
$(5, 27, 2, 22, 12, 4, 3)$	52 [52, 52]	0 [0, 0]	50 [50, 50]	0 [0, 0]	1.9475 [0, 25]	182
$(5, 28, 2, 23, 18, 4, 3)$	61 [61, 61]	0 [0, 0]	45 [45, 45]	0 [0, 0]	0 [0, 0]	200
$(4, 16, 2, 11, 8, 5, 3)$	64 [64, 64]	5 [5, 5]	9 [9, 9]	5 [5, 5]	0 [0, 0]	200
$(5, 19, 2, 14, 8, 4, 2)$	65.4 [64, 66]	12 [12, 12]	5.6 [5, 7]	5.6 [5, 7]	0.0375 [0, 1]	199
$(5, 27, 3, 20, 12, 4, 3)$	31 [31, 31]	0 [0, 0]	68 [68, 68]	0 [0, 0]	1.3917 [0, 7]	189
$(6, 42, 4, 33, 17, 3, 2)$	54 [54, 54]	0 [0, 0]	64 [64, 64]	0 [0, 0]	0.025 [0, 2]	199
$(6, 49, 5, 38, 18, 3, 2)$	57 [57, 57]	0 [0, 0]	80 [80, 80]	0 [0, 0]	0.024 [0, 2]	200

Table 7: Error-section and residual-GCD experiments for irreducible binary Goppa codes. The dimension entries are computed from 200 independent trials per row, with every dirty coordinate used as the hold-out, and report the average followed by the minimum and maximum in brackets. The root entries use the same format and count distinct roots outside the dirty support, without multiplicity. The final column reports the number of those trials in which every dirty coordinate is rejected.

Every measured residual section dimension is at least 31, and hence both r_τ and T_τ are at least 31 throughout the audit. The section-dimension defect is zero in six of the nine parameter rows and lies between 5 and 9 in the remaining three. Thus, the measurements suggest that $\delta_\tau^{(\mathrm{sec})}$ is zero or very small whenever $\Delta_\tau^{(T)}$ is not close to zero.

For these parameter rows, L_{circ} lies between 3 and 5, and the nearby-codeword threshold $1 + (d-s)D/s$ exceeds w in every row. All dirty coordinates are rejected in 1769 of the 1800 rejection trials. Every unrejected dirty coordinate in the remaining 31 trials is explained by a short cycle

$$\sum_{i \in I} \boldsymbol{y}_i' = \mathbf{0}, \qquad \tau \in I, \qquad 2 \leqslant |I| \leqslant 4.$$

Thus, short cycles explain every observed rejection failure.

For most parameter rows, the number of distinct field roots outside the dirty support is quite small: in seven of the nine rows, the average is at most 0.0375. The two rows with the largest

root counts are also the two rows with the smallest values of r_τ^{sec}, namely 31 and 52. In particular, comparing the $(5, 27, 2, 22, 12, 4, 3)$ and $(5, 28, 2, 23, 18, 4, 3)$ rows, both with $(d, s) = (4, 3)$, r_τ^{sec} increases from 52 to 61, and the root count in the latter row is zero. This suggests that larger residual section spaces suppress common field roots and supports Hypothesis A.5. The reported root count excludes dirty evaluation points and therefore does not directly measure ν_τ, which counts all field roots.

End-to-End Decryption/Decoding Tests. We report end-to-end decryption experiments on small instances from four related code families: generalized Reed–Solomon (GRS) codes, generic alternant codes, ordinary Goppa codes over nonbinary base fields, and wild Goppa codes. Definitions of these families are collected in Appendix B, and the results are summarized in Table 8.

We were unable to run the full attack on a binary square-free Goppa instance satisfying all of our parameter constraints. Indeed, the support of a length-n binary Goppa code lies in $\mathbb{F}_{2^m}$, so $n \leqslant 2^m$ and hence $m \geqslant \lceil \log_2 n \rceil$. Our attack also requires

$$\frac{d}{s} \approx 1 + \frac{1}{m}.$$

Since $d > s$ are integers, the smallest gap $d - s = 1$ gives $s \approx m$ and $d \approx m + 1$. Thus, d and s grow on the order of $\log_2 n$, causing the associated polynomial spaces and linear systems to grow rapidly. A search over admissible integer parameters suggests that the smallest compatible binary Goppa instance would already require solving a linear system of dimension approximately 2^{27}, which is beyond the scope of our current implementation. The second-best alternative is to work with families of codes that are similar to Binary Goppa codes. Concretely, we work with alternant, Goppa and wild Goppa families (where binary Goppa is contained in each of those families). A major difference between Goppa and Binary Goppa codes is that we now allow bigger base fields. This permits comparable code lengths with a smaller extension degree m, making the ordinary- and wild-Goppa experiments reported here computationally feasible.

For each row of Table 8, we performed at least 20 independent trials. Every trial recovered the planted error vector of Hamming weight w and hence decrypted successfully. To mirror the error structure of the binary Goppa setting, we sampled

$$\boldsymbol{e} \in \{0, 1\}^n \subseteq \mathbb{F}_q^n$$

uniformly among vectors of Hamming weight w. This choice is consistent with the analysis in Appendix A.2: for the ordinary- and wild-Goppa instances, the normalized dirty points lie on a single one-dimensional error section parametrized by the support coordinate, as in the binary case.

Finally, we provide some reflections about the experiments and discuss which of the data points in Table 8 satisfy the criteria needed for prior polynomial time attacks. Under the standard shortening–Schur-square dimension test for an $[n, k]$ public subcode of GRS_{D+1} [Wie10, CGG+14], only the $(56, 19, 29)$ GRS row and the $\mathbb{F}_9/\mathbb{F}_{81}$ wild-Goppa row are directly distinguishable using $4 \leqslant \ell \leqslant 10$ and $\ell \in \{2, 3\}$ shortened positions respectively. The test gives no interval for the remaining rows, although this is not a general resistance claim. For ordinary Goppa, BMT [BMT24] does not apply as published: our base-field cardinalities are $q \in \{7, 8, 11, 16\}$ rather than $q \in \{2, 3\}$, and the degree-lowering conductor equality required by its generic-alternant filtration fails for Goppa codes. Although the full quadratic wild-Goppa parent codes lie in the parameter regime

targeted by COT [COT17], our public codes are proper subcodes. Under the direct COT-style shortening–Schur-square dimension criterion, only the $\mathbb{F}_9/\mathbb{F}_{81}$ row retains a square code defect; for the other three rows, releasing a subcode reduces the shortened dimensions enough that this criterion does not yield a distinguisher. Thus, among our wild-Goppa experiments, only the $\mathbb{F}_9/\mathbb{F}_{81}$ row appears directly exposed to the initial COT distinguisher. More sophisticated adaptations remain possible and are not ruled out for any of these cases.

Family	Base/Ambient	(n, k, D)	t	(d, s, w)
GRS	$\mathbb{F}_{64}/\mathbb{F}_{64}$	$(56, 19, 29)$	–	$(3, 2, 3)$
GRS	$\mathbb{F}_{67}/\mathbb{F}_{67}$	$(62, 14, 34)$	–	$(5, 3, 4)$
Alternant	$\mathbb{F}_{16}/\mathbb{F}_{256}$	$(113, 25, 69)$	–	$(3, 2, 8)$
Alternant	$\mathbb{F}_{16}/\mathbb{F}_{256}$	$(71, 13, 41)$	–	$(5, 3, 1)$
Ordinary Goppa	$\mathbb{F}_{11}/\mathbb{F}_{121}$	$(109, 23, 65)$	43	$(3, 2, 10)$
Ordinary Goppa	$\mathbb{F}_7/\mathbb{F}_{49}$	$(46, 20, 32)$	13	$(4, 3, 2)$
Ordinary Goppa	$\mathbb{F}_{16}/\mathbb{F}_{256}$	$(109, 23, 65)$	43	$(3, 2, 10)$
Ordinary Goppa	$\mathbb{F}_8/\mathbb{F}_{64}$	$(46, 20, 32)$	13	$(4, 3, 2)$
Wild Goppa	$\mathbb{F}_{11}/\mathbb{F}_{121}$	$(108, 25, 63)$	4	$(3, 2, 12)$
Wild Goppa	$\mathbb{F}_9/\mathbb{F}_{81}$	$(62, 14, 34)$	3	$(5, 3, 4)$
Wild Goppa	$\mathbb{F}_{16}/\mathbb{F}_{256}$	$(125, 27, 73)$	3	$(3, 2, 12)$
Wild Goppa	$\mathbb{F}_8/\mathbb{F}_{64}$	$(62, 14, 34)$	3	$(5, 3, 4)$

Table 8: End-to-end decryption experiments. The t-column gives the degree of the defining Goppa polynomial and applies only to the Goppa-family rows. Each parameter choice was tested for at least twenty independent trials; every trial rejected all dirty coordinates and successfully recovered the message. The runs used random Boolean errors of weight w, in line with the binary Goppa setting. For wild Goppa codes, t is the degree of the irreducible polynomial γ; the code itself uses the polynomial γ^{q-1}. Here D is the degree of the smallest ambient GRS code containing each code, k is the dimension of the code, and n is the block length.

B Definitions of Goppa, Wild Goppa, and Alternant Codes

Let q be a prime power, let $m \geqslant 1$, and let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n) \in \mathbb{F}_{q^m}^n$ have pairwise distinct entries. Throughout, $C^\perp$ denotes the dual of $C \subseteq \mathbb{F}_{q^m}^n$ over $\mathbb{F}_{q^m}$. Here q denotes the base-field size and q^m denotes the ambient-field size, whereas Section 7 uses q_0 and q, respectively.

Alternant codes. For $1 \leqslant r < n$ and $\boldsymbol{\mu} \in (\mathbb{F}_{q^m}^\times)^n$, the q-ary alternant code of degree r, support $\boldsymbol{\alpha}$, and multipliers $\boldsymbol{\mu}$ is

$$\mathsf{Alt}_r(\boldsymbol{\alpha}, \boldsymbol{\mu}) := \mathsf{GRS}_r(\boldsymbol{\alpha}, \boldsymbol{\mu})^\perp \cap \mathbb{F}_q^n.$$

Equivalently, $\boldsymbol{c} \in \mathsf{Alt}_r$ if and only if

$$\sum_{i=1}^{n} c_i \mu_i \alpha_i^j = 0 \quad \text{for every } 0 \leqslant j < r.$$

It satisfies $\dim_{\mathbb{F}_q} \mathsf{Alt}_r \geqslant n - mr$ and $d_{\min}(\mathsf{Alt}_r) \geqslant r + 1$.

Goppa codes. Let $\Gamma(Z) \in \mathbb{F}_{q^m}[Z]$ have degree t and satisfy $\Gamma(\alpha_i) \neq 0$ for every i, and set

$$\boldsymbol{\mu}_\Gamma := \big(\Gamma(\alpha_1)^{-1}, \ldots, \Gamma(\alpha_n)^{-1}\big).$$

The associated q-ary Goppa code is

$$\mathsf{Goppa}_q(\boldsymbol{\alpha}, \Gamma) := \mathsf{GRS}_t(\boldsymbol{\alpha}, \boldsymbol{\mu}_\Gamma)^\perp \cap \mathbb{F}_q^n = \mathsf{Alt}_t(\boldsymbol{\alpha}, \boldsymbol{\mu}_\Gamma).$$

Equivalently,

$$\mathsf{Goppa}_q(\boldsymbol{\alpha}, \Gamma) = \left\{ \boldsymbol{c} \in \mathbb{F}_q^n : \sum_{i=1}^{n} \frac{c_i}{Z - \alpha_i} \equiv 0 \pmod{\Gamma(Z)} \right\}.$$

If Γ is irreducible over $\mathbb{F}_{q^m}$, this is called an irreducible q-ary Goppa code. In all cases,

$$\dim_{\mathbb{F}_q} \mathsf{Goppa}_q(\boldsymbol{\alpha}, \Gamma) \geqslant n - mt, \qquad d_{\min}\big(\mathsf{Goppa}_q(\boldsymbol{\alpha}, \Gamma)\big) \geqslant t + 1.$$

For $q = 2$ we suppress the subscript q, as in the main text.

Wild Goppa codes. Let $\gamma(Z) \in \mathbb{F}_{q^m}[Z]$ be square-free of degree t and satisfy $\gamma(\alpha_i) \neq 0$ for every i. The associated wild Goppa code is

$$\mathsf{WildGoppa}_q(\boldsymbol{\alpha}, \gamma) := \mathsf{Goppa}_q(\boldsymbol{\alpha}, \gamma^{q-1}) = \mathsf{Goppa}_q(\boldsymbol{\alpha}, \gamma^q).$$

Here the exponents denote ordinary polynomial powers; thus t is the degree of the seed polynomial γ, not of γ^{q-1}. Consequently,

$$\dim_{\mathbb{F}_q} \mathsf{WildGoppa}_q(\boldsymbol{\alpha}, \gamma) \geqslant n - m(q-1)t, \qquad d_{\min}\big(\mathsf{WildGoppa}_q(\boldsymbol{\alpha}, \gamma)\big) \geqslant qt + 1.$$

For $q = 2$, this recovers $\mathsf{Goppa}(\boldsymbol{\alpha}, \gamma) = \mathsf{Goppa}(\boldsymbol{\alpha}, \gamma^2)$ and $d_{\min} \geqslant 2t + 1$.

If $m = 2$ and γ has no root in $\mathbb{F}_{q^2}$, then also

$$\mathsf{WildGoppa}_q(\boldsymbol{\alpha}, \gamma) = \mathsf{Goppa}_q(\boldsymbol{\alpha}, \gamma^{q+1}),$$

which gives $d_{\min} \geqslant (q+1)t + 1$.

C Evidence for Key-Recovery Heuristic

In this appendix, we discuss the evidence we have in support of our key-recovery heuristic.

C.1 Structure of the Canonical Solutions

We start by analyzing the canonical solutions arising from a fixed ambient GRS representation. The following calculation depends only on this ambient representation, and therefore applies equally to GRS codes and to alternant codes after extension of scalars, viewed inside an ambient GRS code. Throughout this subsection, let $\mathbb{F}_q$ have characteristic two and assume the no-slack condition $\sum_{j \neq \tau} s_j = dD + 1$ for every $\tau \in [h]$. As in Algorithm 4, write $s_\tau = s$ for $\tau \in [h]$, and set $r := s - 1$, where r is a power of two. Fix an ambient GRS representation and absorb the hidden permutation into its support and multipliers. Thus, $\boldsymbol{y}_j = \lambda_j \boldsymbol{p}(\alpha_j)$, where $\boldsymbol{p}(Z) := \big(f_1(Z), \ldots, f_k(Z)\big)$, the α_j's are distinct, $\lambda_j \in \mathbb{F}_q^\times$, and $\deg(f_i) \leqslant D$ for every $i \in [k]$.

Univariate canonical coefficients. For every $j \in [n]$, define $P_j(Z) := \prod_{i \neq j}(Z - \alpha_i)^{s_i}$. Fix a hold-out $\tau \in [h]$. By the no-slack condition, the order- $< s_j$ Hasse derivatives at the points $\{\alpha_j : j \neq \tau\}$ uniquely determine every polynomial of degree at most dD. The explicit Hermite interpolation formula, written using the Hasse derivatives from Lemma 2.8, gives

$$\phi(\alpha_\tau) = \sum_{j \neq \tau} \sum_{u=0}^{s_j - 1} \eta_{\tau,j,u} \big(\partial^{[u]} \phi\big)(\alpha_j), \tag{24}$$

for every $\phi \in \mathbb{F}_q[Z]$ of degree at most dD, where

$$\eta_{\tau,j,u} = P_\tau(\alpha_\tau)[T^{s_j - 1 - u}] \frac{(\alpha_j + T - \alpha_\tau)^r}{P_j(\alpha_j + T)}. \tag{25}$$

This is the Hasse-derivative form of the standard explicit Hermite coefficients [Spi60].

It will be convenient to include the hold-out block in the coefficient vector. Set $\eta_{\tau,\tau,0} := 1$ and $\eta_{\tau,\tau,u} := 0$ for $1 \leqslant u < s$. Since $\mathbb{F}_q$ has characteristic two, (24) then becomes

$$\sum_{j=1}^{n} \sum_{u=0}^{s_j - 1} \eta_{\tau,j,u} \big(\partial^{[u]} \phi\big)(\alpha_j) = 0.$$

The formula in (25) remains valid for $j = \tau$ under this convention.

To separate the dependence on the hold-out, expand the reciprocals below in $\mathbb{F}_q[T]$, and define

$$b_{j,u} := [T^{s_j - 1 - u}] \frac{1}{P_j(\alpha_j + T)},$$
$$\bar{b}_{j,u} := [T^{s_j - 1 - u - r}] \frac{1}{P_j(\alpha_j + T)},$$

where a coefficient with a negative exponent is zero. Since r is a power of two and $\mathbb{F}_q$ has characteristic two, we have $(\alpha_j + T - \alpha_\tau)^r = \alpha_\tau^r + \alpha_j^r + T^r$. Consequently, for every $j \in [n]$ and $0 \leqslant u < s_j$, we have

$$\eta_{\tau,j,u} = P_\tau(\alpha_\tau) \left(\alpha_\tau^r b_{j,u} + \alpha_j^r b_{j,u} + \bar{b}_{j,u} \right). \tag{26}$$

In particular, if $u \geqslant s_j - s + 1$, then $s_j - 1 - u - r < 0$, and hence $\eta_{\tau,j,u} = P_\tau(\alpha_\tau)(\alpha_\tau - \alpha_j)^r b_{j,u}$.

From univariate to multivariate coefficients. Let $H \in \mathcal{P}^{\mathrm{hom}}_{=d}$, and define $\phi_H(Z) := H\big(\boldsymbol{p}(Z)\big)$. Then $\deg(\phi_H) \leqslant dD$, so (24) applies to ϕ_H. For $j \in [n]$, $u \geqslant 0$, and $\boldsymbol{a} \in \mathcal{A}_{<u+1,k}$, define the chain-rule coefficient

$$\chi_{j,u,\boldsymbol{a}} := [T^u] \prod_{i=1}^{k} \left(\sum_{v \geqslant 1} \big(\partial^{[v]} f_i\big)(\alpha_j) T^v \right)^{a_i}.$$

The Hasse chain rule gives

$$\big(\partial^{[u]} \phi_H\big)(\alpha_j) = \sum_{\boldsymbol{a} \in \mathcal{A}_{<u+1,k}} \chi_{j,u,\boldsymbol{a}} \big(\partial^{[\boldsymbol{a}]} H\big)\big(\boldsymbol{p}(\alpha_j)\big). \tag{27}$$

Moreover, homogeneity and Lemma 3.1 imply that $H(\boldsymbol{y}_\tau) = \lambda_\tau^d H\big(\boldsymbol{p}(\alpha_\tau)\big)$ and

$$\big(\partial^{[\boldsymbol{a}]}H\big)\big(\boldsymbol{p}(\alpha_j)\big) = \lambda_j^{|\boldsymbol{a}|-d}\big(\partial^{[\boldsymbol{a}]}H\big)(\boldsymbol{y}_j).$$

Substituting (27) into the univariate relation therefore transforms the univariate coefficients into the multivariate coefficients

$$c^\star_{\tau,j,\boldsymbol{a}} := \lambda_\tau^d \lambda_j^{|\boldsymbol{a}|-d} \sum_{u=|\boldsymbol{a}|}^{s_j-1} \eta_{\tau,j,u}\chi_{j,u,\boldsymbol{a}}, \qquad (j,\boldsymbol{a}) \in \mathcal{J}_{\boldsymbol{s}}. \tag{28}$$

For $j \neq \tau$, these coefficients satisfy

$$H(\boldsymbol{y}_\tau) = \sum_{j\neq\tau} \sum_{\boldsymbol{a}\in\mathcal{A}_{<s_j,k}} c^\star_{\tau,j,\boldsymbol{a}}\big(\partial^{[\boldsymbol{a}]}H\big)(\boldsymbol{y}_j).$$

The diagonal convention above gives $c^\star_{\tau,\tau,\boldsymbol{0}} = 1$ and $c^\star_{\tau,\tau,\boldsymbol{a}} = 0$ for $\boldsymbol{a} \in \mathcal{A}_{<s,k} \setminus \{\boldsymbol{0}\}$. Hence the full vector $\boldsymbol{c}^\star_\tau := \big(c^\star_{\tau,j,\boldsymbol{a}}\big)_{(j,\boldsymbol{a})\in\mathcal{J}_{\boldsymbol{s}}}$ belongs to $\ker(\boldsymbol{E})$.

The separated form in (26) is preserved by this transformation. Define

$$\begin{aligned} A_\tau &:= \lambda_\tau^d P_\tau(\alpha_\tau), \\ V^\star_{1,(j,\boldsymbol{a})} &:= \lambda_j^{|\boldsymbol{a}|-d} \sum_{u=|\boldsymbol{a}|}^{s_j-1} b_{j,u}\chi_{j,u,\boldsymbol{a}}, \\ V^\star_{2,(j,\boldsymbol{a})} &:= \lambda_j^{|\boldsymbol{a}|-d} \sum_{u=|\boldsymbol{a}|}^{s_j-1} \big(\alpha_j^r b_{j,u} + \bar{b}_{j,u}\big)\chi_{j,u,\boldsymbol{a}}. \end{aligned}$$

Then (28) becomes

$$\boldsymbol{c}^\star_\tau = A_\tau\left(\alpha_\tau^r \boldsymbol{V}_1^\star + \boldsymbol{V}_2^\star\right). \tag{29}$$

For a locator-block index $\boldsymbol{a} \in \mathcal{B}_j$, every summand has $u \geqslant |\boldsymbol{a}| \geqslant s_j - s + 1$, so $\bar{b}_{j,u} = 0$. Thus,

$$V^\star_{2,(j,\boldsymbol{a})} = \alpha_j^{s-1} V^\star_{1,(j,\boldsymbol{a})} \qquad \forall\, j \in [n],\ \boldsymbol{a} \in \mathcal{B}_j. \tag{30}$$

Lemma C.1 (Structure of the ambient-GRS canonical solutions)**.** *Let $\boldsymbol{U}^\star \in \mathbb{F}_q^{h\times 2}$ have rows $\boldsymbol{u}^\star_\tau := A_\tau(\alpha_\tau^{s-1}, 1)$ for $\tau \in [h]$, and let*

$$\boldsymbol{V}^\star := \begin{pmatrix} \boldsymbol{V}_1^\star \\ \boldsymbol{V}_2^\star \end{pmatrix}.$$

Then $\mathsf{rank}(\boldsymbol{U}^\star) = 2$, both rows of $\boldsymbol{V}^\star$ belong to $\ker(\boldsymbol{E})$, and $\boldsymbol{u}^\star_\tau \boldsymbol{V}^\star = \boldsymbol{c}^\star_\tau$ for every $\tau \in [h]$. Moreover, $\boldsymbol{V}^\star \in \mathcal{V}_{\boldsymbol{U}^\star}$ and $\alpha_j \in \mathcal{C}_j(\boldsymbol{U}^\star)$ for every $j \in [n]$.

Proof. Eq. (29) gives $\boldsymbol{u}^\star_\tau \boldsymbol{V}^\star = \boldsymbol{c}^\star_\tau$. Each A_τ is nonzero, and the map $x \mapsto x^{s-1}$ is a Frobenius automorphism of $\mathbb{F}_q$. Since the support entries are distinct, any two rows of $\boldsymbol{U}^\star$ are linearly independent. Thus, $\mathsf{rank}(\boldsymbol{U}^\star) = 2$. Since every $\boldsymbol{c}^\star_\tau$ belongs to $\ker(\boldsymbol{E})$, two of these independent combinations show that $\boldsymbol{V}_1^\star, \boldsymbol{V}_2^\star \in \ker(\boldsymbol{E})$.

The diagonal entries of $\boldsymbol{c}^\star_\tau$ give the hold-out constraints and the normalization in the definition of $\mathcal{V}_{\boldsymbol{U}^\star}$. Therefore, $\boldsymbol{V}^\star \in \mathcal{V}_{\boldsymbol{U}^\star}$. Finally, (30) shows that $\boldsymbol{V}^\star$ witnesses $\alpha_j \in \mathcal{C}_j(\boldsymbol{U}^\star)$ for every $j \in [n]$. □

C.1.1 Binary Goppa Canonical Form

Fix an ambient GRS representation of the binary Goppa code and absorb the hidden permutation into the coordinate ordering. As above, write $\boldsymbol{y}_j = \lambda_j \boldsymbol{p}(\alpha_j)$ for every $j \in [n]$, where the support entries $\alpha_j \in \mathbb{F}_q$ are distinct, the multipliers satisfy $\lambda_j \in \mathbb{F}_q^\times$, and $\boldsymbol{p}(Z) := (f_1(Z), \dots, f_k(Z))$ packages the ambient GRS row polynomials $f_i \in \mathbb{F}_q[Z]_{\leqslant D}$. Thus, for $H \in \mathcal{P}^{\mathrm{hom}}_{=d}$, the associated univariate polynomial is $\phi_H(Z) := H(f_1(Z), \dots, f_k(Z)) = H\big(\boldsymbol{p}(Z)\big)$. Define the resulting composition space by

$$\mathcal{R}_d(\boldsymbol{p}) := \left\{ H\big(\boldsymbol{p}(Z)\big) : H \in \mathcal{P}^{\mathrm{hom}}_{=d} \right\}.$$

For binary Goppa codes, $\mathcal{R}_d(\boldsymbol{p})$ is expected to have codimension t in $\mathbb{F}_q[Z]_{\leqslant dD}$. Equivalently, its expected dimension is $dD + 1 - t$. A similar dimension measurement for the bivariate composition space used in decryption appears in Appendix A.4; the space considered here is its univariate analogue.

For every hold-out $\tau \in [h]$, consider the non-hold-out jet map

$$\mathcal{R}_d(\boldsymbol{p}) \longrightarrow \bigoplus_{j \neq \tau} \mathbb{F}_q^{s_j}, \qquad \phi \longmapsto \Big(\big(\partial^{[u]}\phi\big)(\alpha_j) \Big)_{\substack{j \neq \tau \\ 0 \leqslant u < s_j}}. \tag{31}$$

The no-slack condition (10) makes the target dimension $dD + 1 - t$. We expect, with probability $1 - o(1)$, that $\dim_{\mathbb{F}_q} \mathcal{R}_d(\boldsymbol{p}) = dD + 1 - t$ and every map above is an isomorphism. These auxiliary predictions provide a convenient way to construct candidate relations in the experiments below. They do not prove the multivariate structure required by recovery, which we state directly.

Conjecture C.2 (Binary Goppa canonical form)**.** *Let $q = 2^m$, let $s > 2$, and suppose that $s - 1$ is a power of two. Consider an asymptotic family of square-free binary Goppa instances as in Definition 2.2, sampled according to $\mathcal{D}_{\mathsf{Goppa}}$ from Section 4. For each instance, take the associated ambient GRS representation of degree $D = n - 2t - 1$. Let the multiplicities be as in Algorithm 4 and satisfy (10). With probability $1 - o(1)$ over the sampled instance, there exist multivariate coefficient vectors*

$$\boldsymbol{c}^\star_\tau = \big(c^\star_{\tau,j,\boldsymbol{a}}\big)_{(j,\boldsymbol{a}) \in \mathcal{J}_{\boldsymbol{s}}} \in \ker(\boldsymbol{E}), \qquad \tau \in [h],$$

satisfying the diagonal convention

$$c^\star_{\tau,\tau,\mathbf{0}} = 1, \qquad c^\star_{\tau,\tau,\boldsymbol{a}} = 0 \quad \forall\, \tau \in [h],\ \boldsymbol{a} \in \mathcal{A}_{<s,k} \setminus \{\mathbf{0}\}.$$

Stack these vectors as the rows of $\boldsymbol{C}^\star$. There exist $A_\tau \in \mathbb{F}_q^\times$ and $\boldsymbol{V}_1^\star, \boldsymbol{V}_2^\star \in \mathbb{F}_q^M$ such that, for

$$\boldsymbol{U}^\star := \big(A_\tau(\alpha_\tau^{s-1}, 1)\big)_{\tau \in [h]}, \qquad \boldsymbol{V}^\star := \begin{pmatrix} \boldsymbol{V}_1^\star \\ \boldsymbol{V}_2^\star \end{pmatrix},$$

we have

$$\boldsymbol{C}^\star = \boldsymbol{U}^\star \boldsymbol{V}^\star, \qquad \mathsf{rank}(\boldsymbol{C}^\star) = 2. \tag{32}$$

For the locator blocks, the right factor satisfies

$$V^\star_{2,(j,\boldsymbol{a})} = \alpha_j^{s-1} V^\star_{1,(j,\boldsymbol{a})} \quad \forall\, j \in [n],\ \boldsymbol{a} \in \mathcal{B}_j. \tag{33}$$

Experimental evidence. The conjecture concerns the multivariate coefficients. However, (28) gives an explicit transformation from the univariate interpolation coefficients to the multivariate ones. Thus, it is enough to solve the univariate interpolation systems and then check the conjectured properties after this transformation. For each sampled binary Goppa instance, we used its known ambient row polynomials $\boldsymbol{p} = (f_1, \dots, f_k)$ and row-reduced the coefficient vectors of $H(\boldsymbol{p}(Z))$, as H ranges over the monomial basis of $\mathcal{P}^{\mathrm{hom}}_{=d}$. This gives a basis $\phi_1, \dots, \phi_R$ of the composition space $\mathcal{R}_d(\boldsymbol{p})$ and its exact dimension R. In every reported parameter set we obtained $R = dD + 1 - t$.

We then chose the displayed multiplicity profile so that the modified no-slack condition (10) gives $\sum_{j \neq \tau} s_j = R$ for each hold-out $\tau \in [h]$. With respect to the above basis, the map in (31) is the square $R \times R$ matrix

$$\boldsymbol{J}_\tau := \Big((\partial^{[u]} \phi_\ell)(\alpha_j) \Big)_{\substack{j \neq \tau,\ 0 \leqslant u < s_j \\ \ell \in [R]}} .$$

Thus, the auxiliary isomorphism prediction is tested simply by checking whether $\boldsymbol{J}_\tau$ has rank R. Whenever it did, we solved the square system

$$\phi_\ell(\alpha_\tau) = \sum_{j \neq \tau} \sum_{u=0}^{s_j - 1} \eta^\star_{\tau,j,u} (\partial^{[u]} \phi_\ell)(\alpha_j) \qquad \forall\, \ell \in [R],$$

which is (24) restricted to $\mathcal{R}_d(\boldsymbol{p})$. For each invertible trial, we applied the diagonal convention and (28) to form the full matrix $\boldsymbol{C}^\star$, and checked that it had rank two as required by (32).

For each rank-two matrix, we checked that its column space admitted the canonical left factor $\boldsymbol{U}^\star$ in (32). We then recovered the corresponding right factor $\boldsymbol{V}^\star$ and verified the locator-block identity (33). Neither condition was imposed when solving the interpolation systems.

(m, n, t, k, D, d, s, h)	multiplicity profile	$\dim \mathcal{R}_d$/constraints	invertible maps	rank-two $\boldsymbol{C}^\star$	top-block form
$(6, 53, 6, 17, 40, 4, 3, 4)$	$2^1\, 3^{52}$	155/155	10/10	10/10	10/10
$(5, 24, 2, 14, 19, 6, 5, 4)$	$4^2\, 5^{22}$	113/113	10/10	10/10	10/10
$(6, 58, 4, 34, 49, 8, 5, 4)$	$5^4\, 6^4\, 7^{50}$	389/389	10/10	10/10	10/10
$(4, 13, 1, 9, 10, 11, 9, 4)$	$9^{11}\, 10^2$	110/110	9/10	9/9	9/9
$(7, 107, 10, 37, 86, 5, 3, 4)$	$3^4\, 4^{103}$	421/421	10/10	10/10	10/10

Table 9: Binary Goppa canonical-form experiments. For each trial, the support $\boldsymbol{\alpha}$ was sampled uniformly among valid supports, and the Goppa polynomial was a randomly sampled irreducible polynomial of degree t. In a multiplicity profile, a^b denotes b coordinates of multiplicity a. The third column compares $\dim_{\mathbb{F}_q} \mathcal{R}_d(\boldsymbol{p})$ with the number of non-hold-out jet constraints. The fourth column gives the number of trials in which the non-hold-out jet maps used to construct $\boldsymbol{C}^\star$ were invertible, out of ten trials per parameter set. The fifth column gives the number of invertible trials in which $\boldsymbol{C}^\star$ had rank two, and the last column records the canonical factorization and locator-block form in (32) and (33). All rows use $h = 4$. In total, 49 of the 50 trials had invertible reconstruction maps, and both properties held in all 49 cases. The remaining map had rank deficiency one, so it did not select unique normalized coefficients and was inconclusive for the conjecture.

The rank-two canonical factorization and locator-block form hold across all five parameter sets and every invertible trial, providing direct experimental support for Conjecture C.2. Although the non-hold-out jet map is not always an isomorphism, we expect it to be singular with probability approximately $1/q$, so this failure mode does not matter as q grows. To test this expectation, we

varied m, and consequently k, in the second and fourth parameter settings while keeping their other parameters and multiplicity profiles fixed. For the second setting, the singular rates among 8000 maps were 3.25%, 1.575%, and 0.988% over $\mathbb{F}_{32}$, $\mathbb{F}_{64}$, and $\mathbb{F}_{128}$, respectively; for the fourth setting, they were 8.09% and 3.88% over $\mathbb{F}_{16}$ and $\mathbb{F}_{32}$, respectively.

C.2 Structure of the Kernel

Throughout this subsection, $\mathbb{F}_q$ has characteristic two. We next describe the relations that are expected to span $\ker(\boldsymbol{E})$. A vector $\boldsymbol{c} = (c_{j,\boldsymbol{a}})_{(j,\boldsymbol{a})\in\mathcal{J}_{\boldsymbol{s}}} \in \mathbb{F}_q^M$ belongs to $\ker(\boldsymbol{E})$ precisely when

$$\sum_{j=1}^{n} \sum_{\boldsymbol{a}\in\mathcal{A}_{<s_j,k}} c_{j,\boldsymbol{a}} \big(\partial^{[\boldsymbol{a}]} H\big)(\boldsymbol{y}_j) = 0$$

for every $H \in \mathcal{P}^{\mathrm{hom}}_{=d}$. We first identify the relations already present for random public columns and then add the global relations forced by each structured code family.

Random public matrices. We begin with the random baseline; the nullity contributed by algebraic structure is then measured as the excess over it. Let $\boldsymbol{E}_{\mathrm{rnd}}$ be the jet matrix of a uniformly random full-row-rank public matrix over $\mathbb{F}_q$, with the same field, dimensions, and multiplicity profile as $\boldsymbol{Y}$. For a relation $\boldsymbol{c} \in \ker(\boldsymbol{E}_{\mathrm{rnd}})$, its block support

$$\mathrm{supp}_{\mathrm{blk}}(\boldsymbol{c}) := \big\{\, j \in [n] : c_{j,\boldsymbol{a}} \neq 0 \text{ for some } \boldsymbol{a} \in \mathcal{A}_{<s_j,k} \,\big\}$$

records which coordinate blocks it involves, and we let $\mathcal{K}_{\mathrm{loc}}$ be the span of the relations with $\sum_{j\in\mathrm{supp}_{\mathrm{blk}}(\boldsymbol{c})}(d - s_j + 1) \leqslant d$. These are the relations confined to a single jet block—the one-block projective relations—together with the few-block relations arising when the derivative-evaluation vectors are linearly dependent. Both occur for arbitrary column points and carry no information about the hidden support. We now argue that for a random matrix the kernel is exactly $\mathcal{K}_{\mathrm{loc}}$.

Its dimension reduces to a single nonelementary quantity. Since $\boldsymbol{E}_{\mathrm{rnd}}$ has N_d rows and M columns, rank–nullity gives the exact identity

$$\dim\ker(\boldsymbol{E}_{\mathrm{rnd}}) = M - N_d + \dim\ker(\boldsymbol{E}_{\mathrm{rnd}}^{\top}),$$

whose only nontrivial term $\dim\ker(\boldsymbol{E}_{\mathrm{rnd}}^{\top})$ is the number of degree-d forms whose order-s_j jet vanishes at every column $\boldsymbol{y}_j$—equivalently, the forms singular to order s_j at each of the n points. For points in general position this dimension is governed by the conjectural estimate from [Frö85, Iar97, Cha05]. Writing $[\cdot]_+$ for the truncation that zeroes every coefficient from the first nonpositive one onward, the predicted value and the resulting nullity are

$$\begin{aligned} \mathrm{HF}_{\mathrm{rnd}}(d) &:= [z^d] \left[\frac{\prod_{j=1}^{n} \big(1 - z^{\,d-s_j+1}\big)}{(1-z)^k} \right]_+, \\ \kappa_{\mathrm{rnd}} &:= M - N_d + \mathrm{HF}_{\mathrm{rnd}}(d). \end{aligned} \tag{34}$$

This is obtained by replacing $\dim\ker(\boldsymbol{E}_{\mathrm{rnd}}^{\top})$ in the above identity by $\mathrm{HF}_{\mathrm{rnd}}(d)$, which is the conjectural step.

The closed form exhibits κ_{rnd} as a count of local relations. For the uniform profile $s_j = s$, write $N_u := \binom{k+u-1}{u}$, $L_s := \binom{k+s-1}{s-1}$, and $\ell_{\max} := \lfloor d/(d-s+1) \rfloor$. Expanding (34) when the truncation does not act before degree d, we get

$$\kappa_{\mathrm{rnd}} = n\big(L_s - N_{s-1}\big) + \sum_{\ell=2}^{\ell_{\max}} (-1)^{\ell} \binom{n}{\ell} N_{d-\ell(d-s+1)}. \tag{35}$$

The leading term is the $L_s - N_{s-1}$ one-block relations carried by each of the n blocks. The term indexed by ℓ is the inclusion–exclusion contribution of relations shared by ℓ blocks, and the sum terminates at $\ell_{\max}$ because no local relation can span more than $\lfloor d/(d-s+1) \rfloor$ blocks. We therefore predict

$$\ker(\boldsymbol{E}_{\mathrm{rnd}}) = \mathcal{K}_{\mathrm{loc}}, \qquad \dim \ker(\boldsymbol{E}_{\mathrm{rnd}}) = \kappa_{\mathrm{rnd}},$$

and the random matrices in Table 10 attain this value in every reported instance.

Generic GRS codes. We next count the global relations contributed by one ambient GRS parameterization. Under (9), the complete multiplicity profile satisfies $\sum_{j=1}^{n} s_j = dD + s + 1$. Restricting the jet functionals to $\mathbb{F}_q[Z]_{\leqslant dD}$ therefore produces an s-dimensional univariate relation space. Lifting these relations through the Hasse chain rule gives an s-dimensional canonical subspace $\mathcal{H}_{\boldsymbol{\alpha}} \subseteq \ker(\boldsymbol{E})$. Consequently, we predict

$$\ker(\boldsymbol{E}) = \mathcal{K}_{\mathrm{loc}} \oplus \mathcal{H}_{\boldsymbol{\alpha}}, \qquad \dim \ker(\boldsymbol{E}) = \kappa_{\mathrm{rnd}} + s. \tag{36}$$

The two GRS instances in Table 10 have observed excess nullity $s = 3$.

Generic alternant codes. Let $\mathbb{F}_{q_0} \subseteq \mathbb{F}_q$ be the base field and let $m = [\mathbb{F}_q : \mathbb{F}_{q_0}]$. After extension of scalars to $\mathbb{F}_q$, a generic alternant code is contained in m Frobenius-conjugate GRS codes. For $0 \leqslant a < m$, let $\mathcal{H}^{(a)}$ be the s-dimensional canonical relation space of the ambient GRS representation with support $\boldsymbol{\alpha}^{q_0^a}$. Assuming that these spaces are independent modulo $\mathcal{K}_{\mathrm{loc}}$, we predict

$$\ker(\boldsymbol{E}) = \mathcal{K}_{\mathrm{loc}} \oplus \bigoplus_{a=0}^{m-1} \mathcal{H}^{(a)}, \qquad \dim \ker(\boldsymbol{E}) = \kappa_{\mathrm{rnd}} + ms. \tag{37}$$

Both alternant instances in Table 10 have observed excess nullity $6 = ms$.

Goppa and wild Goppa codes. The Goppa families have the same m Frobenius-conjugate canonical spaces as alternant codes, together with $m - 1$ additional relations. We derive these additional relations first for a square-free binary Goppa code with monic Goppa polynomial Γ of degree t and ambient degree $D = n - 2t - 1$. Set $\Pi_{\boldsymbol{\alpha}}(Z) := \prod_{j=1}^{n}(Z - \alpha_j)$. For the ambient parameterization $\boldsymbol{p}(Z) = (f_1(Z), \ldots, f_k(Z))$, the multiplier identity in Fact 2.3 gives

$$\Gamma(Z)^2 f_i(Z) = \sum_{j=1}^{n} Y_{i,j} \frac{\Pi_{\boldsymbol{\alpha}}(Z)}{Z - \alpha_j}. \tag{38}$$

Since Γ is monic and $\deg(f_i) \leqslant D$, comparison of the coefficients of Z^{n-1} in (38) gives $[Z^D] f_i = \sum_{j=1}^{n} Y_{i,j} \in \mathbb{F}_2$. The point at infinity of this parameterization is therefore represented by $\boldsymbol{p}_\infty :=$

$\big([Z^D]f_1, \ldots, [Z^D]f_k\big) \in \mathbb{F}_2^k$. In particular, $\boldsymbol{p}_\infty^{2^a} = \boldsymbol{p}_\infty$ for every $0 \leqslant a < m$, so all m Frobenius-conjugate parameterizations have the same point at infinity. The analogous leading-coefficient point for the Goppa and wild Goppa families over $\mathbb{F}_{q_0}$ lies in $\mathbb{F}_{q_0}^k$ and is likewise fixed by the q_0-Frobenius action.

For each conjugate ambient GRS representation, interpolation expresses $H(\boldsymbol{p}_\infty)$ as a linear combination of the finite jet evaluations forming $\boldsymbol{E}$. Let $\boldsymbol{c}_\infty^{(a)} \in \mathbb{F}_q^M$ denote the coefficient vector for the a-th representation, so that

$$H(\boldsymbol{p}_\infty) = \sum_{(j,\boldsymbol{b}) \in \mathcal{J}_{\boldsymbol{s}}} c_{\infty,j,\boldsymbol{b}}^{(a)} \big(\partial^{[\boldsymbol{b}]} H\big)(\boldsymbol{y}_j)$$

for every $H \in \mathcal{P}_{=d}^{\mathrm{hom}}$. Subtracting the reconstruction for the original representation from the reconstruction for the a-th conjugate gives

$$\boldsymbol{E}\big(\boldsymbol{c}_\infty^{(a)} - \boldsymbol{c}_\infty^{(0)}\big)^\top = \boldsymbol{0}, \qquad 1 \leqslant a < m.$$

We expect these $m-1$ relations to be independent modulo $\mathcal{K}_{\mathrm{loc}}$ and the canonical spaces. Let $\mathcal{H}_\infty$ denote their span, for which we therefore predict $\dim \mathcal{H}_\infty = m-1$. This gives

$$\ker(\boldsymbol{E}) = \mathcal{K}_{\mathrm{loc}} \oplus \bigoplus_{a=0}^{m-1} \mathcal{H}^{(a)} \oplus \mathcal{H}_\infty, \qquad \dim\ker(\boldsymbol{E}) = \kappa_{\mathrm{rnd}} + ms + m - 1. \tag{39}$$

We cannot directly audit this nullity prediction for binary Goppa codes because the smallest instances satisfying the modified no-slack and kernel conditions already produce jet matrices beyond the range of our implementation. We instead test Goppa and wild Goppa codes over larger base fields $\mathbb{F}_{q_0}$. For these instances, the univariate composition space has full rank $dD+1$, so there is no t-dimensional rank deficiency and we use the original no-slack condition (9). These experiments audit the predicted Goppa-family excess $ms+m-1$, but they do not directly audit the modified binary Goppa no-slack condition. The Goppa row in Table 10 has excess $7 = ms+m-1$, and both wild Goppa rows have excess $11 = ms+m-1$.

Combined prediction and experiments. Combining the preceding counts gives the predicted nullity

$$\kappa_{\mathrm{pred}} := \kappa_{\mathrm{rnd}} + \begin{cases} 0, & \text{for a random public matrix,} \\ s, & \text{for a generic GRS code,} \\ ms, & \text{for a generic alternant code,} \\ ms+m-1, & \text{for a Goppa or wild Goppa code.} \end{cases} \tag{40}$$

family	(q, n, k, D, d)	multiplicity profile	random nullity	structured nullity	observed excess
GRS	$(64, 44, 22, 32, 4)$	3^{44}	1958	1961	$3 = s$
GRS	$(64, 64, 28, 47, 4)$	3^{64}	3872	3875	$3 = s$
$\mathbb{F}_8$-alternant, $m = 2$	$(64, 44, 22, 32, 4)$	3^{44}	1958	1964	$6 = ms$
$\mathbb{F}_{16}$-alternant, $m = 2$	$(256, 66, 14, 39, 5)$	$4^1\, 3^{65}$	1160	1166	$6 = ms$
$\mathbb{F}_8$-Goppa, $m = 2$	$(64, 44, 22, 32, 4)$	3^{44}	1958	1965	$7 = ms + m - 1$
$\mathbb{F}_4$-wild Goppa, $m = 3$	$(64, 63, 27, 46, 4)$	$2^1\, 3^{62}$	3628	3639	$11 = ms + m - 1$
$\mathbb{F}_4$-wild Goppa, $m = 3$	$(64, 64, 28, 47, 4)$	3^{64}	3872	3883	$11 = ms + m - 1$

Table 10: Experimental kernel dimensions. Each structured instance is paired with a random full-row-rank public matrix having the same ambient field, dimensions, degree, and multiplicity profile. All instances satisfy (9) and the kernel condition, and the experiments use $h = 4$ and $s = 3$. In the family column, $\mathbb{F}_{q_0}$ is the base field, while q in the parameter tuple is the ambient-field size and $m = [\mathbb{F}_q : \mathbb{F}_{q_0}]$. In a multiplicity profile, a^b denotes b coordinates of multiplicity a. The random-nullity column gives $\dim \ker(\boldsymbol{E}_{\mathrm{rnd}})$, the structured-nullity column gives $\dim \ker(\boldsymbol{E})$, and the last column gives their difference. The predicted differences are s, ms, and $ms + m - 1$ for GRS, alternant, and Goppa-family codes, respectively. For every structured instance, the exhaustive locator-block tests with the correct normalized branch returned a singleton at every coordinate.

The random nullity agrees with (34) in every row, and the structured excess agrees exactly with (40). Thus, the canonical spaces account for the entire observed excess, with no unpredicted global kernel dimension in these experiments. The singleton locator-block outcomes further indicate that the predicted extra nullity does not produce another accepted canonical branch.

From nullity to recovery. A correct nullity count does not by itself imply that the recovery algorithm succeeds, because the additional kernel relations could produce another rank-one locator block. We must therefore determine how many hold-outs are needed to remove the effect of the code-related nullity and verify that the remaining relations do not create another accepted locator.

For a GRS code, the canonical space $\mathcal{H}_{\boldsymbol{\alpha}}$ represents a single support, but we must still account for the projective freedom of its representation. Applying a fractional-linear transformation $x \mapsto (g_{11}x + g_{12})/(g_{21}x + g_{22})$ to every support point, together with the corresponding change of multipliers, gives an equivalent ambient GRS code. We interpret such a transformation on $\mathbb{F}_q \cup \{\infty\}$, sending a support point to ∞ when its denominator vanishes. The rank-one and cross-ratio arguments remain valid in this case, so this possibility is compatible with the present analysis. After recovering the transformed support, one may compose with another fractional-linear transformation whose unique input mapped to ∞ lies outside that support, thereby obtaining an equivalent representation in which every locator belongs to $\mathbb{F}_q$. Such a choice always exists because the support has $n \leqslant q$ points, whereas $\mathbb{F}_q \cup \{\infty\}$ has $q + 1$ points. Since a fractional-linear transformation is uniquely determined by the images of three distinct points, three hold-outs fix this freedom completely. Thus, for a generic GRS code, $h \geqslant 3$ is sufficient to select a normalized representative of its canonical solution.

For an alternant code, however, there are m canonical branches, represented by the canonical spaces associated with the Frobenius-conjugate supports. With only three hold-outs, each conjugate support admits its own fractional-linear transformation that maps those hold-outs to the chosen normalized values, so all m canonical branches may survive. A fourth hold-out imposes an invariant

condition that cannot be absorbed by these transformations. Indeed, the cross-ratio

$$\rho := \frac{(\alpha_1 - \alpha_3)(\alpha_2 - \alpha_4)}{(\alpha_1 - \alpha_4)(\alpha_2 - \alpha_3)}$$

is preserved by fractional-linear transformations, whereas the corresponding cross-ratio of the a-th conjugate support is $\rho^{q_0^a}$. A nontrivial conjugate branch can therefore survive the fourth hold-out only if $\rho^{q_0^a} = \rho$, which requires ρ to lie in the proper subfield fixed by that Frobenius automorphism. For a generic support this event has small probability, so the fourth hold-out is expected to leave only the target canonical branch. The same argument applies to the Frobenius-conjugate canonical spaces of Goppa-family codes.

For later use, let $\boldsymbol{\alpha}^{(a)}$ denote the support $\boldsymbol{\alpha}^{q_0^a}$ after the unique fractional-linear transformation that sends its first three hold-out locators to the chosen normalized values; write $\boldsymbol{\alpha}^{(0)} = \boldsymbol{\alpha}$. Thus, when only three hold-outs are fixed, $\boldsymbol{\alpha}^{(0)}, \dots, \boldsymbol{\alpha}^{(m-1)}$ are the normalized Frobenius-conjugate supports that remain compatible with the same choice of projective coordinates. For a GRS code, we use the convention $m = 1$.

It remains to bound the number local relations, which we expect to behave generically. We first estimate the dimension of their projection onto a locator block and then study how this projected space could produce a wrong locator. Let $\pi_j : \mathbb{F}_q^M \to \mathbb{F}_q^{|\mathcal{B}_j|}$ be the projection onto the j-th locator block, and write

$$\mathcal{R}_j^{\mathrm{loc}} := \pi_j(\mathcal{K}_{\mathrm{loc}}), \qquad r_j^{\mathrm{loc}} := \dim \mathcal{R}_j^{\mathrm{loc}}, \qquad b_j := |\mathcal{B}_j|.$$

To estimate this dimension, let $\boldsymbol{s}^{(j)}$ be obtained from $\boldsymbol{s}$ by replacing s_j with $s_j - s + 1$. This deletes exactly the columns in $\mathcal{B}_j$. Since the kernel of the reduced jet matrix is $\ker(\boldsymbol{E}_{\mathrm{rnd}}) \cap \ker(\pi_j)$, rank–nullity gives the prediction

$$\widehat{r}_j^{\mathrm{loc}} := \kappa_{\mathrm{rnd}}(\boldsymbol{s}) - \kappa_{\mathrm{rnd}}(\boldsymbol{s}^{(j)}) \tag{41}$$

under $\ker(\boldsymbol{E}_{\mathrm{rnd}}) = \mathcal{K}_{\mathrm{loc}}$. In the asymptotic regime of Corollary 4.2, it gives

$$\frac{\widehat{r}_j^{\mathrm{loc}}}{b_j} = O\left(\frac{s^2}{n}\right) = O\left(\frac{\log^2 n}{n}\right) = o(1).$$

We next consider the local freedom relevant to a wrong locator. Fix a canonical branch modulo $\mathcal{K}_{\mathrm{loc}}$, and let $\boldsymbol{V}^\star$, with rows $\boldsymbol{V}_1^\star$ and $\boldsymbol{V}_2^\star$, denote its canonical second factor as in Lemma C.1. Any remaining local perturbation has the form

$$\boldsymbol{V}_1 = \boldsymbol{V}_1^\star + \boldsymbol{\delta}_1, \qquad \boldsymbol{V}_2 = \boldsymbol{V}_2^\star + \boldsymbol{\delta}_2, \qquad \boldsymbol{\delta}_1, \boldsymbol{\delta}_2 \in \mathcal{K}_{\mathrm{loc}}.$$

Write $\boldsymbol{w}_j := \pi_j(\boldsymbol{V}_1^\star)$, so that $\pi_j(\boldsymbol{V}_2^\star) = \alpha_j^{s-1}\boldsymbol{w}_j$. If a finite candidate $x \neq \alpha_j$ passes the locator test, then

$$\pi_j(\boldsymbol{\delta}_2 - x^{s-1}\boldsymbol{\delta}_1) = \left(x^{s-1} - \alpha_j^{s-1}\right)\boldsymbol{w}_j. \tag{42}$$

The left-hand side lies in $\mathcal{R}_j^{\mathrm{loc}}$, because $\boldsymbol{\delta}_2 - x^{s-1}\boldsymbol{\delta}_1 \in \mathcal{K}_{\mathrm{loc}}$, while the scalar on the right is nonzero because $z \mapsto z^{s-1}$ is an automorphism of $\mathbb{F}_q$. Hence every wrong locator requires $\boldsymbol{w}_j \in \mathcal{R}_j^{\mathrm{loc}}$; the candidate $x = \infty$ gives the same condition.

For a nonzero canonical block $\boldsymbol{w}_j$ in generic position relative to an r_j^{loc}-dimensional subspace of $\mathbb{F}_q^{b_j}$, the probability of this event is

$$\frac{q^{r_j^{\mathrm{loc}}}-1}{q^{b_j}-1} = O\Big(q^{r_j^{\mathrm{loc}}-b_j}\Big).$$

Thus, with probability $1-O(q^{r_j^{\mathrm{loc}}-b_j})$, the generic local relations produce no wrong projective locator. For Goppa codes, we likewise expect the projected directions from $\mathcal{H}_\infty$ to be in generic position relative to the canonical locator block, and hence not to create another locator branch.

Conclusion of the nullity analysis. The calculations in Eqs. (34), (36), (37) and (39) lead to the prediction (40), and Table 10 matches that prediction in every reported instance. This nullity prediction explains the observed kernel sizes, but its correctness is not required for key recovery. Conditional correctness uses only the following assertion about rank-one locator blocks in the affine space belonging to the correct normalized matrix $\boldsymbol{U}^\star$.

Conjecture C.3 (Rank-one rigidity)**.** *Consider a generic GRS, alternant, or binary Goppa instance satisfying the corresponding no-slack condition, such that*

$$\mathsf{rank}(\boldsymbol{E}) < N_d, \qquad \dim \pi_j\big(\ker(\boldsymbol{E})\big) < |\mathcal{B}_j| \quad \forall\, j \in [n],$$

and let $\boldsymbol{U}^\star$ be the correct normalized rank-two matrix. With probability $1-o(1)$, the following hold for every $j \in [n]$:

1. *If $h = 3$, then*
$$\mathcal{C}_j(\boldsymbol{U}^\star) = \Big\{\alpha_j^{(a)} : 0 \leqslant a < m\Big\}.$$
Equivalently, only the target locator and its normalized Frobenius conjugates occur as row ratios of rank-one blocks $\boldsymbol{W}_j(\boldsymbol{V})$ with $\boldsymbol{V} \in \mathcal{V}_{\boldsymbol{U}^\star}$.
2. *If $h = 4$, then*
$$\mathcal{C}_j(\boldsymbol{U}^\star) = \{\alpha_j\}.$$
Equivalently, only the target locator occurs as the row ratio of such a rank-one block.

Implicit recovery test for binary Goppa codes. One can inspect that running our recovery algorithm even on small binary Goppa instances is infeasible: the smallest instances satisfying the modified no-slack condition (10) and the kernel conditions already require the kernel of a jet matrix with 749,398 rows and 1,020,604 columns. We therefore propose a different test that runs the recovery in a controlled setting where the kernel vectors are generated *implicitly* rather than computed from $\boldsymbol{E}$. Since the support and the Goppa polynomial are known in the experiment, the canonical relations of the m conjugate ambient representations and the $m-1$ relations of $\mathcal{H}_\infty$ are obtained by univariate Hermite interpolation on $\mathcal{R}_d(\boldsymbol{p})$ followed by the chain-rule lift (28), while $\mathcal{K}_{\mathrm{loc}}$ is generated explicitly from the one-block Euler–Hasse identities and the few-block derivative-evaluation dependencies described above. Under the decomposition (39), the locator test in Algorithm 4 uses only the projection of $\ker(\boldsymbol{E})$ onto the hold-out blocks, reducing each candidate check to a small rank test. For a key with $(m,t,n,k,D,d,s) = (7,10,107,37,86,5,3)$

and multiplicity profile $3^4\,4^{103}$, the local relations have exactly the rank predicted by (41) on every hold-out block. As expected, with three hold-outs the candidate set at every coordinate is exactly the seven normalized Frobenius conjugates of its locator, and with four hold-outs it is the singleton $\{\alpha_j\}$ at all 103 non-hold-out coordinates. In short, rather than computing $\pi_j(\ker(\boldsymbol{E}))$ from the full jet matrix, we run the locator test on the projection predicted by (39). The resulting candidate sets provide evidence for Conjecture C.3, conditional on that kernel decomposition; an unpredicted kernel direction could only enlarge them.

End-to-end Recovery Tests. We tested Algorithm 4 end-to-end for every structured parameter set in Table 10 except the last one, rather than only measuring kernel dimensions or testing the univariate coefficient model. Every trial recovered the target support and returned a valid ambient GRS representation as expected. With the assistance of Steve Weis from Anthropic, we additionally ran the binary Goppa variant of Algorithm 4 on the full multivariate recovery system for a binary Goppa key with $(m, t, n) = (6, 6, 64)$, whose support lies in $\mathbb{F}_q = \mathbb{F}_{64}$. The implementation by Weis is an optimized variant that may rely on additional conjectures, incorporates the efficiency improvements in Remark 7.3, and uses evaluation at the public point $\boldsymbol{p}_\infty$ as one additional interpolation constraint to relax the parameter conditions, but it preserves the canonical recovery branch, so a successful optimized run also gives a successful branch of the unoptimized algorithm. With three hold-outs, the candidate set at every coordinate consisted exactly of the six Frobenius conjugates of the true locator. The consistent values at a fourth hold-out were likewise exactly its six Frobenius conjugates, so these values could be read from the recovery system instead of guessed over $\mathbb{F}_q$. After fixing any one of them, every coordinate had exactly the corresponding conjugate locator as its unique accepted value. Thus, the recovery algorithm succeeded end-to-end on this binary Goppa instance, recovering the complete support and yielding a working decoder.